

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
SAN DIEGO REGION**

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**ORDER NO. R9-2021-0204
AMENDING**

**ORDER NO. R9-2019-0169 as amended by Order No. R9-2020-0193
NPDES NO. CA0108031**

**WASTE DISCHARGE REQUIREMENTS
FOR THE FALLBROOK PUBLIC UTILITY DISTRICT,
FALLBROOK WATER RECLAMATION PLANT AND
SANTA MARGARITA GROUNDWATER TREATMENT PLANT
DISCHARGE TO THE PACIFIC OCEAN THROUGH THE OCEANSIDE OCEAN
OUTFALL**

The California Regional Water Quality Control Board, San Diego Region (San Diego Water Board), finds that:

Background

1. The Fallbrook Public Utility District (FPUD) is the owner and operator of the Fallbrook Water Reclamation Plant (Fallbrook WRP) and Santa Margarita Groundwater Treatment Plant (Santa Margarita GTP) (separately and collectively referred to herein as Facility). The Facility is described in Section II of Order No. R9-2019-0169.
2. Order No. R9-2019-0169 was adopted by the San Diego Water Board on February 12, 2020 and amended by Order No. R9-2020-0193 on August 12, 2020. Order No. R9-2019-0169 is a National Pollutant Discharge Elimination System (NPDES) permit which establishes waste discharge requirements for the Facility for the discharge of tertiary-treated wastewater and brine waste through the Oceanside Ocean Outfall to the Pacific Ocean. The Oceanside Ocean Outfall is owned and operated by the City of Oceanside.

Purpose of this Order

3. This Order makes modifications to Order No. R9-2019-0169 as directed by the California State Water Resources Control Board (State Water Board) on remand pursuant to Order WQ 2021-0005, as well as to more accurately describe the FPUD's land outfall structure, update the monitoring and reporting program to be consistent with recently adopted NPDES permits in the San Diego Region, and correct non-substantive typographical errors.

Legal Authorities

4. Order No. R9-2019-0169 is not being reopened for any other purpose than the revisions contained herein. Except as contradicted or superseded by the findings and directives set forth in this Order, all the previous findings and directives of Order No. R9-2019-0169 shall remain in full force and effect.

California Environmental Quality Act

5. This action is exempt from the requirement of preparation of environmental documents under the California Environmental Quality Act (Public Resources Code, division 13, chapter 3, section 21000 et seq.) in accordance with section 13389 of the Water Code.

Public Participation

6. The San Diego Water Board has notified all known interested parties of its intent to adopt this Order.
7. The San Diego Water Board in a public meeting on December 8, 2021, heard and considered all comments pertaining to the adoption of this Order.
8. Any person aggrieved by this action of the San Diego Water Board may petition the State Water Resources Control Board (State Water Board) to review the action in accordance with Water Code section 13320 and California Code of Regulations, title 23, sections 2050 et seq. The State Water Board must receive the petition by 5:00 p.m., 30 days after the adoption date of this Order. Copies of the law and regulations applicable to filing petitions may be found on the Internet at: https://www.waterboards.ca.gov/public_notices/petitions/water_quality or will be provided upon request.

IT IS HEREBY ORDERED:

9. This Order amends Order No. R9-2019-0169 as amended by Order No. R9-2020-0193, NPDES No. CA0108031 as described in the revised version included as Attachment 1 to this Order. Added text to Order No. R9-2019-0169 is displayed in redline text and deleted text is displayed as strikeout text. Modifications to Order No. R9-2019-0169 were made to the following sections:

<u>Pages No</u>	<u>Section No.</u>
-	Cover Pages
Entire Order	Modified all Websites Throughout the Order to Include “s” in “https”
7	Section IV.A.2
7 to 10	Section IV.A.2, Table 8
17	Section VI.C.2.b.iv
32 to 33	Section VII.L
A-1 to A-4	Table for abbreviations
A-7	Definition for chronic toxicity
B-2	Replaced map

<u>Pages No</u>	<u>Section No.</u>
B-4	New map added
E-8 to E-11	Attachment E, Section III.B.2, Table E-4
E-12	Attachment E, section III.C, Monitoring Questions 1 through 3
E-13 to E-14	Attachment E, Section III.C.4
E-14 to E-15	Attachment E, Sections III.C.5.a and d
E-16	Attachment E, Section III.C.7
E-16	Attachment E, Section III.C.8
E-17 to E-18	Attachment E, Sections III.C.9.a, c, and e
E-19 to E-20	Attachment E, Section IV.A, Table E-6
E-20	Attachment E, Section IV.B, Monitoring Question 6
E-21	Attachment E, Section IV.B.1, Table E-7
E-21 to E-22	Attachment E, Section IV.B.2
E-22 to E-23	Attachment E, Section IV.B.3
E-25 to E-26	Attachment E, Section IV.C.1, Table E-8
E-26	Attachment E, Section IV.C.2.c
E-27	Attachment E, Section IV.C.3.b
E-31 to E-32	Attachment E, Section IV.D.2, Table E-9
E-32 to E-33	Attachment E, Section IV.E
E-37	Attachment E, Section VI.A
E-44 to E-45	Attachment E, Section VII.D, Table E-11
F-7 to F-10	Attachment F, Section II.B
F-10 to F-30	Attachment F, tables renumbered
F-27	Attachment F, Section IV.C.4.f Table F-12
F-27 to F-28	Attachment F, Section IV.C.4.g
F-28 to F-30	Attachment F, Section IV.C.4.g, Table F-13
F-30 to F-34	Attachment F, Sections IV.C.5.b and d
F-42	Attachment F, Section VII.A.2.c
F-43	Attachment F, Section VII.A.2.i
F-43 to F-44	Attachment F, Section VII.A.3
F-45	Attachment F, Section VII.B.1
F-45 to F-47	Attachment F, Sections VII.B.2.a, b, c, e, and f
F-48 to F-49	Attachment F, Section VII.B.3
F-50 to F-51	Attachment F, Section VII.B.5
F-53	Attachment F, Section VII.D.1

10. The amended version of Order No. R9-2019-0169 included as an attachment to this Order shall become effective on December 8, 2021.
11. San Diego Water Board staff is directed to prepare and post a conformed copy of Order No. R9-2019-0169 incorporating the revisions made by this Order.

I, David W. Gibson, Executive Officer, do hereby certify that this Order is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, San Diego Region, on **December 8, 2021**.

David W. Gibson
Executive Officer

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**ORDER NO. R9-2019-0169
AS AMENDED BY ORDER NO. S. R9-2020-0193 AND R9-2021-0204
NPDES NO. CA0108031**

**WASTE DISCHARGE REQUIREMENTS
FOR THE FALLBROOK PUBLIC UTILITY DISTRICT,
FALLBROOK WATER RECLAMATION PLANT AND
SANTA MARGARITA GROUNDWATER TREATMENT PLANT
DISCHARGE TO THE PACIFIC OCEAN THROUGH THE OCEANSIDE OCEAN OUTFALL**

The following Discharger is subject to waste discharge requirements (WDRs) set forth in this Order:

Table 1. Discharger Information

Discharger	Name of Facility	Facility Address
Fallbrook Public Utility District	Fallbrook Water Reclamation Plant (Fallbrook WRP)	1425 S. Alturas Road Fallbrook, CA 92028
Fallbrook Public Utility District	Santa Margarita Groundwater Treatment Plant (Santa Margarita GTP)	1425 S. Alturas Road Fallbrook, CA 92028

Table 2. Discharge Location

Discharge Point	Effluent Description	Discharge Point Latitude (North)	Discharge Point Longitude (West)	Receiving Water
001	Dechlorinated tertiary-treated wastewater and brine waste	33° 09' 46" N	117° 23' 29" W	Pacific Ocean

Table 3. Administrative Information

Information	Date/Classification
This Order was adopted on:	February 12, 2020
This Order shall become effective on:	April 1, 2020
This Order shall expire on:	March 31, 2025
The Discharger shall file a Report of Waste Discharge (ROWD) as an application for reissuance of WDRs in accordance with title 23, California Code of Regulations (CCR), and an application for reissuance of a National Pollutant Discharge Elimination System (NPDES) permit no later than:	180 days prior to the Order expiration date
The U.S. Environmental Protection Agency (USEPA) and the California Regional Water Quality Control Board, San Diego Region have classified this discharge as follows:	Major

I, David W. Gibson, Executive Officer, do hereby certify that this Order with all attachments is a full, true, and correct copy of the Order adopted by the California Regional Water Quality Control Board, San Diego Region, on the date indicated above, as amended by Order No. R9-2020-0192 on August 12, 2020 and by Order No. R9-2021-0204 on December 8, 2021.

David W. Gibson, Executive Officer

CONTENTS

I.	Facility Information.....	3
II.	Findings	3
III.	Discharge Prohibitions.....	4
IV.	Effluent Limitations and Discharge Specifications	4
	A. Effluent Limitations and Performance Goals.....	4
	1. Effluent Limitations	4
	2. Performance Goals.....	7
	B. Land Discharge Specifications – Not Applicable	11
	C. Recycling Specifications – Not Applicable	11
V.	Receiving Water Limitations	11
	A. Surface Water Limitation	11
	B. Groundwater Limitations – Not Applicable	12
VI.	Provisions	12
	A. Standard Provisions.....	12
	B. Monitoring and Reporting Program (MRP) Requirements	13
	C. Special Provisions	13
	1. Reopener Provisions	13
	2. Special Studies, Technical Reports and Additional Monitoring Requirements....	15
	3. Best Management Practices and Pollution Prevention	17
	4. Construction, Operation and Maintenance Specifications	18
	5. Special Provisions for Wastewater Facilities	20
	6. Other Special Provisions – Not Applicable	28
	7. Compliance Schedules – Not Applicable	28
VII.	Compliance Determination	28

TABLES

Table 1.	Discharger Information	1
Table 2.	Discharge Location	1
Table 3.	Administrative Information	1
Table 4.	Effluent Limitations at Monitoring Location M-001	5
Table 5.	Effluent Limitations at Monitoring Location M-002	6
Table 6.	Effluent Limitations at Monitoring Location M-002	6
Table 7.	Effluent Limitations at Monitoring Location M-003 ¹	6
Table 8.	Performance Goals at Monitoring Location M-002.....	7

ATTACHMENTS

Attachment A – Abbreviations and Definitions.....	A-1
Attachment B – Maps	B-1
Attachment C – Flow Schematics.....	C-1
Attachment D – Standard Provisions.....	D-1
Attachment E – Monitoring and Reporting Program	E-1
Attachment F – Fact Sheet	F-1
Attachment G – Discharge Prohibitions contained in the Ocean Plan and Basin Plan	G-1

I. FACILITY INFORMATION

Information describing the Fallbrook Water Reclamation Plant (Fallbrook WRP), Santa Margarita Groundwater Treatment Plant (Santa Margarita GTP), and the associated sanitary sewer system, pump stations, and land outfalls (collectively referred to as the Facilities) is summarized in Table 1 and in sections I and II of the Fact Sheet (Attachment F). Section I of the Fact Sheet also includes information regarding the Fallbrook Public Utility District (Discharger's) permit application.

II. FINDINGS

The California Regional Water Quality Control Board, San Diego Region (San Diego Water Board), finds:

- A. Legal Authorities.** This Order serves as WDRs pursuant to article 4, chapter 4, division 7 of the California Water Code (Water Code) (commencing with section 13260). This Order is also issued pursuant to section 402 of the federal Clean Water Act (CWA) and implementing regulations adopted by the USEPA and chapter 5.5, division 7 of the Water Code (commencing with section 13370). This Order shall serve as an NPDES permit authorizing the Discharger to discharge into waters of the United States at the discharge location described in Table 2 subject to the WDRs in this Order.
- B. Background and Rationale for Requirements.** The San Diego Water Board developed the requirements in this Order based on information submitted as part of the application, through monitoring and reporting programs, and other available information. The Fact Sheet (Attachment F), which contains background information and rationale for the requirements in this Order, is hereby incorporated into and constitutes Findings for this Order. Attachments A through E, and G are also incorporated into this Order.
- C. Provisions and Requirements Implementing State Law.** The provisions/requirements in subsections IV.B, IV.C, and V.B, VI.A.2.a, VI.C.1.b, VI.C.1.c, and VI.C.4.a-d are included to implement State law only. These provisions/requirements are not required or authorized under the federal CWA; consequently, violations of these provisions/requirements are not subject to the enforcement remedies that are available for NPDES violations.
- D. Executive Officer Delegation of Authority.** The San Diego Water Board by prior resolution has delegated all matters that may legally be delegated to its Executive Officer to act on its behalf pursuant to Water Code section 13223. Therefore, the Executive Officer is authorized to act on the San Diego Water Board's behalf on any matter within this Order, unless such delegation is unlawful under Water Code section 13223 or this Order explicitly states otherwise.
- E. Notification of Interested Parties.** The San Diego Water Board has notified the Discharger and interested agencies and persons of its intent to prescribe WDRs for the discharge and has provided them with an opportunity to submit their written comments and recommendations. The San Diego Water Board has also provided an opportunity for the Discharger and interested agencies and persons to submit oral

comments and recommendations at a public hearing. Details of the notification are provided in the Fact Sheet (Attachment F).

- F. Consideration of Public Comment.** The San Diego Water Board, in a public meeting, heard and considered all comments pertaining to the discharge. Details of the Public Hearing are provided in the Fact Sheet.

THEREFORE, IT IS HEREBY ORDERED, that this Order supersedes Order No. R9-2012-0004 except for enforcement purposes, and, in order to meet the provisions contained in division 7 of the Water Code (commencing with section 13000) and regulations adopted thereunder, and the provisions of the CWA and regulations and guidelines adopted thereunder, the Discharger shall comply with the requirements in this Order. The Discharger is hereby authorized to discharge subject to WDRs in this Order at the discharge location described in Table 2 to the Pacific Ocean off the coast of San Diego County. This action in no way prevents the San Diego Water Board from taking enforcement action for past violations of Order No. R9-2012-0004.

III. DISCHARGE PROHIBITIONS

- A. The discharge of waste from the Fallbrook WRP not treated by a secondary treatment process, not in compliance with the effluent limitations specified in section IV.A of this Order, and/or to a location other than Discharge Point No. 001, unless specifically regulated by this Order or separate WDRs, is prohibited.
- B. The discharge of waste from the Santa Margarita GTP not in compliance with the effluent limitations specified in section IV.A of this Order, and/or to a location other than Discharge Point No. 001, unless specifically regulated by this Order or separate WDRs, is prohibited.
- C. The Discharger must comply with any applicable Discharge Prohibitions contained in the *Water Quality Control Plan for Ocean Waters of California, California Ocean Plan* (Ocean Plan), incorporated into this Order as if fully set forth herein and summarized in Attachment G, as a condition of this Order.
- D. The Discharger must comply with any applicable Discharge Prohibitions contained in chapter 4 of the *Water Quality Control Plan for the San Diego Basin* (Basin Plan), incorporated into this Order as if fully set forth herein and summarized in Attachment G, as a condition of this Order.
- E. The bypassing of untreated wastes is prohibited, except as allowed by federal Standard Provisions I.G or I.H of this Order. (Attachment D).

IV. EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS

A. Effluent Limitations and Performance Goals

1. Effluent Limitations

- a. The Discharger shall maintain compliance with the following effluent limitations for the Fallbrook WRP, with compliance measured at Monitoring Location M-001, as described in the Monitoring and Reporting Program (MRP, Attachment E). The mass emission rate (MER) limitation, in lbs/day, was calculated based on the following equation: $MER \text{ (lbs/day)} = 8.34 \times Q \times$

C, where Q is the permitted flow for the Fallbrook WRP (2.7 MGD) and C is the concentration (mg/L).

Table 4. Effluent Limitations at Monitoring Location M-001

Parameter	Units	Average Monthly	Average Weekly	Instantaneous Minimum	Instantaneous Maximum
Carbonaceous Biochemical Oxygen Demand 5-day @ 20°C (CBOD ₅)	milligram per liter (mg/L)	25	40	--	--
CBOD ₅	pounds per day (lbs/day)	560	900	--	--
CBOD ₅	% Removal	≥85	--	--	--
Total Suspended Solids (TSS)	mg/L	30	45	--	--
TSS	lbs/day	680	1,000	--	--
TSS	% Removal	≥85	--	--	--
pH	standard units	--	--	6.0	9.0
Oil and Grease	mg/L	25	40	--	75
Oil and Grease	lbs/day	560	900	--	1,700
Settleable Solids	milliliter per liter (ml/L)	1.0	1.5	--	3.0
Turbidity	nephelometric turbidity unit (NTU)	75	100	--	225

- b. The Discharger shall maintain compliance with the following effluent limitations at Discharge Point No. 001, with compliance measured at Monitoring Location M-002, as described in the MRP (Attachment E). Scientific "E" notation is used to express certain values. In scientific "E" notation, the number following the "E" indicates that position of the decimal point in the value. Negative numbers after the "E" indicate that the value is less than 1, and positive numbers after the "E" indicate that the value is greater than 1. In this notation a value of 6.1 E-02 represents 6.1 x 10⁻² or 0.061, 6.1E+02 represents 6.1 x 10² or 610, and 6.1E+00 represents 6.1 x 10⁰ or 6.1. The MER limitation, in lbs/day, was calculated based on the following equation: MER (lbs/day) = 8.34 x Q x C, where Q is the permitted flow for the Facilities (2.7 MGD) and C is the concentration (mg/L).

Table 5. Effluent Limitations at Monitoring Location M-002

Parameter	Unit	Six-Month Median	Average Monthly	Maximum Daily	Instantaneous Maximum
Total Chlorine Residual	microgram per liter (µg/L)	1.8E+02	--	7.0E+02	5.3E+03
Total Chlorine Residual	lbs/day	4.0E+00	--	1.6E+01	1.2E+02
Heptachlor	µg/L	--	4.4E-03	--	--
Heptachlor	lbs/day	--	9.9E-05	--	--
Heptachlor Epoxide	µg/L	--	1.8E-03	--	--
Heptachlor Epoxide	lbs/day	--	4.0E-05	--	--
TCDD Equivalents	µg/L	--	3.4E-07	--	--
TCDD Equivalents	lbs/day	--	7.7E-09	--	--

- c. The Discharger shall maintain compliance with the following effluent limitations at Discharge Point No. 001, with compliance measured at Monitoring Location M-002, as described in the MRP (Attachment E):

Table 6. Effluent Limitations at Monitoring Location M-002

Parameter	Unit	Average Monthly (May to October)	Average Monthly (November to April)
Flow	million gallons per day (MGD)	2.7	3.6

- d. The Discharger shall maintain compliance with the following effluent limitations for the Santa Margarita GTP, with compliance measured at Monitoring Location M-003, as described in the MRP (Attachment E). The mass emission rate (MER) limitation, in lbs/day, was calculated based on the following equation: $MER \text{ (lbs/day)} = 8.34 \times Q \times C$, where Q is the maximum expected flow for the Santa Margarita GTP (0.65 MGD) and C is the concentration (mg/L).

Table 7. Effluent Limitations at Monitoring Location M-003¹

Parameter	Units	Average Monthly	Average Weekly	Instantaneous Minimum	Instantaneous Maximum
TSS	mg/L	60	--	--	--
TSS	lbs/day	325	--	--	--
pH	standard units	--	--	6.0	9.0
Oil and Grease	mg/L	25	40	--	75
Oil and Grease	lbs/day	135	217	--	407
Settleable Solids	milliliter per liter (ml/L)	1.0	1.5	--	3.0

Parameter	Units	Average Monthly	Average Weekly	Instantaneous Minimum	Instantaneous Maximum
Turbidity	nephelometric turbidity unit (NTU)	75	100	--	225

2. Performance Goals

Parameters that do not have reasonable potential to cause or contribute to an exceedance of water quality objectives, or for which reasonable potential to cause or contribute to an exceedance of water quality objectives cannot be determined, are referred to as performance goal parameters and are assigned the performance goals listed in Table 8 below. Performance goal parameters shall be monitored at Monitoring Location M-002, as described in the MRP (Attachment E). ~~The San Diego Water Board will use the results for informational purposes only, not compliance determinations.~~ The performance goals in Table 8 are not water quality-based effluent limitations (WQBELs) and are not enforceable, as such. However, the Discharger shall maintain, if not improve, the effluent quality to levels at or below the performance goals. The Discharger shall report all exceedances of performance goals in the cover letter of the applicable self-monitoring report (SMR). Any two consecutive exceedances of the performance goals shall trigger an investigation into the cause of the exceedance. If the exceedance persists in three successive monitoring events, the Discharger shall submit a written Performance Goal Exceedance Report to the San Diego Water Board within 30 days on the nature of the exceedance, the results of the investigation including the cause of the exceedance, the corrective actions taken, any proposed corrective measures, and a timetable for implementation, if necessary. The San Diego Water Board may reopen this Order to include effluent limitations for parameters that exceed performance goals. Scientific "E" notation is used to express certain values. In scientific "E" notation, the number following the "E" indicates that position of the decimal point in the value. Negative numbers after the "E" indicate that the value is less than 1, and positive numbers after the "E" indicate that the value is greater than 1. In this notation a value of 6.1 E-02 represents 6.1×10^{-2} or 0.061, 6.1E+02 represents 6.1×10^2 or 610, and 6.1E+00 represents 6.1×10^0 or 6.1.

Table 8. Performance Goals at Monitoring Location M-002

Parameter	Units	Six-Month Median	Average Monthly	Maximum Daily	Instantaneous Maximum
Arsenic, Total Recoverable	µg/L	4.43E+02	--	2.56E+03	6.78E+03
Cadmium, Total Recoverable	µg/L	8.80E+01	--	3.52E+02	8.80E+02
Chromium (VI), Total Recoverable (note 1)	µg/L	1.76E+02	--	7.04E+02	1.76E+03
Copper, Total Recoverable	µg/L	9.00E+01	--	8.82E+02	2.47E+03

Parameter	Units	Six-Month Median	Average Monthly	Maximum Daily	Instantaneous Maximum
Lead, Total Recoverable	µg/L	1.76E+02	--	7.04E+02	1.76E+03
Mercury, Total Recoverable	µg/L	3.48E+00	--	1.40E+01	3.52E+01
Nickel, Total Recoverable	µg/L	4.40E+02	--	1.76E+03	4.40E+03
Selenium, Total Recoverable	µg/L	1.32E+03	--	5.28E+03	1.32E+04
Silver, Total Recoverable	µg/L	4.77E+01	--	2.32E+02	6.02E+02
Zinc, Total Recoverable	µg/L	1.06E+03	--	6.34E+03	1.69E+04
Cyanide, Total	µg/L	8.80E+01	--	3.52E+02	8.80E+02
Ammonia Nitrogen, Total (as N)	µg/L	5.28E+04	--	2.11E+05	5.28E+05
Chronic Toxicity (Test of Significant Toxicity) (Notes 3 and 4)	<u>Toxicity</u> <u>Units</u> <u>Chronic</u> <u>(TUc)</u> “Pass”/ “Fail”	--	--	<u>8.8E+01</u> “Pass”	--
Phenolic Compounds (non-chlorinated)	µg/L	2.64E+03	--	1.06E+04	2.64E+04
Chlorinated Phenolics	µg/L	8.80E+01	--	3.52E+02	8.80E+02
Endosulfan	µg/L	7.92E-01	--	1.58E+00	2.38E+00
Endrin	µg/L	1.76E-01	--	3.52E-01	5.28E-01
HCH (BHC)	µg/L	3.52E-01	--	7.04E-01	1.06E+00
Radioactivity (alpha and beta particles)	picocuries per liter (pCi/L)	(note 2)			
Acrolein	µg/L	--	1.94E+04	--	--
Antimony	µg/L	--	1.06E+05	--	--
Bis(2-chloroethoxy) Methane	µg/L	--	3.87E+02	--	--
Bis(2-chloroisopropyl) Ether	µg/L	--	1.06E+05	--	--
Chlorobenzene	µg/L	--	5.02E+04	--	--
Chromium (III), Total Recoverable (note 1)	µg/L	--	1.67E+07	--	--
Di-n-butyl Phthalate	µg/L	--	3.08E+05	--	--
Dichlorobenzenes ¹	µg/L	--	4.49E+05	--	--
Diethyl Phthalate	µg/L	--	2.90E+06	--	--
Dimethyl Phthalate	µg/L	--	7.22E+07	--	--
4,6-dinitro-2-methylphenol	µg/L	--	1.94E+04	--	--
2,4-dinitrophenol	µg/L	--	3.52E+02	--	--
Ethylbenzene	µg/L	--	3.61E+05	--	--

Parameter	Units	Six-Month Median	Average Monthly	Maximum Daily	Instantaneous Maximum
Fluoranthene	µg/L	--	1.32E+03	--	--
Hexachlorocyclopentadiene	µg/L	--	5.10E+03	--	--
Nitrobenzene	µg/L	--	4.31E+02	--	--
Thallium, Total Recoverable	µg/L	--	1.76E+02	--	--
Toluene	µg/L	--	7.48E+06	--	--
Tributyltin	µg/L	--	1.23E-01	--	--
1,1,1-trichloroethane	µg/L	--	4.75E+07	--	--
Acrylonitrile	µg/L	--	8.80E+00	--	--
Aldrin	µg/L	--	1.94E-03	--	--
Benzene	µg/L	--	5.19E+02	--	--
Benzidine	µg/L	--	6.07E-03	--	--
Beryllium, Total Recoverable	µg/L	--	2.90E+00	--	--
Bis(2-chloroethyl) Ether	µg/L	--	3.96E+00	--	--
Bis(2-ethylhexyl) Phthalate	µg/L	--	3.1E+02	--	--
Carbon Tetrachloride	µg/L	--	7.92E+01	--	--
Chlordane	µg/L	--	2.02E-03	--	--
Chlorodibromomethane (Dibromochloromethane)	µg/L	--	7.57E+02	--	--
Chloroform	µg/L	--	1.14E+04	--	--
Dichlorodiphenyltrichloroethane (DDT)	µg/L	--	1.50E-02	--	--
1,4-dichlorobenzene	µg/L	--	1.58E+03	--	--
3,3'-dichlorobenzidine	µg/L	--	7.13E-01	--	--
1,2-dichloroethane	µg/L	--	2.46E+03	--	--
1,1-dichloroethylene	µg/L	--	7.92E+01	--	--
Dichlorobromomethane	µg/L	--	5.5E+02	--	--
Dichloromethane (Methylene Chloride)	µg/L	--	3.96E+04	--	--
1,3-dichloropropene (1,3-Dichloropropylenes)	µg/L	--	7.83E+02	--	--
Dieldrin	µg/L	--	3.52E-03	--	--
2,4-dinitrotoluene	µg/L	--	2.29E+02	--	--
1,2-diphenylhydrazine	µg/L	--	1.41E+01	--	--
Halomethanes	µg/L	--	1.14E+04	--	--
Hexachlorobutadiene	µg/L	--	1.23E+03	--	--
Hexachloroethane	µg/L	--	2.20E+02	--	--

Parameter	Units	Six-Month Median	Average Monthly	Maximum Daily	Instantaneous Maximum
Isophorone	µg/L	--	6.42E+04	--	--
N-nitrosodimethylamine	µg/L	--	6.42E+02	--	--
N-nitrosodi-N-propylamine	µg/L	--	3.34E+01	--	--
N-nitrosodiphenylamine	µg/L	--	2.20E+02	--	--
Polynuclear Aromatic Hydrocarbons (PAHs)	µg/L	--	7.74E-01	--	--
Polychlorinated Biphenyls (PCBs)	µg/L	--	1.67E-03	--	--
1,1,2,2-tetrachloroethane	µg/L	--	2.02E+02	--	--
Tetrachloroethylene (Tetrachloroethene)	µg/L	--	1.76E+02	--	--
Toxaphene	µg/L	--	1.85E-02	--	--
Trichloroethylene (Trichloroethene)	µg/L	--	2.38E+03	--	--
1,1,2-trichloroethane	µg/L	--	8.27E+02	--	--
2,4,6-trichlorophenol	µg/L	--	2.55E+01	--	--
Vinyl Chloride	µg/L	--	3.17E+03	--	--

Note 1 for Table 8: Dischargers may, at their option, apply this performance goal as a total chromium performance goal.

Note 2 for Table 8: Not to exceed limits specified in title 17, division 1, chapter 5, subchapter 4, group 3, article 3, section 30253 of the CCR, reference to section 30253 is prospective, including future changes to any incorporated provisions of federal law, as the changes take effect.

Note 3 for Table 8: As specified in section VII.L of this Order and section III.C of the MRP (Attachment E).

Note 4 for Table 8: The chronic toxicity performance goal is protective of both the numeric acute and chronic toxicity Ocean Plan water quality objectives. The performance goal shall be implemented using *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to West Coast Marine and Estuarine Organisms* (EPA/600/R-95/136, 1995). Chronic toxicity expressed as Toxic Units Chronic (TUC) = 100/NOEC, where NOEC (No Observed Effect Concentration; also referred to as the No Observed Effect Level or NOEL) is expressed as the maximum percent effluent or receiving water that causes no observable effect on a test organism.
~~The chronic toxicity performance goal is protective of both the numeric acute and chronic toxicity Ocean Plan water quality objectives. The performance goal will be implemented using Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to West Coast Marine and Estuarine Organisms (EPA/600/R-95/136, 1995); current USEPA guidance in the National Pollutant Discharge Elimination System Test of Significant Toxicity Implementation Document (EPA 833-R-10-003, June 2010) (https://www3.epa.gov/npdes/pubs/wet_final_tst_implementation2010.pdf); and USEPA Regions 8, 9, and 10, Toxicity Training Tool (January 2010).~~

B. Land Discharge Specifications – Not Applicable

C. Recycling Specifications – Not Applicable

V. RECEIVING WATER LIMITATIONS

A. Surface Water Limitation

The receiving water limitations set forth below for ocean waters are based on water quality objectives contained in the Basin Plan and Ocean Plan and are a required part of this Order. The discharge of waste shall not cause or contribute to violation of these limitations in the Pacific Ocean. Compliance with these limitations shall be determined from samples collected at stations representative of the area outside of the zone of initial dilution (ZID).

1. Bacterial Characteristics

- a. Within a zone bounded by the shoreline and a distance of three nautical miles from the shoreline, including all kelp beds, the following bacterial objectives shall be maintained throughout the water column. The ZID for the ocean outfall is excluded.
 - i. Fecal Coliform
 - (a) Thirty-day geometric mean of fecal coliform density not to exceed 200 CFU per 100 milliliters (mL) calculated based on the five most recent samples from each site.
 - (b) Single sample maximum not to exceed 400 CFU per 100 mL.
 - ii. Enterococci
 - (a) Six-week rolling geometric mean not to exceed 30 CFU per 100 mL, calculated weekly.
 - (b) Statistical threshold value (STV) of 110 CFU per 100 mL not to be exceeded by more than 10 percent of samples collected in a calendar month, calculated in a static manner.
- b. The ZID of any wastewater outfall shall be excluded from designation as kelp beds for purposes of bacterial standards. Adventitious assemblages of kelp plants on waste discharge structures (e.g., outfall pipes and diffusers) do not constitute kelp beds for purposes of bacterial standards.
- c. At all areas where shellfish may be harvested for human consumption, as determined by the San Diego Water Board, the median total coliform density (CFU) shall not exceed 70 per 100 ml throughout the water column, and not more than 10 percent of the samples shall exceed 230 per 100 ml.

2. Physical Characteristics

- a. Floating particulates and grease and oils shall not be visible.
- b. The discharge of waste shall not cause aesthetically undesirable discoloration of the ocean surface.

- c. Natural light shall not be significantly reduced at any point outside the initial dilution zone as a result of the discharge of waste.
- d. The rate of deposition of inert solids and the characteristics of inert solids in the ocean sediments shall not be changed such that benthic communities are degraded.
- e. Trash shall not be present in ocean waters, along shorelines or adjacent areas in amounts that adversely affect beneficial uses or cause nuisance.

3. Chemical Characteristics

- a. The dissolved oxygen concentration shall not at any time be depressed more than 10 percent from that which occurs naturally, as the result of the discharge of oxygen demanding waste materials.
- b. The pH shall not be changed at any time more than 0.2 standard units from that which occurs naturally.
- c. The dissolved sulfide concentration of waters in and near sediments shall not be significantly increased above that present under natural conditions.
- d. The concentration of substances set forth in chapter II, Table 3 of the Ocean Plan, shall not be increased in marine sediments to levels that would degrade indigenous biota.
- e. The concentration of organic materials in marine sediments shall not be increased to levels that would degrade marine life.
- f. Nutrient materials shall not cause objectionable aquatic growths or degrade indigenous biota.

4. Biological Characteristics

- a. Marine communities, including vertebrate, invertebrate, and plant species, shall not be degraded.
- b. The natural taste, odor, color of fish, shellfish, or other marine resources used for human consumption shall not be altered.
- c. The concentration of organic materials in fish, shellfish, or other marine resources used for human consumption shall not bioaccumulate to levels that are harmful to human health.

5. Radioactivity

Discharge of radioactive waste shall not degrade marine life.

B. Groundwater Limitations – Not Applicable

VI. PROVISIONS

A. Standard Provisions

- 1. **Federal Standard Provisions.** The Discharger shall comply with all applicable Standard Provisions included in Attachment D of this Order.

2. **San Diego Water Board Standard Provisions.** The Discharger shall comply with the following provisions. In the event that there is any conflict, duplication, or overlap between provisions specified by this Order, the more stringent provision shall apply.
 - a. The Facilities shall be supervised and operated by persons possessing certificates of appropriate grade pursuant to title 23, division 3, chapter 26 of the CCR. The Facilities shall be provided with a sufficient number of qualified personnel to operate the Facilities effectively so as to achieve the required level of treatment at all times.
 - b. The expiration date of this Order is contained in Table 3 of this Order. After the expiration date, the terms and conditions of this Order are automatically continued pending issuance of a new permit, provided that all requirements of USEPA's NPDES regulations at title 40 of the Code of Federal Regulations (40 CFR) section 122.6 and the State's regulations at title 23, division 3, chapter 9, article 3, section 2235.4 of the CCR regarding the continuation of expired permits and WDRs are met.
 - c. A copy of this Order shall be posted at a prominent location and shall be available to site personnel, San Diego Water Board, State Water Resources Control Board (State Water Board), and USEPA or their authorized representative at all times.

B. Monitoring and Reporting Program (MRP) Requirements

1. The Discharger shall comply with the MRP, and future revisions thereto, in Attachment E of this Order.
2. Notifications required to be provided under this Order to the San Diego Water Board shall be made to:
E-mail - SanDiego@waterboards.ca.gov or
Telephone – (619) 516-1990, or
Facsimile – (619) 516-1994.

C. Special Provisions

1. Reopener Provisions

- a. This Order may be reopened for modification to include an effluent limitation if monitoring establishes that the discharge causes, has the reasonable potential to cause, or contributes to an excursion above a performance goal(s) set forth in section IV.A.2 of this Order or as otherwise described in Table 3 of the Ocean Plan. (40 CFR section 122.44(d)(1))
- b. This Order may be reopened for modification of the monitoring and reporting requirements and/or special studies requirements, at the discretion of the San Diego Water Board. Such modification(s) may include, but is (are) not limited to, revision(s) (i) to implement recommendations from the Southern California Coastal Water Research Project (SCCWRP); (ii) to develop, refine, implement, and/or coordinate a regional monitoring program; (iii) to develop

and implement improved monitoring and assessment programs in keeping with San Diego Water Board Resolution No. R9-2012-0069, *Resolution in Support of a Regional Monitoring Framework*; and/or (iv) to add provisions to require the Discharger to evaluate and provide information on cost and values of the MRP (Attachment E).

- c. This Order may be modified, revoked and reissued, or terminated for cause in accordance with the provisions of 40 CFR parts 122, 124, and 125 at any time prior to its expiration under any of the following circumstances:
 - i. Violation of any terms or conditions of this Order. (Water Code section 13381(a));
 - ii. Obtaining this Order by misrepresentation or failure to disclose fully all relevant facts. (Water Code section 13381(b)); and
 - iii. A change in any condition that requires either a temporary or permanent reduction or elimination of the authorized discharge. (Water Code section 13381(c)).
- d. The filing of a request by the Discharger for modifications, revocation and reissuance, or termination of this Order does not stay any condition of this Order. Notification by the Discharger of planned operational or facility changes, or anticipated noncompliance with this Order does not stay any condition of this Order. (40 CFR section 122.41(f))
- e. If any applicable toxic effluent standard or prohibition (including any schedule of compliance specified in such effluent standard or prohibition) is promulgated under CWA section 307(a) for a toxic pollutant and that standard or prohibition is more stringent than any limitation on the pollutant in this Order, the San Diego Water Board may institute proceedings under these regulations to modify or revoke and reissue this Order to conform to the toxic effluent standard or prohibition. (40 CFR section 122.44(b)(1))
- f. This Order may be reopened and modified for consistency with any new water quality control plan, policy, law, or regulation. (40 CFR section 122.62(a)(3).)
- g. This Order may be reopened and modified to revise effluent limitations as a result of future Ocean Plan, Basin Plan, and/or other statewide Water Quality Control Plan amendments; or the adoption of a total maximum daily load (TMDL) for the receiving water. (40 CFR section 122.62(a)(2))
- h. This Order may be reopened upon submission by the Discharger of new information, the adequacy of which shall be determined by the San Diego Water Board, to provide for dilution credits or a mixing zone, as may be appropriate. (40 CFR section 122.62(a)(2))
- i. This Order may also be reopened and modified, revoked and reissued, or terminated in accordance with the provisions of 40 CFR sections 122.44, 122.62 to 122.64, and 125.62. Causes for taking such actions include, but

are not limited to, failure to comply with any condition of this Order, and endangerment to human health or the environment resulting from the permitted activity.

- j. The performance goals, contained in section IV.A.2 of this Order, may be re-evaluated and modified during this Order term, or this Order may be modified to incorporate WQBELs, in accordance with the requirements set forth at 40 CFR sections 122.62 and 124.5.

2. Special Studies, Technical Reports and Additional Monitoring Requirements

a. Spill Prevention and Response Plans

- i. For purposes of this section of the Order, a spill is a discharge that occurs at or downstream of the Fallbrook WRP's headworks, or influent intake location for the Santa Margarita GTP, including the associated pumps stations and land outfalls, in violation of Discharge Prohibitions III.A and III.B of this Order. A spill may include a discharge of treated or untreated wastewater, or material other than treated or untreated wastewater that causes, may cause, or is caused by significant operational failure, and/or endangers or may endanger human health or the environment. The term "spill" as used in this section of the Order does not include sanitary sewer overflows from the sewage collection system that are covered under section VI.C.5.d of this Order.
- ii. The Discharger shall maintain a Spill Prevention Plan (SPP) and a Spill Response Plan (SRP) for the Facilities, in an up-to-date condition and shall amend the SPP/SRP whenever there is a change (e.g., in the design, construction, operation, or maintenance of the sewerage system or sewerage facilities) which materially affects the potential for spills and the response required for each potential spill. The Discharger shall review and amend the SPP/SRP as appropriate after each spill from the Facilities. The SPP/SRP and any amendments thereto shall be subject to the approval of the San Diego Water Board and shall be modified as directed by the San Diego Water Board. The Discharger shall submit the SPP/SRP and any amendments thereto to the San Diego Water Board upon request of the San Diego Water Board. The Discharger shall ensure that the up-to-date SPP/SRP is readily available to the Facilities personnel at all times and that the sewerage system personnel are familiar with it.

b. Spill Reporting Requirements

The Discharger shall report spills, as defined in section VI.C.2.a.i above, in accordance with the following procedures:

- i. If a spill results in a discharge of treated or untreated wastewater that is equal to or exceeds 1,000 gallons, and/or results in a discharge to a drainage channel and/or surface water, or results in a discharge to a

storm drain that was not fully captured and returned to the sanitary sewer system, the Discharger shall:

- (a) Report the spill to the San Diego Water Board by email at SanDiego@waterboards.ca.gov within 24 hours from the time the Discharger becomes aware of the spill. If email communication is not possible, report the spill by telephone (619-516-1990) within 24 hours from the time the Discharger becomes aware of the spill. The report shall include a description of the spill and its cause; the spill material; the duration of the spill including exact dates and times; the estimated spill volume and its destination; if the spill has not been terminated, the anticipated time it is expected to continue; and steps taken or planned to reduce and/or eliminate the spill.
 - (b) Submit a written report by email at SanDiego@waterboards.ca.gov, as well as any additional pertinent information, to the San Diego Water Board no later than five days from the time the Discharger becomes aware of the spill. The written report must be signed and certified as required by section V of the Standards Provisions (Attachment D).
 - (c) The San Diego Water Board may waive the above-required written report under this provision on a case-by-case basis if the email or oral report has been received within 24 hours.
- ii. If a spill results in a discharge of treated or untreated wastewater less than 1,000 gallons and the discharge does not reach a drainage channel or surface waters, or results in a discharge to a storm drain that was fully captured and returned to the wastewater treatment facility, the Discharger is not required to notify the San Diego Water Board within 24 hours, or provide a 5-day written report.
 - iii. For spills of material other than treated or untreated wastewater that cause, may cause, or are caused by significant operational failure, and/or endangers or may endanger human health or the environment, the Discharger shall:
 - (a) Notify the San Diego Water Board by email at SanDiego@waterboards.ca.gov within 24 hours from the time the Discharger becomes aware of the spill. If email communication is not possible, report the spill by telephone (619-516-1990) within 24 hours from the time the Discharger becomes aware of the spill. The report shall include a description of the spill and its cause; the spill material; the duration of the spill including exact dates and times; the estimated spill volume and its destination; if the spill has not been terminated, the anticipated time it is expected to continue; and steps taken or planned to reduce and/or eliminate the spill.
 - (b) Submit a written report by email at SanDiego@waterboards.ca.gov, as well as any additional pertinent information, to the San Diego

Water Board no later than five days from the time the Discharger becomes aware of the spill. The written report must be signed and certified as required by section V of the Standards Provisions (Attachment D).

- (c) The San Diego Water Board may waive the above-required written report under this provision on a case-by-case basis if the email or oral report has been received within 24 hours.
- iv. For all spills, the Discharger shall include a detailed summary of spills in the monthly ~~Self-Monitoring Report (SMR)~~ for the month in which the spill occurred. If no spills occurred during the calendar month, the Discharger shall report no spills in the monthly SMR for that calendar month.
- v. The spill reporting requirements contained in this Order do not relieve the Discharger of responsibilities to report spills to other agencies, such as the California Office of Emergency Services and the County of San Diego Department of Environmental Health Services.

3. Best Management Practices and Pollution Prevention

Pollutant Minimization Program (PMP) - The goal of the PMP shall be to reduce all potential sources of a pollutant through pollutant minimization (control) strategies, including pollution prevention measures as appropriate, to maintain the effluent concentration at or below the effluent limitation. Pollution prevention measures may be particularly appropriate for persistent bioaccumulative priority pollutants where there is evidence that beneficial uses are being impacted. The San Diego Water Board may consider cost-effectiveness when establishing the requirements of a PMP. The completion and implementation of a Pollution Prevention Plan (PPP), if required pursuant to Water Code section 13263.3(d), shall be considered as fulfilling the PMP requirements.

- a. Reporting protocols in the MRP (Attachment E) describe sample results that are to be reported as Detected, But Not Quantified (DNQ) or Not Detected (ND). Definitions for a reported Minimum Level (ML) and Method Detection Limit (MDL) are provided in the Ocean Plan and in Abbreviations and Definitions (Attachment A). These reporting protocols and definitions are used in determining the need to conduct a PMP, as follows:
 - i. The Discharger shall develop and conduct a PMP as further described below if all of the following conditions are true:
 - (a) The calculated effluent limitation is less than the reported Minimum Level.
 - (b) The concentration of the pollutant is reported as DNQ.
 - (c) There is evidence showing that the pollutant is present in the effluent above the calculated effluent limitation.

- ii. Alternatively, the Discharger shall develop and conduct a PMP as further described below if all of the following conditions are true:
 - (a) The calculated effluent limitation is less than the Method Detection Limit.
 - (b) The concentration of the pollutant is reported as ND.
 - (c) There is evidence showing that the pollutant is present in the effluent above the calculated effluent limitation.
- b. The PMP shall include, but not be limited to, the following actions and submittals acceptable to the San Diego Water Board:
 - i. An annual review and semi-annual monitoring of potential sources of the reportable pollutant(s), which may include fish tissue monitoring and other bio-uptake sampling;
 - ii. Quarterly monitoring for the reportable pollutant(s) in the influent to the wastewater treatment system;
 - iii. Submittal of a control strategy designed to proceed toward the goal of maintaining concentrations of the reportable pollutant(s) in the effluent at or below the effluent limitation;
 - iv. Implementation of appropriate cost-effective control measures for the reportable pollutant(s), consistent with the control strategy; and
 - v. An annual status report that shall be sent to the San Diego Water Board including:
 - (a) All PMP monitoring results for the previous year;
 - (b) A list of potential sources of the reportable pollutant(s);
 - (c) A summary of all actions undertaken pursuant to the control strategy; and
 - (d) A description of actions to be taken in the following year.

4. Construction, Operation and Maintenance Specifications

- a. All proposed new treatment facilities and expansions of existing treatment facilities shall be completely constructed and operable prior to initiation of the discharge from the new or expanded facilities. The Discharger shall submit a certification report for each new treatment facility, expansion of an existing treatment facility, and design capacity re-ratings, prepared by the design engineer. For design capacity re-ratings, the certification report shall be prepared by the engineer who evaluated the treatment facility design capacity. The signature and engineering license number of the engineer preparing the certification report shall be affixed to the report. If reasonable, the certification report shall be submitted prior to beginning construction of new treatment facilities or expansions of existing treatment facilities.

- i. The certification report shall:
 - (a) Identify the design capacity of the treatment facility, including the daily and 30-day design capacity;
 - (b) Certify the adequacy of each component of the treatment facility; and
 - (c) Contain a requirement-by-requirement analysis, based on acceptable engineering practices, of the process and physical design of the facility to ensure compliance with this Order.
- ii. The Discharger shall not initiate a discharge from a treatment facility at a daily flow rate in excess of its previously approved design capacity until:
 - (a) The certification report is received by the San Diego Water Board,
 - (b) The San Diego Water Board has received written notification of completion of construction (new or expanded treatment facilities only),
 - (c) An inspection of the facility has been made by the San Diego Water Board or its designated representatives (new or expanded treatment facilities only), and
 - (d) The San Diego Water Board has provided the Discharger with written authorization to initiate discharge from a new or expanded treatment facility or at a daily flow rate in excess of its previously approved design capacity.
- b. All waste treatment, containment, and disposal facilities shall be protected against a 100-year storm event as defined by the San Diego County Flood Control District (FCD).
- c. All waste treatment, containment, and disposal facilities shall be protected against erosion, overland runoff, and other impacts resulting from a 100-year, 24-hour storm event as defined by the San Diego County FCD.
- d. All waste treatment, containment, and disposal facilities shall be protected against regional impacts due to climate change (e.g., sea level rise and floods).
- e. The Discharger shall provide and maintain in good working order a sufficient alternate power source(s) to assure that, in the event of the loss, reduction, or failure of electrical power, all waste treatment, containment, and disposal facilities are in compliance with the terms and conditions of this Order. In addition to a sufficient alternate power source(s), backup systems may also include auxiliary power generators, retention storage capacity, emergency operation procedures, and other contingencies to ensure continuous operation of all critical devices and systems used in the conveyance, storage, treatment, and recycling of municipal wastewater in the event of the loss, reduction, or failure of electrical power. All equipment shall be located to minimize failure due to moisture, liquid spray, flooding, sea level rise, and

other physical phenomena. The alternate power source(s) shall be designed to permit inspection and maintenance and shall provide for periodic testing.

5. Special Provisions for Wastewater Facilities

a. Ensuring Adequate Treatment Plant Capacity

Four years prior to reaching POTW design capacity, the Discharger shall submit a Treatment Plant Capacity Report to the San Diego Water Board showing how flow volumes will be prevented from exceeding existing capacity or how capacity will be increased. A notification and copy of the report shall be sent to appropriate local elected officials, local permitting agencies, and the press. The required technical report shall be reviewed, approved, and jointly submitted to the San Diego Water Board by all planning and building departments having jurisdiction in the area served by the POTW. Opportunities for public participation and involvement are required during the preparation and development of the technical report. The report shall be accompanied by a statement outlining how interested persons were involved in the preparation of the technical report.

b. Source Control Program Requirements

- i. The Discharger shall conduct an Industrial Waste Survey (IWS) of all the industrial users (IUs) in the service area of the Fallbrook WRP to determine whether any IUs are subject to pretreatment standards specified in 40 CFR part 403. The Discharger shall also perform a priority pollutant scan of the influent to the Fallbrook WRP. The IWS and priority pollutant monitoring is required during the 12-month period beginning July 1, 2020 through June 30, 2021. Based on results of the IWS, the priority pollutant scan, and the requirements of 40 CFR part 403, the Discharger shall submit a certification report indicating whether the Fallbrook WRP receives pollutants from any IU that would require the Discharger to establish a pretreatment program in accordance with 40 CFR part 403. The certification report, along with results of the IWS and priority pollutant monitoring, shall be submitted to the San Diego Water Board by December 1, 2021.
- ii. If, during the IWS and priority pollutant monitoring or at any other time during term of this permit, if the Discharger becomes aware of an IU in its service area that would require development of a pretreatment program pursuant to 40 CFR part 403, the Discharger shall notify the San Diego Water Board and request a modification of this Order to include pretreatment program requirements. In such circumstances, the Discharger shall develop and implement a pretreatment program in accordance with the requirements of CWA sections 307(b) and (c) and 402(b)(8) and 40 CFR part 403. The Discharger shall ensure compliance with applicable federal and local pretreatment standards by the IUs within its service area.

- iii. The San Diego Water Board may amend this Order, at any time, to require the Discharger to develop and implement an industrial pretreatment program pursuant to the requirements of 40 CFR part 403 if the San Diego Water Board finds that the Fallbrook WRP receives pollutants from an IU that is subject to pretreatment standards, or if other circumstances so warrant.

c. Sludge (Biosolids) Disposal Requirements

i. General Requirements

- (a) All biosolids generated by the Discharger during the treatment of wastewater shall be used or disposed of in compliance with applicable portions of: 40 CFR part 503-for biosolids that are land applied, placed on a surface disposal site (dedicated land disposal site, monofill, or sludge-only parcel at a municipal landfill), or incinerated; 40 CFR part 258-for biosolids disposed of in a municipal solid waste landfill (with other materials); and 40 CFR part 257-for all biosolids use and disposal practices not covered under 40 CFR parts 258 or 503. The preparer of the biosolids is required under 40 CFR 503.7 to ensure that the applicable requirements in 40 CFR 503 are met when the sewage sludge is applied to the land.

Requirements for biosolids that are applied for the purpose of enhancing plant growth or for land reclamation are set forth in 40 CFR part 503, subpart B (land application). Requirements for biosolids that are placed on land for the purpose of disposal are set forth in 40 CFR part 503, subpart C (surface disposal).

The Discharger shall take all reasonable steps to ensure that all biosolids produced at the Fallbrook WRP are used or disposed of in accordance with these rules, whether the Discharger uses or disposes of the biosolids itself, or transfers their biosolids to another party for further treatment, use, or disposal. The Discharger is responsible for informing subsequent preparers, applicators, and disposers of requirements they must meet under these rules.

- (b) The Discharger shall take all reasonable steps to prevent or minimize any biosolids use or disposal which has a likelihood of adversely affecting human health or the environment.
- (c) No biosolids shall be allowed to enter wetlands or other waters of the United States.
- (d) Biosolids treatment, storage, use, or disposal shall not contaminate groundwater.
- (e) Biosolids treatment, storage, use, or disposal shall not create a nuisance condition such as objectionable odors or flies.

- (f) The Discharger shall take all reasonable steps to ensure that haulers transporting biosolids offsite for treatment, storage, use, or disposal are contractually required to take all necessary measures to keep the biosolids contained. Trucks hauling biosolids that are not classified Class A with respect to pathogens, as defined at 40 CFR section 503.32(a), shall be cleaned as necessary after loading and after unloading, so as to have no biosolids on the exterior of the truck, or wheels. Trucks hauling biosolids that are not Class A shall be tarped. All haulers must have and implement spill clean-up procedures. Trucks hauling biosolids that are not Class A shall not be used for hauling food or feed crops after unloading the biosolids unless the Discharger submits a hauling description, to be approved by USEPA, describing how trucks will be thoroughly cleaned prior to adding food or feed.
 - (g) If biosolids are stored for over two years from the time they are generated, the Discharger must ensure compliance with all requirements for surface disposal under 40 CFR part 503, subpart C, or must submit a written notification to USEPA, State Water Board, and San Diego Water Board with the information specified under 40 CFR section 503.20(b), demonstrating the need for longer temporary storage. During storage of any length for non-Class A biosolids, whether on the Fallbrook WRP site or offsite, adequate procedures must be taken to restrict access by the public and domestic animals.
 - (h) Any biosolids treatment, disposal, or storage site shall have facilities adequate to divert surface runoff from adjacent areas, to protect the site boundaries from erosion, and to prevent any conditions that would cause drainage from the materials to escape from the site. Adequate protection is defined as protection from at least a 100-year storm event as defined by the San Diego County FCD and the highest tidal stage which may occur.
 - (i) If the biosolids are land applied, there shall be adequate screening at the Fallbrook WRP headworks and/or at the biosolids treatment units to ensure that all pieces of metal, plastic, glass, and other inert objects are removed.
- ii. Inspection and Entry
- The USEPA, San Diego Water Board, State Water Board, or an authorized representative thereof, upon the presentation of credentials, shall be allowed by the Discharger directly, or through contractual arrangements with their biosolids management contractors, to:
- (a) Enter upon all premises where biosolids produced by the Discharger are treated, stored, used, or disposed of, by either the Discharger or

another party to whom the Discharger transfers biosolids for further treatment, storage, use, or disposal;

- (b) Have access to and copy any records that must be kept by either the Discharger or another party to whom the Discharger transfers biosolids for further treatment, storage, use, or disposal, under the conditions of this Order or 40 CFR part 503; and
- (c) Inspect any facilities, equipment (including monitoring and control equipment), practices, or operations used in biosolids treatment, storage, use, or disposal by either the Discharger or another party to whom the Discharger transfers biosolids for further treatment, storage, use, or disposal.

iii. Monitoring

Biosolids shall be monitored for the following constituents, at the frequency stipulated in Table 1 of 40 CFR section 503.16:

- arsenic,
- cadmium,
- chromium,
- copper,
- lead,
- mercury,
- molybdenum,
- nickel,
- selenium,
- zinc, and
- total solids.

If biosolids are removed for use or disposal on a routine basis, sampling should be scheduled at regular intervals throughout the year. If biosolids are stored for an extended period prior to use or disposal, sampling may occur at regular intervals, or samples of the accumulated stockpile may be collected prior to use or disposal, corresponding to the tons accumulated in the stockpile over that period.

Monitoring shall be conducted using the methods in *Test Methods for Evaluating Solid Waste, Physical/Chemical Methods* (SW-846), or as otherwise required under 40 CFR section 503.8(b). All results must be reported on a 100% dry weight basis and records of all analyses must state on each page of the analytical results whether the reported results are expressed on an "as-is" or a "100% dry weight" basis.

iv. Pathogen and Vector Control

- (a) Prior to land application, the permittee shall demonstrate that biosolids meet Class A or Class B pathogen reduction levels by one of the methods listed under 40 CFR section 503.32.
- (b) Prior to disposal on a surface disposal site, the Discharger shall demonstrate that biosolids meet Class B pathogen reduction levels, or ensure that the site is covered at the end of each operating day. If pathogen reduction is demonstrated using a "Process to Further Reduce Pathogens" or one of the "Processes to Significantly Reduce Pathogens," the Discharger shall maintain daily records of the operating parameters used to achieve this reduction. If pathogen reduction is demonstrated by testing for fecal coliform and/or pathogens, samples must be collected at the frequency specified in Table 1 of 40 CFR section 503.16. If Class B is demonstrated using fecal coliform, at least seven grab samples must be collected during each monitoring period and a geometric mean calculated from these samples. The following holding times between sample collection and analysis shall not be exceeded: fecal coliform-24 hours when cooled to four °C; Salmonella spp. bacteria-24 hours when cooled to four °C; enteric viruses-two weeks when frozen; and helminth ova-one month when cooled to four °C.
- (c) For biosolids that are land applied or placed on a surface disposal site, the Discharger shall track and keep records of the operational parameters used to achieve the Vector Attraction Reduction requirements under 40 CFR section 503.33(b).

v. Surface Disposal

If biosolids are placed on a surface disposal site (dedicated land disposal site or monofill), a qualified groundwater scientist shall develop a groundwater monitoring program for the site, or shall certify that the placement of biosolids on the site will not contaminate an aquifer.

vi. Landfill Disposal

Biosolids placed in a municipal landfill shall be tested by the Paint Filter Test (Method 9095) at the frequency specified in Table 1 of 40 CFR section 503.16, or more often if necessary, to demonstrate that there are no free liquids.

vii. Notifications

The Discharger, either directly or through contractual arrangements with their biosolids management contractors, shall comply with the following notification requirements.

(a) Notification of Noncompliance

The Discharger shall notify USEPA, State Water Board, and San

Diego Water Board (for both Discharger and use or disposal site) of any noncompliance with the biosolids within 24 hours, if the noncompliance may endanger human health or the environment. For other instances of noncompliance with the biosolids, the Discharger shall notify USEPA, State Water Board, and San Diego Water Board of the noncompliance in writing within five working days of becoming aware of the noncompliance. The Discharger shall require their biosolids management contractors to notify USEPA, State Water Board, and San Diego Water Board of any noncompliance within these same time frames.

(b) Interstate Notification

If biosolids are shipped to another State or tribal land, the Discharger shall send 60 days prior notice of the shipment to the permitting authorities in the receiving State or tribal land, and the USEPA.

(c) Land Application Notification

Prior to using any biosolids from the Fallbrook WRP (other than composted biosolids) at a new or previously unreported site, the Discharger shall notify USEPA, State Water Board, and San Diego Water Board. This notification shall include a description and topographic map of the proposed site(s), names and addresses of the applier and site owner, and a listing of any State or local permits which must be obtained. It shall also include a description of the crops or vegetation to be grown, proposed loading rates, and a determination of agronomic rates.

Within a given monitoring period, if any biosolids do not meet the applicable metals concentration limits specified under 40 CFR section 503.13, then the Discharger (or its contractor) must pre-notify USEPA, State Water Board, and San Diego Water Board, and determine the cumulative metals loading at that site to date, as required by 40 CFR section 503.12.

The Discharger shall notify the applier of all subject requirements under 40 CFR part 503, including the requirement for the applier to certify that management practices, site restrictions, and applicable vector attraction reduction requirements have been met. The Discharger shall require the applier to certify at the end of 38 months, following application of Class B biosolids, that harvesting restrictions in effect for up to 38 months have been met.

(d) Surface Disposal Notification

Prior to disposal at a new or previously unreported site, the Discharger shall notify USEPA, State Water Board, and San Diego Water Board. The notice shall include a description and topographic

map of the proposed site, depth to groundwater, whether the site is lined or unlined, site operator and site owner, and any State or local permits. It shall also describe procedures for ensuring grazing and public access restrictions for three years following site closure. The notice shall include a groundwater monitoring plan or description of why groundwater monitoring is not required.

viii. Reporting

The Discharger shall submit an annual biosolids report to the State Water Board's CIWQS program website (https://www.waterboards.ca.gov/water_issues/programs/ciwqs/), to the USEPA Biosolids Coordinator (CDX NeT electronic reporting system), and, if applicable, to the Arizona Department of Environmental Quality Biosolids Program Coordinator by February 19 of each year for the period covering the previous calendar year. The report shall include:

- (a) The amount of biosolids generated that year, in dry metric tons, and the amount accumulated from previous years.
- (b) Results of all pollutant monitoring required under section VI.C.5.d.iii of this Order. Results must be reported on a 100% dry weight basis.
- (c) Demonstrations of pathogen and vector attraction reduction methods, as required under 40 CFR sections 503.17 and 503.27, and certifications.
- (d) Names, mailing addresses, and street addresses of persons who received biosolids for storage, further treatment, disposal in a municipal landfill, or other use or disposal method not covered above, and volumes delivered to each.
- (e) The following information must be submitted by the Discharger as an attachment to the CDX NeT electronic reporting system, unless the Discharger requires its biosolids management contractors to report this information directly to the USEPA Biosolids Coordinator:
 - (1) For land application sites:
 - Locations of land application sites (with field names and numbers) used that calendar year, size of each field applied to, applier, and site owner;
 - Volumes applied to each field (in wet tons and dry metric tons), nitrogen applied, and calculated plant available nitrogen;
 - Crops planted, dates of planting and harvesting;
 - For biosolids exceeding 40 CFR section 503.13 Table 3 metals concentrations, the locations of sites where the biosolids were applied and cumulative metals loading at the sites to date;

- Certifications of management practices at 40 CFR section 503.14; and
- Certifications of site restrictions at 40 CFR section 503.32(b)(5).

(2) For surface disposal sites:

- Locations of sites, site operator and site owner, size of parcel on which biosolids were disposed;
- Results of any required groundwater monitoring;
- Certifications of management practices at 40 CFR section 503.24; and
- For closed sites, the date of site closure and certifications of management practices for three years following site closure.

ix. All reports shall be submitted to:

State Water Board's CIWQS program website

(https://www.waterboards.ca.gov/water_issues/programs/ciwqs/)

Regional Biosolids Coordinator

U.S. Environmental Protection Agency

EPA's CDX NeT electronic reporting system

If applicable,

Biosolids Program Coordinator

Arizona Department of Environmental Quality

Mail Code: 5415B-1

1110 West Washington Street

Phoenix, AZ 85007

d. **Sewage Collection System**

The Discharger is subject to the requirements of, and must separately comply with State Water Board Order No. 2006-0003-DWQ, *Statewide General Waste Discharge Requirements for Sanitary Sewer Systems* (Statewide General SSO Order), including monitoring and reporting requirements as amended by State Water Board Order No. WQ 2013-0058-EXEC and any subsequent amendment/reissuance order. The Discharger is also subject to the requirements of, and must separately comply with the San Diego Water Board Order No. R9-2007-0005, *Waste Discharge Requirements for Sewage Collection Agencies in the San Diego Region* (Regional General SSO Order), and any subsequent amendment/reissuance order.

Regardless of the coverage obtained under Order No. 2006-0003-DWQ or Order No. R9-2007-0005, the Discharger's collection system is part of the treatment system that is subject to this Order. As such, pursuant to federal regulations, the Discharger must report any noncompliance (40 CFR sections 122.41(l)(6) and (7)), properly operate and maintain its collection system (40

CFR section 122.41(e)), and mitigate or prevent any discharge from the collection system in violation of this Order [40 CFR section 122.41(d)].

e. **Resource Recovery from Anaerobically Digestible Material**

If the Discharger plans to receive hauled-in anaerobically digestible material for injection into an anaerobic digester, the Discharger shall notify the San Diego Water Board and develop and implement Standard Operating Procedures (SOPs) for this activity. The SOPs shall be developed prior to receiving hauled-in anaerobically digestible material. The SOPs shall address material handling, including unloading, screening, or other processing prior to anaerobic digestion; transportation; spill prevention; and spill response. In addition, the SOPs shall address avoidance of the introduction of materials that could cause interference, pass-through, or upset of the treatment processes; avoidance of prohibited material; vector control; odor control; operation and maintenance; and the disposition of any solid waste segregated from introduction to the digester. The Discharger shall train its staff on the SOPs and shall maintain records for a minimum of three years for each load received, describing the hauler, waste type, and quantity received. In addition, the Discharger shall maintain records for a minimum of three years for the disposition, location, and quantity of cumulative pre-digestion-segregated solid waste hauled offsite.

6. **Other Special Provisions – Not Applicable**

7. **Compliance Schedules – Not Applicable**

VII. COMPLIANCE DETERMINATION

Compliance with the effluent limitations contained in section IV of this Order will be determined as specified below:

A. Compliance with Average Monthly Effluent Limitation (AMEL)

If the average of daily discharges over a calendar month exceeds the AMEL for a given parameter, an alleged violation will be flagged and the Discharger will be considered out of compliance for each day of that month for that parameter (e.g., resulting in 31 days of noncompliance in a 31-day month). The average of daily discharges over the calendar month that exceeds the AMEL for a parameter will be considered out of compliance for the month only. If only a single sample is taken during the calendar month and the analytical result for that sample exceeds the AMEL, the Discharger will be considered out of compliance for that calendar month. For any one calendar month during which no sample is taken, no compliance determination can be made for that calendar month.

B. Compliance with Average Weekly Effluent Limitation (AWEL)

If the average of daily discharges over a calendar week (Sunday through Saturday) exceeds the AWEL for a given parameter, an alleged violation will be flagged and the Discharger will be considered out of compliance for each day of that week for that parameter, resulting in seven days of noncompliance. The average of daily

discharges over the calendar week that exceeds the AWEL for a parameter will be considered out of compliance for that week only. If only a single sample is taken during the calendar week and the analytical result for that sample exceeds the AWEL, the Discharger will be considered out of compliance for that calendar week. For any one calendar week during which no sample is taken, no compliance determination can be made for that calendar week.

C. Compliance with Maximum Daily Effluent Limitation (MDEL)

The MDEL shall apply to flow weighted 24-hour composite samples, or grab samples, as specified in the MRP (Attachment E). If a daily discharge exceeds the MDEL for a given parameter, an alleged violation will be flagged and the Discharger will be considered out of compliance for that parameter for that one day only within the reporting period. For any one day during which no sample is taken, no compliance determination can be made for that day.

D. Compliance with Instantaneous Minimum Effluent Limitation

The instantaneous minimum effluent concentration limitation shall apply to grab sample determinations. If the analytical result of a single grab sample is lower than the instantaneous minimum effluent limitation for a parameter, an alleged violation will be flagged and the Discharger will be considered out of compliance for that parameter for that single sample. Non-compliance for each sample will be considered separately (e.g., the results of two grab samples taken within a calendar day that both are lower than the instantaneous minimum effluent limitation would result in two instances of noncompliance with the instantaneous minimum effluent limitation).

E. Compliance with Instantaneous Maximum Effluent Limitation

The instantaneous maximum effluent concentration limitation shall apply to grab sample determinations. If the analytical result of a single grab sample is higher than the instantaneous maximum effluent limitation for a parameter, an alleged violation will be flagged and the Discharger will be considered out of compliance for that parameter for that single sample. Non-compliance for each sample will be considered separately (e.g., the results of two grab samples taken within a calendar day that both exceed the instantaneous maximum effluent limitation would result in two instances of noncompliance with the instantaneous maximum effluent limitation).

F. Compliance with Six-Month Median Effluent Limitation

If the median of daily discharges over any 180-day period exceeds the six-month median effluent limitation for a given parameter, an alleged violation will be flagged and the Discharger will be considered out of compliance for each day of that 180-day period for that parameter. The next assessment of compliance will occur after the next sample is taken. If only a single sample is taken during a given 180-day period and the analytical result for that sample exceeds the six-month median, the Discharger will be considered out of compliance for the 180-day period. For any 180-day period during which no sample is taken, no compliance determination can be made for the six-month median limitation.

G. Mass and Concentration Limitations

Compliance with mass and concentration effluent limitations for the same parameter shall be determined separately with their respective limitations. When the concentration of a constituent in an effluent sample is determined to be ND or DNQ, the corresponding MER determined from that sample concentration shall also be reported as “ND” or “DNQ.”

H. Percent Removal

Compliance with percent removal requirements for average monthly percent removal of CBOD₅ and TSS shall be determined separately for each wastewater treatment facility discharging through an outfall. For each wastewater treatment facility, the monthly average percent removal is the average of the calculated daily discharge percent removals only for days on which the constituent concentration is monitored in both the influent and effluent of the wastewater treatment facility at the locations specified in the MRP (Attachment E) within a calendar month.

The percent removal for each day shall be calculated according to the following equation:

$$\text{Daily discharge percent removal} = \frac{\text{Influent concentration} - \text{Effluent concentration}}{\text{Influent concentration}} \times 100\%$$

I. Ocean Plan Provisions for Table 3 Constituents

Sufficient sampling and analysis shall be required to determine compliance with the effluent limitation.

1. Compliance with Single-constituent Effluent Limitations

The Discharger shall be deemed out of compliance with an effluent limitation or discharge specification if the concentration of the constituent in the monitoring sample is greater than the effluent limitation or discharge specification and greater than or equal to the ML.

2. Compliance with Effluent Limitations Expressed as a Sum of Several Constituents

The Discharger is out of compliance with an effluent limitation that applies to the sum of a group of chemicals (e.g., PCBs) if the sum of the individual pollutant concentrations is greater than the effluent limitation. Individual pollutants of the group will be considered to have a concentration of zero if the constituent is reported as ND or DNQ.

3. Multiple Sample Data Reduction

The concentration of the pollutant in the effluent may be estimated from the result of a single sample analysis or by a measure of central tendency (arithmetic mean, geometric mean, median, etc.) of multiple sample analyses when all sample results are quantifiable (i.e., greater than or equal to the reported ML). When one or more sample results are reported as ND or DNQ, the central tendency concentration of the pollutant shall be the median (middle) value of the

multiple samples. If, in an even number of samples, one or both of the middle values is ND or DNQ, the median will be the lower of the two middle values.

4. Mass Emission Rate (MER)

The MER, in lbs/day, shall be obtained from the following calculation for any calendar day:

$$\text{MER (lbs/day)} = 8.34 \times Q \times C$$

In which Q and C are the flow rate in MGD and the constituent concentration in mg/L, respectively, and 8.34 is a conversion factor (lbs/gallon of water). If a composite sample is taken, then C is the concentration measured in the composite sample and Q is the average flow rate occurring during the period over which the samples are composited.

J. Bacterial Standards and Analysis

1. The geometric mean used for determining compliance with bacterial standards is calculated with the following equation:

$$\text{Geometric Mean} = (C_1 \times C_2 \times \dots \times C_n)^{1/n}$$

Where n is the number of days samples were collected during the period and C is the concentration of bacteria (CFU/100 mL) found on each day of sampling.

2. The STV used for determining compliance with bacterial standards shall not be exceeded by more than 10 percent of the samples collected in a calendar month, collected in a static manner.
3. Sample dilutions for fecal coliform bacterial analyses should be performed so the range of values extends from 2 to 16,000 CFU. Sample dilutions for enterococci bacterial analyses shall range from 1 to 10,000 CFU per 100 mL. The detection methods used for each analysis shall be reported with the results of the analysis. Detection methods used for fecal coliform shall be those listed in 40 CFR part 136 an Alternative Test Procedure approved by USEPA. Detection methods used for enterococci shall be those presented in USEPA publication USEPA 600/4-85/076, *Test Methods for Escherichia coli and Enterococci in Water by Membrane Filter Procedure*, listed under 40 CFR part 136, and any other method approved by the San Diego Water Board.

K. Single Operational Upset (SOU)

A SOU that leads to simultaneous violations or more than one pollutant parameter shall be treated as a single violation and limits the Discharger's liability in accordance with the following conditions:

1. A SOU is broadly defined as a single unusual event that temporarily disrupts the usually satisfactory operation of a system in such a way that it results in violation of multiple pollutant parameters.
2. A Discharger may assert SOU to limit liability only for those violations which the Discharger submitted notice of the upset as required in section I.H of Attachment D.

3. For purposes outside of Water Code section 13385(h) and (i), determination of compliance and civil liability (including any more specific definition of SOU), the requirements for Dischargers to assert the SOU limitation of liability, and the manner of counting violations, shall be in accordance with the USEPA Memorandum *Issuance of Guidance Interpreting Single Operational Upset* (September 27, 1989).
4. For purposes of Water Code section 13385(h) and (i), determination of compliance and civil liability (including any more specific definition of SOU), the requirements for Dischargers to assert the SOU limitation of liability, and the manner of counting violations shall be in accordance with Water Code section 13385(f)(2).

L. Chronic Toxicity

Chronic toxicity is used to measure the acceptability of waters for supporting a healthy marine biota until approved methods are developed to evaluate biological response. Compliance with the chronic toxicity performance goal established in section IV.A.2 of this Order for Discharge Point No. 001 shall be determined using critical life stage toxicity tests in accordance with procedures prescribed by the Ocean Plan (2019) and restated in the MRP (Attachment E). Chronic toxicity shall be expressed as Toxic Units Chronic (TUc), where:

$$\text{TUc} = 100 / \text{NOEC}$$

where NOEC is the No Observed Effect Concentration (also referred to as the No Observed Effect Level or NOEL) and is expressed as the maximum percent of effluent that causes no observable effect on a test organism, as determined by the result of a critical life stage toxicity test. The performance goal for chronic toxicity is exceeded when a chronic toxicity test results in greater than or equal to 88 TUc. The performance goal for chronic toxicity is set at the IWC for the discharge (1.15% effluent¹). The discharge is subject to determination of "Pass" or "Fail" from a chronic toxicity test using the Test of Significant Toxicity (TST) statistical t-test approach described in National Pollutant Discharge Elimination System Test of Significant Toxicity Implementation Document (EPA 833-R-10-003, 2010), Appendix A, Figure A-1 and Table A-1, and Appendix B, Table B-1. The null hypothesis (Ho) for the TST statistical approach is:

Mean discharge "in-stream" waste concentration (IWC) response $\leq 0.75 \times$ Mean control response.

A test result that rejects this null hypothesis is reported as "Pass". A test result that does not reject this null hypothesis is reported as "Fail". This is a t-test (formally Student's t-test), a statistical analysis comparing two sets of replicate observations—in the case of WET test, only two test concentrations (i.e., a control and IWC). The purpose of this statistical test is to determine if the means of the two sets of observations are different (i.e., if the IWC or receiving water concentration differs from the control (the test result is "Pass" or "Fail")). The Welch's t-test employed by the

¹ IWC = 1/minimum initial dilution factor (Dm) = 1/87 = 0.0115 = 1.15%

~~TST statistical approach is an adaptation of Student's t-test and is used with two samples having unequal variances.~~

~~The performance goal for chronic toxicity is exceeded when a chronic toxicity test, analyzed using the TST statistical approach, results in "Fail".~~

~~The performance goal for chronic toxicity is set at the IWC for the discharge (1.15% effluent²) and expressed in units of the TST statistical approach ("Pass" or "Fail"). All monitoring for the performance goal for chronic toxicity shall be reported using the IWC effluent concentration and negative control, expressed in units of the TST. The TST hypothesis (see above) is statistically analyzed using the IWC and a negative control. Effluent toxicity tests shall be run using *Short-Term Methods for Estimating the Chronic Toxicity of Effluent and Receiving Waters to West Coast Marine Estuarine Organisms* (EPA/600/R-95/136, 1995). The San Diego Water Board's review of reported toxicity test results will include review of concentration response patterns as appropriate (see section IV.C.5 of the Fact Sheet (Attachment F)). As described in the laboratory audit directives to the San Jose Creek Water Quality Laboratory from the State Water Board dated August 07, 2014, and from USEPA dated December 24, 2013, the Percent Minimum Significant Difference (PMSD) criteria only apply to compliance reporting for the no-observed effect concentration (NOEC) and the sublethal statistical endpoints of the NOEC, and therefore are not used to interpret TST results. SOPs used by the toxicity testing laboratory to identify and report valid, invalid, anomalous, or inconclusive effluent (and receiving water) toxicity test measurement results from the TST statistical approach, including those that incorporate a consideration of concentration response patterns, must be submitted to the San Diego Water Board (40 CFR section 122.41(h)). The San Diego Water Board will make a determination as to whether a toxicity test result is compliant, and may consult with the Discharger, USEPA, the State Water Board's Quality Assurance (QA) Officer, or the State Water Board, Division of Drinking Water Environmental Laboratory Accreditation Program (ELAP), as needed.~~

² ~~IWC = 1/minimum initial dilution factor (Dm) = 1/87 = 0.0115 = 1.15%~~

ATTACHMENT A – ABBREVIATIONS AND DEFINITIONS

Part 1. – Abbreviations

Abbreviation	Definition
40 CFR	Title 40 of the Code of Federal Regulations
AMEL ¹	Average Monthly Effluent Limitation
AQUA	Aquaculture
ASBS ¹	Areas of Special Biological Significance
AUV	Autonomous Underwater Vehicle
AWEL ¹	Average Weekly Effluent Limitation
Basin Plan	<i>Water Quality Control Plan for the San Diego Basin</i>
BIOL	Preservation of Biological Habitats of Special Significance
BOD ₅	Biochemical Oxygen Demand (5-Day @ 20°C)
BPJ	Best Professional Judgement
BRI	Benthic Response Index
°C	Degrees Celsius
CBOD ₅	Carbonaceous Biochemical Oxygen Demand (5-Day @ 20°C)
CCAP	Climate Change Action Plan
CCR	California Code of Regulations
CEQA	California Environmental Quality Act
CFR	Code of Federal Regulations
CFU	Colony Forming Units
City	City of Oceanside
CIWQS	California Integrated Water Quality System
CO ₂	Carbon Dioxide
COMM	Commercial and Sport Fishing
CSM	Conceptual Site Model
CTD	Conductivity-Temperature-Depth
CWA	Clean Water Act
DDT ¹	Dichlorodiphenyltrichloroethane
Discharger	Fallbrook Public Utility District
DMR ¹	Discharge Monitoring Report
DNQ ¹	Detected, But Not Quantified
EC25	Effects Concentration at 25 Percent
ELAP	Environmental Laboratory Accreditation Program
eSMR	Electronic Self-Monitoring Reports
°F	Degrees Fahrenheit
Facilities	The Fallbrook Water Reclamation Plant, Santa Margarita Groundwater Treatment Plant, and the associated sanitary sewer system, pump stations, and land outfalls.
FCD	Flood Control District
Fallbrook WRP	Fallbrook Water Reclamation Plant
GPS	Global Positioning System

Abbreviation	Definition
HCH ¹	Hexachlorocyclohexane
Ho	Hypothesis
HSA	Hydrologic Subarea
HU	Hydrologic Unit
IND	Industrial Service Supply
IU	Industrial User
IWC ¹	"In-Stream" Waste Concentration
lbs/day	Pounds per Day
LC	Lethal Concentration
LC 50	Percent Waste Giving 50 Percent Survival of Test Organisms
LSWTP	La Salina Wastewater Treatment Plant
MAR	Marine Habitat
MBGPF	Mission Basin Groundwater Purification Facility
MCB Camp Pendleton	Marine Corps Base, Camp Pendleton
MDEL ¹	Maximum Daily Effluent Limitation
MDL ¹	Method Detection Limit
MEC	Maximum Effluent Concentration
MER	Mass Emission Rate
MF	Microfiltration
mg/kg	Milligram per Kilogram
mg/L	Milligram per Liter
MGD	Million Gallons per Day
MIGR	Migration of Aquatic Organisms
ML ¹	Minimum Level
ml	Milliliter
ml/L	Milliliter per Liter
MRP	Monitoring and Reporting Program
NAV	Navigation
ND ¹	Not Detected
ng/kg	Nanogram per Kilogram
NOAA's	National Oceanic and Atmospheric Administration's
NOEC	No Observed Effect Concentration
NOEL	No Observed Effect Level
NPDES	National Pollutant Discharge Elimination System
NTU	Nephelometric Turbidity Unit
Ocean Plan	<i>Water Quality Control Plan for Ocean Waters of California, California Ocean Plan</i>
OOO	Oceanside Ocean Outfall
PAHs ¹	Polynuclear Aromatic Hydrocarbons
PCBs ¹	Polychlorinated Biphenyls
pCi/L	Picocuries per Liter
PMP ¹	Pollutant Minimization Program
PMSD	Percent Minimum Significant Difference

Abbreviation	Definition
POTWs	Publicly-Owned Treatment Works
PPP	Pollution Prevention Plan
ppt	Parts per Thousand
psu	Practical Salinity Unit
QA	Quality Assurance
QAPP	Quality Assurance Project Plan
QC	Quality Control
RARE	Rare, Threatened, or Endangered Species
REC-1	Contact Water Recreation
REC-2	Non-Contact Water Recreation
RCRA	Resource Conservation and Recovery Act
Regional General SSO Order	California Regional Water Quality Control Board Region 9, San Diego Region Order No. R9-2007-0005, <i>Waste Discharge Requirements for Sewage Collection Agencies in the San Diego Region</i>
RL	Reporting Level
ROTV	Remotely Operated Towed Vehicle
ROWD	Report of Waste Discharge
RPA	Reasonable Potential Analysis
San Diego Water Board	California Regional Water Quality Control Board, San Diego Region
Santa Margarita GTP	Santa Margarita Groundwater Treatment Plant
SCCWRP	Southern California Coastal Water Research Project
SHELL	Shellfish Harvesting
SLRWRF	San Luis Rey Water Reclamation Facility
SMR	Self-Monitoring Report
SOPs	Standard Operating Procedures
SOU	Single Operational Upset
SPP	Spill Prevention Plan
SPWN	Spawning, Reproduction, and/or Early Development
SRP	Spill Response Plan
SSMPs	Sanitary Sewer Management Plans
SSO ¹	Sanitary Sewer Overflow
State Water Board	State Water Resources Control Board
Statewide General SSO Order	State Water Board Order No. 2006-0003-DWQ, <i>Statewide General Waste Discharge Requirements for Sanitary Sewer Systems</i>
STV	Statistical Threshold Value
TAC	Test Acceptability Criteria
TBELs	Technology-Based Effluent Limitations
TCDD ¹	Tetrachlorodibenzodioxin
TDS	Total Dissolved Solids
TIE ¹	Toxicity Identification Evaluation

FALLBROOK PUBLIC UTILITY DISTRICT
 FALLBROOK WATER RECLAMATION PLANT
 SANTA MARGARITA GROUNDWATER TREATMENT PLANT

ORDER NO. R9-2019-0169
 Amended by Order No. R9-2020-0193 and R9-2021-0204
 NPDES NO. CA0108031

Abbreviation	Definition
TMDL	Total Maximum Daily Load
TRE ¹	Toxicity Reduction Evaluation
TSD	Technical Support Document
TSS	Total Suspended Solids
TST	Test of Significant Toxicity
TUa ¹	Toxic Units Acute
TUc ¹	Toxic Units Chronic
UF	Ultrafiltration
µg	Microgram
µg/kg	Microgram per Kilogram
µg/L	Microgram per Liter
UM3	USEPA Modeling Application Visual Plumes
µmhos/cm	Micromhos per Centimeter
U.S.C.	United States Code
USEPA	United States Environmental Protection Agency
U.S.	United States
Water Code	California Water Code
WDRs	Waste Discharge Requirements
WET	Whole Effluent Toxicity
WILD	Wildlife Habitat
WQBELs	Water Quality-Based Effluent Limitations
ZID	Zone of Initial Dilution

¹ See Part 2 of Attachment A (Glossary of Common Terms) for further definition.

Part 2. – Definitions

30-day average

The arithmetic mean of pollutant parameter values of samples collected in a period of 30 consecutive days.

6-Month Median Effluent Limitation

The highest allowable moving median of all daily discharges for any 180-day period.

Acute Toxicity

a. Acute Toxicity (TUa)

Expressed in Toxic Units Acute (TUa)

$$TUa = \frac{100}{\frac{96\text{-hr LC}}{50\%}}$$

b. Lethal Concentration 50% (LC 50)

LC 50 (percent waste giving 50% survival of test organisms) shall be determined by static or continuous flow bioassay techniques using standard marine test species as specified in Ocean Plan Appendix III. If specific identifiable substances in wastewater can be demonstrated by the discharger as being rapidly rendered harmless upon discharge to the marine environment, but not as a result of dilution, the LC 50 may be determined after the test samples are adjusted to remove the influence of those substances.

When it is not possible to measure the 96-hour LC 50 due to greater than 50 percent survival of the test species in 100 percent waste, the toxicity concentration shall be calculated by the expression:

$$TUa = \frac{\log (100 - S)}{1.7}$$

where:

S = percentage survival in 100% waste. If S > 99, TUa shall be reported as zero.

Anaerobically Digestible Material

Inedible kitchen grease as defined in section 19216 of the Food and Agricultural Code and food material as defined in title 14, division 7, chapter 3.1, article 1, section 17582(a) (20) of the CCR.

Antidegradation

Policies which ensure protection of water quality for a particular body where the water quality exceeds levels necessary to protect fish and wildlife propagation and recreation on and in the water. This also includes special protection of waters designated as outstanding natural resource waters.

Areas of Special Biological Significance (ASBS)

Those areas designated by the State Water Resources Control Board (State Water Board) as ocean areas requiring protection of species or biological communities to the extent that

alteration of natural water quality is undesirable. All Areas of Special Biological Significance are also classified as a subset of STATE WATER QUALITY PROTECTION AREAS.

Average Monthly Effluent Limitation (AMEL)

The highest allowable average of daily discharges over a calendar month, calculated as the sum of all daily discharges measured during a calendar month divided by the number of daily discharges measured during that month.

Average Weekly Effluent Limitation (AWEL)

The highest allowable average of daily discharges over a calendar week (Sunday through Saturday), calculated as the sum of all daily discharges measured during a calendar week divided by the number of daily discharges measured during that week.

Beneficial Uses

The uses of water necessary for the survival or wellbeing of man, plants, and wildlife. These uses of water serve to promote the tangible and intangible economic, social, and environmental goals. "Beneficial Uses" of the waters of the State that may be protected against include, but are not limited to, domestic, municipal, agricultural and industrial supply; power generation; recreation; aesthetic enjoyment; navigation; and preservation and enhancement of fish, wildlife, and other aquatic resources or preserves. In the Basin Plan, existing beneficial uses are uses that were attained in the surface or ground water on or after November 28, 1975; and potential beneficial uses are uses that would probably develop in future years through the implementation of various control measures. "Beneficial Uses" are equivalent to "Designated Uses" under federal law. [Water Code section 13050(f)].

Bioaccumulation

The accumulation of contaminants in the tissues of organisms through any route, including respiration, ingestion, or direct contact with contaminated water, sediment, food, or dredged material.

Biosolids

Nutrient-rich organic materials resulting from the treatment of sewage sludge. When treated and processed, sewage sludge becomes biosolids which can be safely recycled and applied as fertilizer to sustainably improve and maintain productive soils and stimulate plant growth.

Brine

Brine is the byproduct of desalinated water having a salinity concentration greater than a desalination facility's intake source water.

Bypass

The intentional diversion of waste streams from any portion of a treatment facility. (40 CFR section 122.41(m)(1)(i).)

Chlordane

Shall mean the sum of chlordane-alpha, chlordane-gamma, chlordene-alpha, chlordene-gamma, nonachlor-alpha, nonachlor-gamma, and oxychlordane.

Chlorinated Phenolics

The sum of 4-chloro-3-methylphenol, 2-chlorophenol, pentachlorophenol, 2,4,5-trichlorophenol, and 2,4,6-trichlorophenol.

Chronic Toxicity

Chronic toxicity is the measure of the sub-lethal effects of a discharge or ambient water sample (e.g. reduced growth or reproduction.). Certain chronic toxicity tests include an additional measurement of lethality. Compliance with the performance goal for chronic toxicity in this Order is demonstrated by conducting chronic toxicity tests for the effluent as described in section VII.L of this Order and section III.C of the MRP (Attachment E). Chronic Toxicity performance goal is expressed as Toxic Units Chronic (TUC) and is calculated by the following equation:

$$TUC = 100/NOEC$$

where NOEC is the No Observed Effect Concentration (also referred to as the No Observed Effect Level or NOEL) or the maximum percent effluent that causes no observable effect on a test organism, as determined by the result of a critical life stage toxicity test.

Composite Sample

A composite sample is defined as a combination of at least eight sample aliquots of at least 100 ml, collected at periodic intervals during the operating hours of a facility over a 24-hour period. For volatile pollutants, aliquots must be combined in the laboratory immediately before analysis. Unless otherwise authorized by the San Diego Water Board, the composite must be flow proportional; either the time interval between each aliquot or the volume of each aliquot must be proportional to either the stream flow at the time of sampling or the total stream flow since the collection of the previous aliquot. Aliquots may be collected manually or automatically. The 100 ml minimum volume of an aliquot does not apply to automatic self-purging samplers. If one day is defined as a 24-hour period other than a calendar day, the analytical result for the 24-hour period will be considered as the result for the calendar day in which the 24-hour period ends.

Daily Discharge

Daily Discharge is defined as either: (1) the total mass of the constituent discharged over the calendar day (12:00 am through 11:59 pm) or any 24-hour period that reasonably represents a calendar day for purposes of sampling (as specified in the permit), for a constituent with limitations expressed in units of mass; or (2) the unweighted arithmetic mean measurement of the constituent over the day for a constituent with limitations expressed in other units of measurement (e.g., concentration).

The daily discharge may be determined by the analytical results of a composite sample taken over the course of one day (a calendar day or other 24-hour period defined as a day) or by the arithmetic mean of analytical results from one or more grab samples taken over the course of the day.

A grab sample is an individual sample of at least 100 ml collected at a randomly selected time over a period not exceeding 15 minutes.

Degrade

Degradation shall be determined by comparison of the waste field and reference site(s) for characteristic species diversity, population density, contamination, growth anomalies, debility, or supplanting of normal species by undesirable plant and animal species. Degradation occurs if there are significant differences in any of three major biotic groups, namely, demersal fish,

benthic invertebrates, or attached algae. Other groups may be evaluated where benthic species are not affected, or are not the only ones affected.

Desalination Facility

Desalination Facility is an industrial facility that processes water to remove salts and other components from the source water to produce water that is less saline than the source water.

Detected, But Not Quantified (DNQ)

Sample results that are less than the reported ML, but greater than or equal to the laboratory's MDL. Sample results reported as DNQ are estimated concentrations.

Dichlorobenzenes

Shall mean the sum of 1,2- and 1,3-dichlorobenzene.

Dichlorodiphenyltrichloroethane (DDT)

Shall mean the sum of 4,4'DDT, 2,4'DDT, 4,4'DDE, 2,4'DDE, 4,4'DDD, and 2,4'DDD.

Discharge of a Pollutant

Discharge of a pollutant means: (a) Any addition of any "pollutant" or combination of pollutants to "waters of the United States" from any "point source," or (b) Any addition of any pollutant or combination of pollutants to the waters of the "contiguous zone" or the ocean from any point source other than a vessel or other floating craft which is being used as a means of transportation. This definition includes additions of pollutants into waters of the United States from: surface runoff which is collected or channeled by man; discharges through pipes, sewers, or other conveyances owned by a State, municipality, or other person which do not lead to a treatment works; and discharges through pipes, sewers, or other conveyances, leading into privately owned treatment works. This term does not include an addition of pollutants by any "indirect discharger." "Discharge" when used without qualification means the "discharge of a pollutant." (40 CFR 122.2)

Discharge Monitoring Reports (DMRs)

The DMRs means the U.S. Environmental Protection Agency (USEPA) uniform national form, including any subsequent additions, revisions, or modifications for the reporting of self-monitoring results by permittees. DMRs must be used by "approved States" as well as by USEPA. USEPA will supply DMRs to any approved State upon request. The USEPA national forms may be modified to substitute the State agency name, address, logo, and other similar information, as appropriate, in place of USEPA's.

Downstream Ocean Waters

Waters downstream with respect to ocean currents.

Dredged Material

Any material excavated or dredged from the navigable waters of the United States, including material otherwise referred to as "spoil."

Enclosed Bays

Indentations along the coast that enclose an area of oceanic water within distinct headlands or harbor works. Enclosed bays include all bays where the narrowest distance between headlands or outermost harbor works is less than 75 percent of the greatest dimension of the enclosed portion of the bay. This definition includes but is not limited to: Humboldt Bay,

Bodega Harbor, Tomales Bay, Drakes Estero, San Francisco Bay, Morro Bay, Los Angeles Harbor, Upper and Lower Newport Bay, Mission Bay, and San Diego Bay.

Endosulfan

The sum of endosulfan-alpha and -beta and endosulfan sulfate.

Estuaries and Coastal Lagoons are waters at the mouths of streams that serve as mixing zones for fresh and ocean waters during a major portion of the year. Mouths of streams that are temporarily separated from the ocean by sandbars shall be considered as estuaries. Estuarine waters will generally be considered to extend from a bay or the open ocean to the upstream limit of tidal action but may be considered to extend seaward if significant mixing of fresh and salt water occurs in the open coastal waters. The waters described by this definition include but are not limited to the Sacramento-San Joaquin Delta as defined by section 12220 of the California Water Code, Suisun Bay, Carquinez Strait downstream to Carquinez Bridge, and appropriate areas of the Smith, Klamath, Mad, Eel, Noyo, and Russian Rivers.

Facilities

The Fallbrook Water Reclamation Plant, Santa Margarita Groundwater Treatment Plant, and the associated sanitary sewer system, pump stations, and land outfalls.

Halomethanes

The sum of bromoform, bromomethane (methyl bromide) and chloromethane (methyl chloride).

HCH

The sum of the alpha, beta, gamma (lindane) and delta isomers of hexachlorocyclohexane.

Initial Dilution

The process that results in the rapid and irreversible turbulent mixing of wastewater with ocean water around the point of discharge.

For a submerged buoyant discharge, characteristic of most municipal and industrial wastes that are released from the submarine outfalls, the momentum of the discharge and its initial buoyancy act together to produce turbulent mixing. Initial dilution in this case is completed when the diluting wastewater ceases to rise in the water column and first begins to spread horizontally.

For shallow water submerged discharges, surface discharges, and non-buoyant discharges, characteristic of cooling water wastes and some individual discharges, turbulent mixing results primarily from the momentum of discharge. Initial dilution, in these cases, is considered to be completed when the momentum induced velocity of the discharge ceases to produce significant mixing of the waste, or the diluting plume reaches a fixed distance from the discharge to be specified by the San Diego Water Board, whichever results in the lower estimate for initial dilution.

Instantaneous Maximum Effluent Limitation

The highest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous maximum limitation).

Instantaneous Minimum Effluent Limitation

The lowest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous minimum limitation).

Interference

A discharge which, alone or in conjunction with a discharge or discharges from other sources, both:

- (1) Inhibits or disrupts the POTW, its treatment processes or operations, or its sludge processes, use or disposal; and
- (2) Therefore is a cause of a violation of any requirement of the POTW's NPDES permit (including an increase in the magnitude or duration of a violation) or of the prevention of sewage sludge use or disposal in compliance with the following statutory provisions and regulations or permits issued thereunder (or more stringent State or local regulations): section 405 of the CWA, the Solid Waste Disposal Act (SWDA) (including title II, more commonly referred to as the Resource Conservation and Recovery Act (RCRA), and including State regulations contained in any State sludge management plan prepared pursuant to subtitle D of the SWDA), the Clean Air Act, the Toxic Substances Control Act, and the Marine Protection, Research and Sanctuaries Act.

Kelp Beds

For purposes of the bacteriological standards of the Ocean Plan, are significant aggregations of marine algae of the genera Macrocystis and Nereocystis. Kelp beds include the total foliage canopy of Macrocystis and Nereocystis plants throughout the water column.

Mariculture

The culture of plants and animals in marine waters independent of any pollution source.

Material

(a) In common usage: (1) the substance or substances of which a thing is made or composed (2) substantial; (b) For purposes of the Ocean Plan relating to waste disposal, dredging and the disposal of dredged material and fill, MATERIAL means matter of any kind or description which is subject to regulation as waste, or any material dredged from the navigable waters of the United States. See also, DREDGED MATERIAL.

Maximum Daily Effluent Limitation (MDEL)

The highest allowable daily discharge of a pollutant.

Method Detection Limit (MDL)

The minimum concentration of a substance that can be reported with 99 percent confidence that the measured concentration is distinguishable from method blank results, as defined in 40 CFR part 136, Attachment B.

Minimum Level (ML)

The concentration at which the entire analytical system must give a recognizable signal and acceptable calibration point. The ML is the concentration in a sample that is equivalent to the concentration of the lowest calibration standard analyzed by a specific analytical procedure, assuming that all the method specified sample weights, volumes, and processing steps have been followed.

Natural Light

Reduction of natural light may be determined by the San Diego Water Board by measurement of light transmissivity or total irradiance, or both, according to the monitoring needs of the San Diego Water Board.

Not Detected (ND)

Those sample results less than the laboratory's MDL.

Ocean Waters

The territorial marine waters of the state as defined by California law to the extent these waters are outside of enclosed bays, estuaries, and coastal lagoons. If a discharge outside the territorial waters of the state could affect the quality of the waters of the state, the discharge may be regulated to assure no violation of the Ocean Plan will occur in ocean waters.

Pass Through

A discharge which exits the POTW into waters of the United States in quantities or concentrations which, alone or in conjunction with a discharge or discharges from other sources, is a cause of a violation of any requirement of the POTW's NPDES permit (including an increase in the magnitude or duration of a violation).

Percent Removal

A percentage expression of the removal efficiency across a treatment plant for a given pollutant parameter, as determined from the average values of the raw wastewater influent pollutant concentrations to the facility and the average values of the effluent pollutant concentrations for a given time period.

PAHs (polynuclear aromatic hydrocarbons)

The sum of acenaphthylene, anthracene, 1,2-benzanthracene, 3,4-benzofluoranthene, benzo[k]fluoranthene, 1,12-benzoperylene, benzo[a]pyrene, chrysene, dibenzo[ah]anthracene, fluorene, indeno[1,2,3-cd]pyrene, phenanthrene and pyrene.

PCBs (polychlorinated biphenyls)

The sum of chlorinated biphenyls whose analytical characteristics resemble those of Aroclor-1016, Aroclor-1221, Aroclor-1232, Aroclor-1242, Aroclor-1248, Aroclor-1254 and Aroclor-1260.

Phenolic Compounds (non-chlorinated)

The sum of 2,4-dimethylphenol, 4,6-Dinitro-2-methylphenol, 2,4-dinitrophenol, 2-methylphenol, 4-methylphenol, 2-nitrophenol, 4-nitrophenol, and phenol.

Pollutant

Pollutant means dredged spoil, solid waste, incinerator residue, filter backwash, sewage, garbage, sewage sludge, munitions, chemical wastes, biological materials, radioactive materials (except those regulated under the Atomic Energy Act of 1954, as amended (42 U.S.C. 2011 et seq.)), heat, wrecked or discarded equipment, rock, sand, cellar dirt and industrial, municipal, and agricultural waste discharged into water. It does not mean: (a) Sewage from vessels; or (b) Water, gas, or other material which is injected into a well to facilitate production of oil or gas, or water derived in association with oil and gas production and disposed of in a well, if the well used either to facilitate production or for disposal purposes is approved by authority of the State in which the well is located, and if the State determines that the injection or disposal will not result in the degradation of ground or surface water resources.

Pollutant Minimization Program (PMP)

A program to reduce all potential sources of a pollutant through pollutant minimization (control) strategies, including pollution prevention measures, in order to maintain the effluent concentration at or below the effluent limitation.

Recycled Water

Recycled water means water which, as a result of treatment of waste, is suitable for a direct beneficial use or a controlled use that would not otherwise occur and is therefore considered a valuable resource.

Reported Minimum Level

The reported ML (also known as the Reporting Level or RL) is the ML (and its associated analytical method) chosen by the Discharger for reporting and compliance determination from the MLs included in this Order, including an additional factor if applicable as discussed herein. The MLs included in this Order correspond to approved analytical methods for reporting a sample result that are selected by the San Diego Water Board either from Appendix II of the Ocean Plan in accordance with section III.C.5.a. of the Ocean Plan or established in accordance with section III.C.5.b. of the Ocean Plan. The ML is based on the proper application of method-based analytical procedures for sample preparation and the absence of any matrix interferences. Other factors may be applied to the ML depending on the specific sample preparation steps employed. For example, the treatment typically applied in cases where there are matrix-effects is to dilute the sample or sample aliquot by a factor of ten. In such cases, this additional factor must be applied to the ML in the computation of the reported ML.

Sanitary Sewer Overflow (SSO)

An SSO is any overflow, spill, release, discharge or diversion of untreated or partially treated wastewater from a sanitary sewer system. SSOs include: (i) Overflows or releases of untreated or partially treated wastewater that reach waters of the United States; (ii) Overflows or releases of untreated or partially treated wastewater that do not reach waters of the United States; and (iii) Wastewater backups into buildings and on private property that are caused by blockages or flow conditions within the publicly-owned portion of a sanitary sewer system.

Sanitary Sewer System

Any system of pipes, pump stations, sewer lines, or other conveyances, upstream of a wastewater treatment plant headworks used to collect and convey wastewater to the publicly-owned treatment facility. Temporary storage and conveyance facilities (such as vaults, temporary piping, construction trenches, wet wells, impoundments, tanks, etc.) are considered to be part of the sanitary sewer system, and discharges into these temporary storage facilities are not considered to be SSOs.

Severe Property Damage

Substantial physical damage to property, damage to the treatment facilities, which causes them to become inoperable, or substantial and permanent loss of natural resources that can reasonably be expected to occur in the absence of a bypass. Severe property damage does not mean economic loss caused by delays in production. (40 CFR section 122.41(m)(1)(ii))

Shellfish

Organisms identified by the California Department of Health Services as shellfish for public health purposes (i.e., mussels, clams and oysters).

Significant Difference

Defined as a statistically significant difference in the means of two distributions of sampling results at the 95 percent confidence level.

Six-Month Median Effluent Limitation

See 6-Month Median above for definition of this term.

Sludge

Any solid, semisolid, or liquid waste generated from a municipal, commercial, or industrial wastewater treatment plant, water supply treatment plant, or air pollution control facility or any other such waste having similar characteristics and effect.

State Water Quality Protection Areas (SWQPAs)

Non-terrestrial marine or estuarine areas designated to protect marine species or biological communities from an undesirable alteration in natural water quality. All AREAS OF SPECIAL BIOLOGICAL SIGNIFICANCE (ASBS) that were previously designated by the State Water Board in Resolutions 74-28, 74-32, and 75-61 are now also classified as a subset of State Water Quality Protection Areas and require special protections afforded by the Ocean Plan.

Statistical Threshold Value (STV)

A set value that approximates the 90th percentile of the water quality distribution for a bacterial population.

TCDD Equivalents

The sum of the concentrations of chlorinated dibenzodioxins (2,3,7,8-CDDs) and chlorinated dibenzofurans (2,3,7,8-CDFs) multiplied by their respective toxicity factors, as shown in the table below.

Isomer Group	Toxicity Equivalence Factor
	1.0
2,3,7,8-tetra CDD	
2,3,7,8-penta CDD	0.5
2,3,7,8-hexa CDDs	0.1
2,3,7,8-hepta CDD	0.01
octa CDD	0.001
2,3,7,8 tetra CDF	0.1
1,2,3,7,8 penta CDF	0.05
2,3,4,7,8 penta CDF	0.5
2,3,7,8 hexa CDFs	0.1
2,3,7,8 hepta CDFs	0.01
octa CDF	0.001

Thirty-Day Average

See 30-day average above for definition of this term.

Toxicity Identification Evaluation (TIE)

A set of procedures conducted to identify the specific chemical(s) responsible for toxicity. These procedures are performed in three phases (characterization, identification, and confirmation) using aquatic organism toxicity tests.

Toxicity Reduction Evaluation (TRE)

A study conducted in a step-wise process designed to identify the causative agents of effluent or ambient toxicity, isolate the sources of toxicity, evaluate the effectiveness of toxicity control options, and then confirm the reduction in toxicity. The first steps of the TRE consist of the collection of data relevant to the toxicity, including additional toxicity testing, and an evaluation of facility operations and maintenance practices, and best management practices. A Toxicity Identification Evaluation (TIE) may be required as part of the TRE, if appropriate. (A TIE is a set of procedures to identify the specific chemical(s) responsible for toxicity. These procedures are performed in three phases (characterization, identification, and confirmation) using aquatic organism toxicity tests.)

Trash

Trash means all improperly discarded solid material from any production, manufacturing, or processing operation including, but not limited to, products, product packaging, or containers constructed of plastic, steel, aluminum, glass, paper, or other synthetic or natural materials.

Upset

Upset means an exceptional incident in which there is unintentional and temporary noncompliance with technology-based permit effluent limitations because of factors beyond the reasonable control of the Discharger. An upset does not include noncompliance to the extent caused by operational error, improperly designed treatment facilities, inadequate treatment facilities, lack of preventive maintenance, or careless or improper operation.

Waste

As used in the Ocean Plan, waste includes a Discharger's total discharge, of whatever origin, i.e., gross, not net, discharge.

Water Quality Control Plans

There are two types of water quality control plans - Basin Plans and Statewide Plans. Regional Boards adopt Basin Plans for each region based upon surface water hydrologic basin boundaries. The Regional Basin Plans designates or describes (1) existing and potential beneficial uses of ground and surface water; (2) water quality objectives to protect the beneficial uses; (3) implementation programs to achieve these objectives; and (4) surveillance and monitoring activities to evaluate the effectiveness of the water quality control plan. The Statewide Plans address water quality concerns for surface waters that overlap Regional Board boundaries, are statewide in scope, or are otherwise considered significant and contain the same four elements. Statewide Water Quality Control Plans include the Ocean Plan, the Enclosed Bays and Estuaries Plan, the Inland Surface Waters Plan, and the Thermal Plan. A water quality control plan consists of a designation or establishment for the waters within a specified area of (1) beneficial uses to be protected, (2) water quality objectives, and (3) a

program of implementation needed for achieving water quality objectives [Water Code section 13050(j)].

Water Quality Objectives

Numerical or narrative limits on constituents or characteristics of water designed to protect designated beneficial uses of the water. [Water Code section 13050(h)]. California's water quality objectives are established by the State and Regional Water Boards in the Water Quality Control Plans.

Water Quality Standards

Provisions of State or federal law which consist of a designated use or uses for waters of the United States and water quality criteria for such waters based upon such uses. Water quality standards are to protect the public health or welfare, enhance the quality of water and serve the purposes of the Clean Water Act [40 CFR section 131.3(i)]. Under State law, the Water Boards establish beneficial uses and water quality objectives in their water quality control plans or basin plans. Together with an antidegradation policy, these beneficial uses and water quality objectives serve as water quality standards under the Clean Water Act. In Clean Water Act parlance, state beneficial uses are called “designated uses” and state water quality objectives are called “criteria.” Throughout this Order, the relevant term is used depending on the statutory scheme.

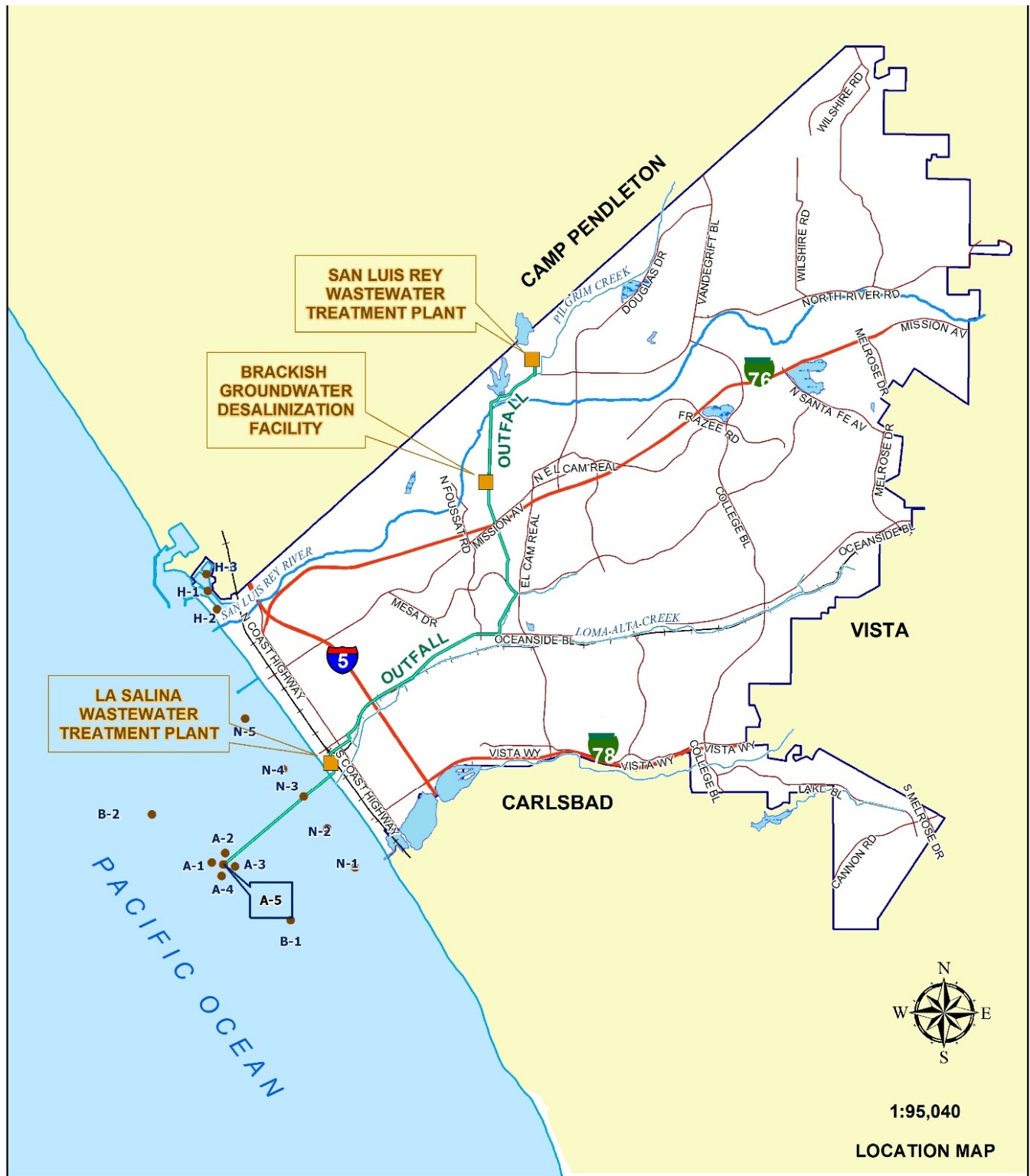
FALLBROOK PUBLIC UTILITY DISTRICT
FALLBROOK WATER RECLAMATION PLANT
SANTA MARGARITA GROUNDWATER TREATMENT PLANT

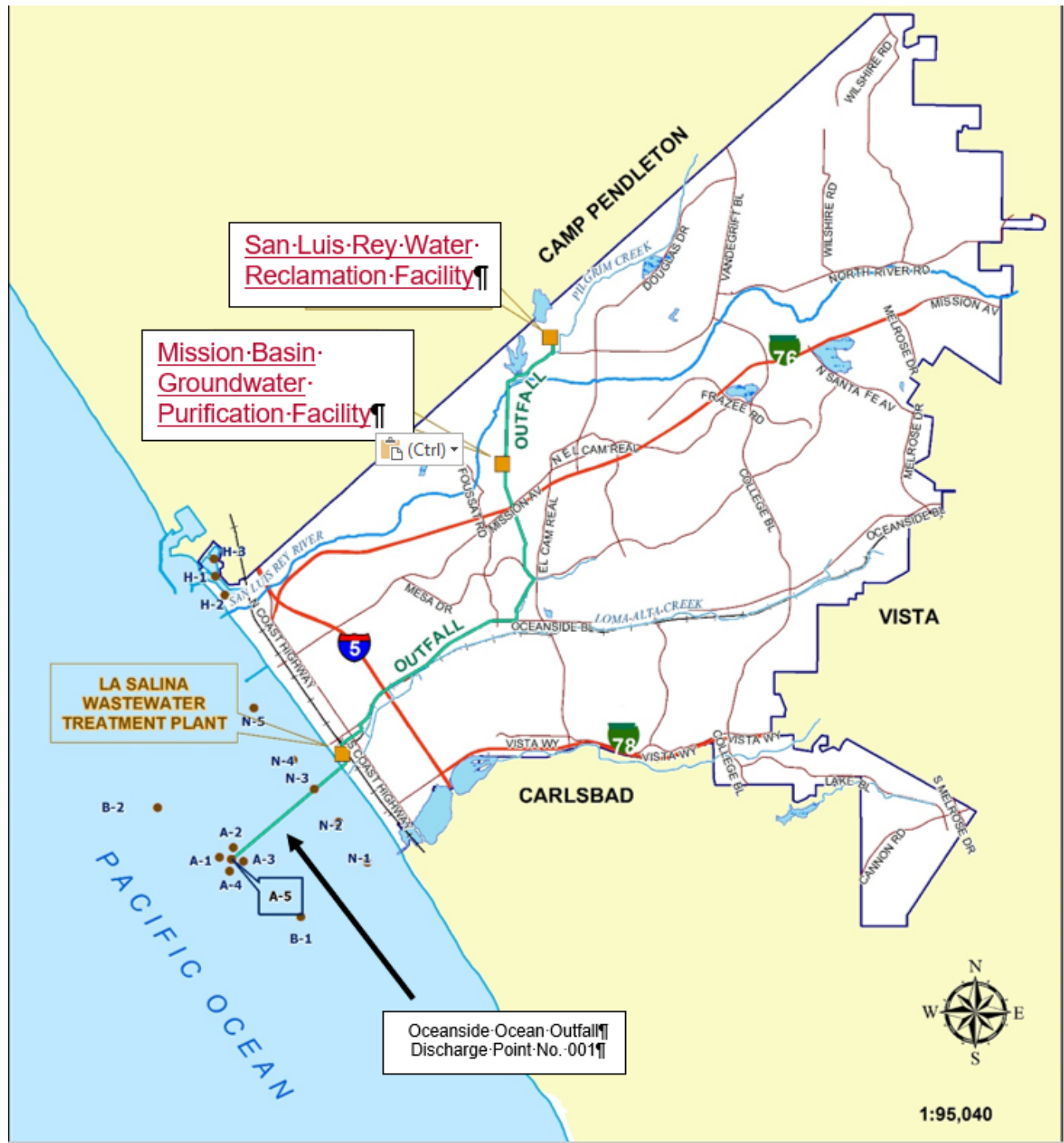
Attachment 1 to Order No. R9-2021-0204
ORDER NO. R9-2019-0169
Amended by Order No. R9-2020-0193 and R9-2021-0204
NPDES NO. CA0108031

ATTACHMENT B – MAPS



Facilities and Discharge Point Location

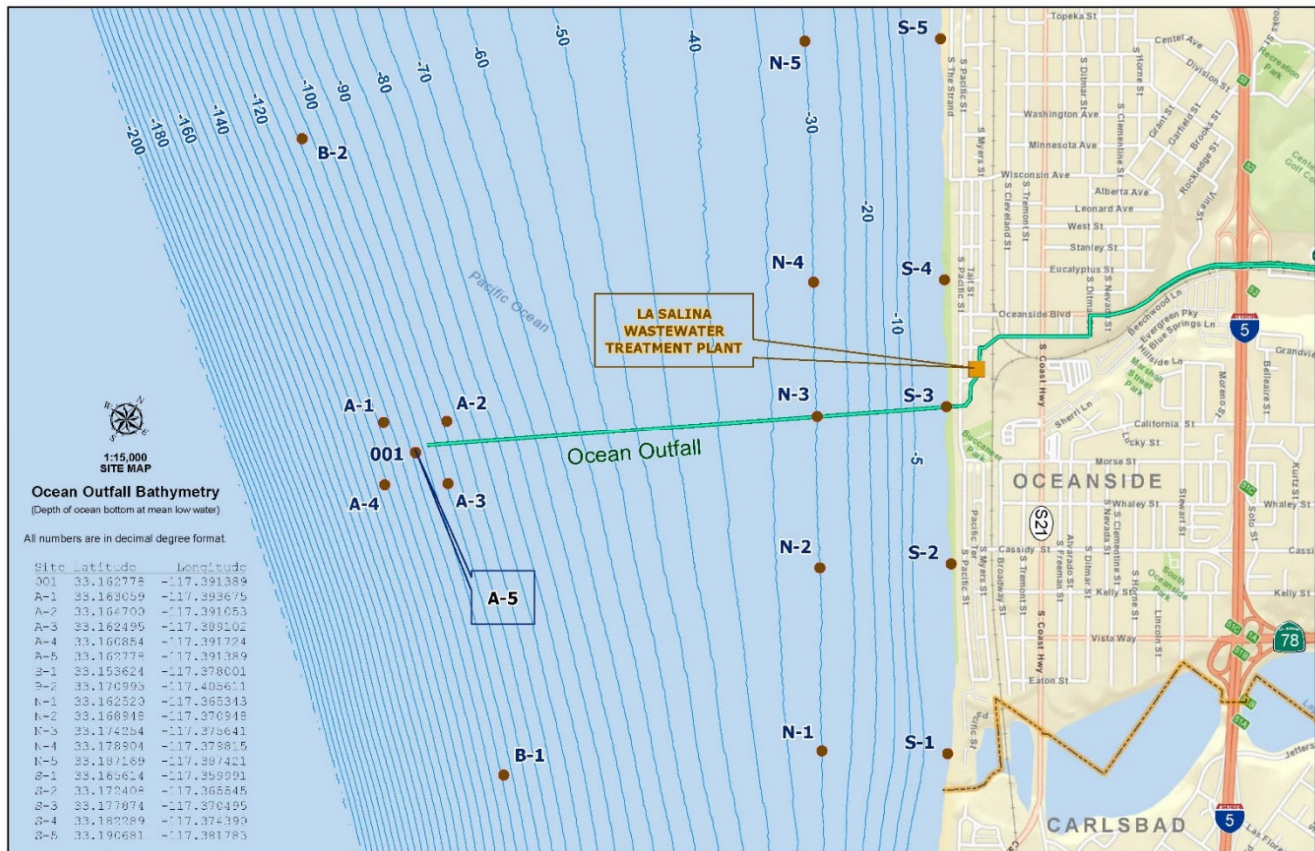




Oceanside Ocean Outfall Location

FALLBROOK PUBLIC UTILITY DISTRICT
FALLBROOK WATER RECLAMATION PLANT
SANTA MARGARITA GROUNDWATER TREATMENT PLANT

Attachment 1 to Order No. R9-2021-0204
ORDER NO. R9-2019-0169
Amended by Order No. R9-2020-0193 and R9-2021-0204
NPDES NO. CA0108031



Oceanside Ocean Outfall Ocean Monitoring Locations

FALLBROOK PUBLIC UTILITY DISTRICT
FALLBROOK WATER RECLAMATION PLANT
SANTA MARGARITA GROUNDWATER TREATMENT PLANT

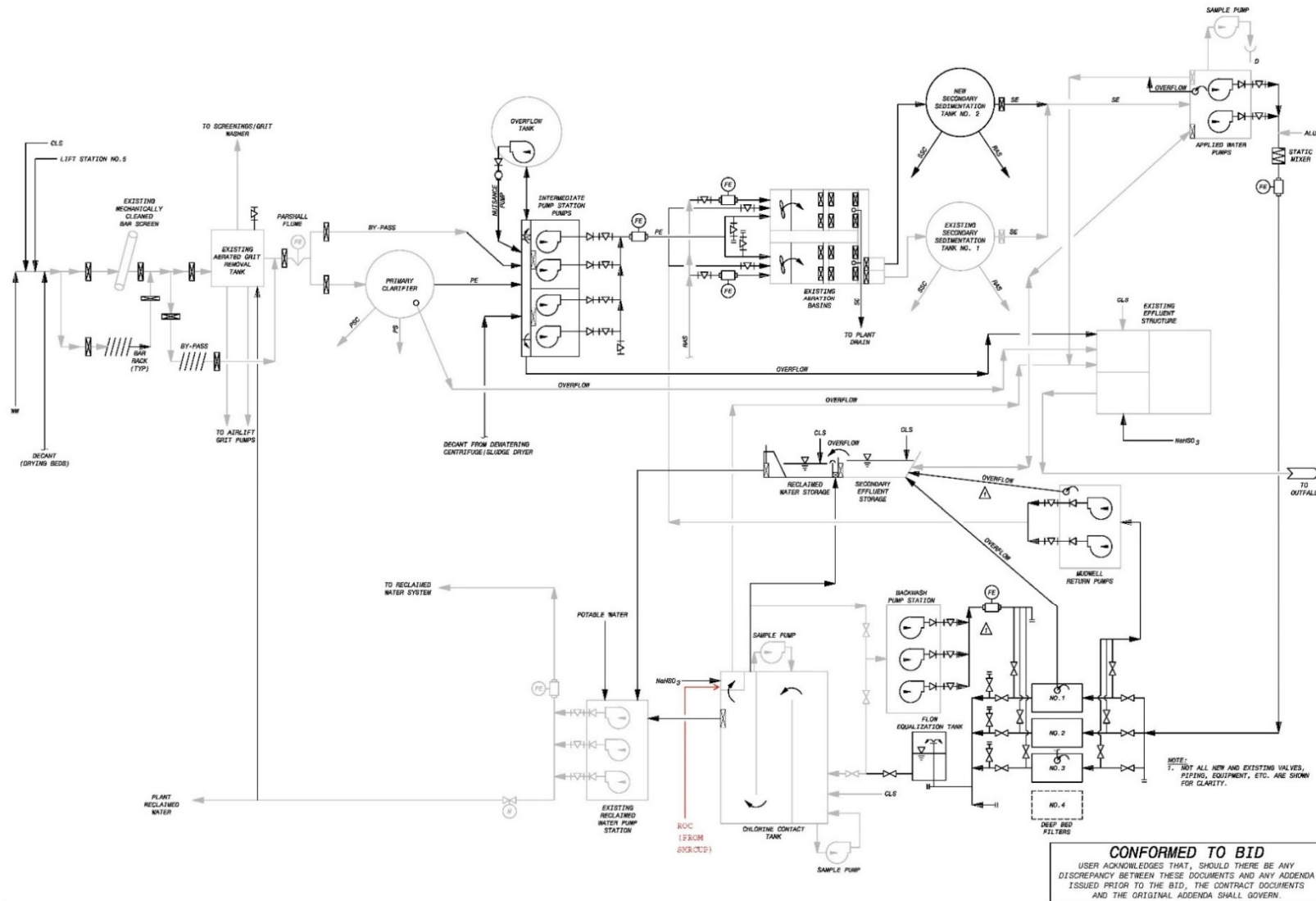
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ORDER NO. R9-2019-0169
Amended by Order No. R9-2020-0193 and R9-2021-0204
NPDES NO. CA0108031



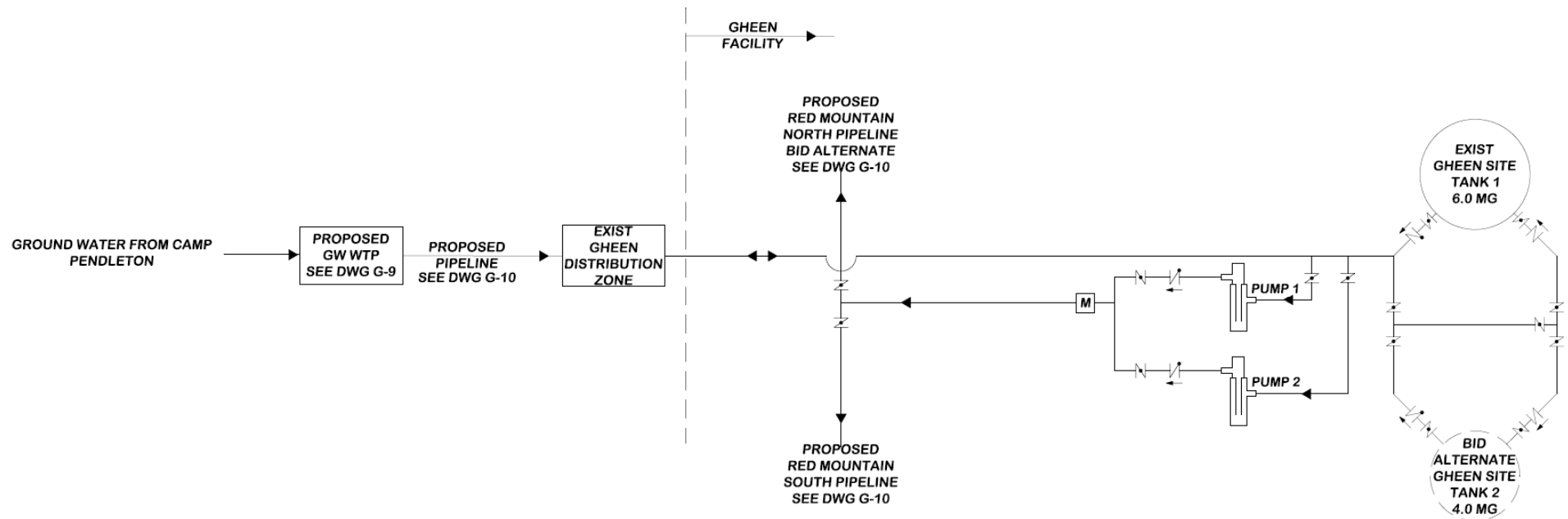
Fallbrook Public Utility District's connection to the San Luis Rey Land Outfall

ATTACHMENT C – FLOW SCHEMATICS

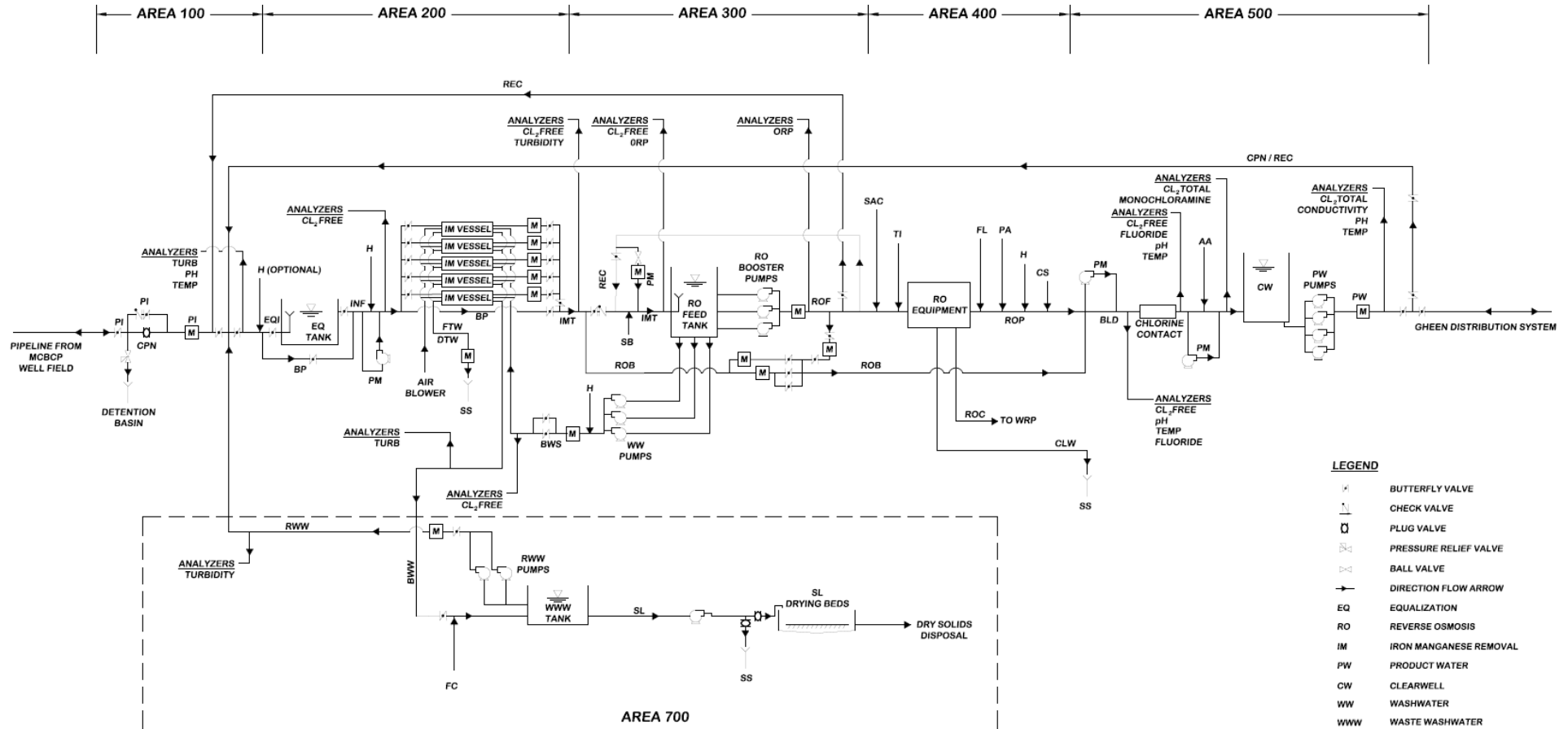
Fallbrook Water Reclamation Plant (Fallbrook WRP)



Santa Margarita Groundwater Treatment Plant (Santa Margarita GTP)



Santa Margarita GTP



FOR AREA 600 - CHEMICAL FACILITIES SEE 600 M-1

- LEGEND**
- BUTTERFLY VALVE
 - CHECK VALVE
 - PLUG VALVE
 - PRESSURE RELIEF VALVE
 - BALL VALVE
 - DIRECTION FLOW ARROW
 - EQ EQUALIZATION
 - RO REVERSE OSMOSIS
 - IM IRON MANGANESE REMOVAL
 - PW PRODUCT WATER
 - CW CLEARWELL
 - WW WASHWATER
 - WWW WASTE WASHWATER
 - FTW FILTER TO WASTE
 - MCBP MARINE CORP BASE CAMP PENDLETON
- *SEE G-11 PIPE SCHEDULE FOR ADDITIONAL ABBREVIATIONS
**THIS PLANT PROCESS FLOW DIAGRAM IS PROVIDED AS A SCHEMATIC REPRESENTATION. SEE NON-GENERAL SHEETS FOR EQUIPMENT.

ATTACHMENT D – STANDARD PROVISIONS

I. STANDARD PROVISIONS – PERMIT COMPLIANCE

A. Duty to Comply

1. The Discharger must comply with all of the terms, requirements, and conditions of this Order. Any noncompliance constitutes a violation of the Clean Water Act (CWA) and the California Water Code (Water Code) and is grounds for enforcement action; permit termination, revocation and reissuance, or modification; denial of a permit renewal application; or a combination thereof. (title 40 of the Code of Federal Regulations (40 CFR) sections 122.41(a); Water Code, sections 13261, 13263, 13265, 13268, 13000, 13001, 13304, 13350, 13385.)
2. The Discharger shall comply with effluent standards or prohibitions established under section 307(a) of the CWA for toxic pollutants within the time provided in the regulations that establish these standards or prohibitions, even if this Order has not yet been modified to incorporate the requirement. (40 CFR section 122.41(a)(1).)

B. Need to Halt or Reduce Activity Not a Defense

It shall not be a defense for a Discharger in an enforcement action that it would have been necessary to halt or reduce the permitted activity in order to maintain compliance with the conditions of this Order. (40 CFR section 122.41(c).)

C. Duty to Mitigate

The Discharger shall take all reasonable steps to minimize or prevent any discharge in violation of this Order that has a reasonable likelihood of adversely affecting human health or the environment. (40 CFR section 122.41(d).)

D. Proper Operation and Maintenance

The Discharger shall at all times properly operate and maintain all facilities and systems of treatment and control (and related appurtenances) which are installed or used by the Discharger to achieve compliance with the conditions of this Order. Proper operation and maintenance also includes adequate laboratory controls and appropriate quality assurance procedures. This provision requires the operation of backup or auxiliary facilities or similar systems that are installed by a Discharger only when necessary to achieve compliance with the conditions of this Order. (40 CFR section 122.41(e).)

E. Property Rights

1. This Order does not convey any property rights of any sort or any exclusive privileges. (40 CFR section 122.41(g).)
2. The issuance of this Order does not authorize any injury to persons or property or invasion of other private rights, or any infringement of State or local law or regulations. (40 CFR section 122.5(c).)

F. Inspection and Entry

The Discharger shall allow the California Regional Water Quality Control Board, San Diego Region (San Diego Water Board), State Water Resources Control Board (State Water Board), U.S. Environmental Protection Agency (USEPA), and/or their authorized representatives (including an authorized contractor acting as their representative), upon the presentation of credentials and other documents, as may be required by law, to (33 U.S.C. section 1318(a)(4)(b); 40 CFR section 122.41(i); Water Code, sections 13267, 13383):

1. Enter upon the Discharger's premises where a regulated facility or activity is located or conducted, or where records are kept under the conditions of this Order (33 U.S.C. section 1318(a)(4)(b)(i); 40 CFR section 122.41(i)(1); Water Code, sections 13267, 13383);
2. Have access to and copy, at reasonable times, any records that must be kept under the conditions of this Order (33 U.S.C. section 1318(a)(4)(b)(ii); 40 CFR section 122.41(i)(2); Water Code, sections 13267, 13383);
3. Inspect and photograph, at reasonable times, any facilities, equipment (including monitoring and control equipment), practices, or operations regulated or required under this Order (33 U.S.C. section 1318(a)(4)(b)(ii); 40 CFR section 122.41(i)(3); Water Code, sections 13267, 13383); and
4. Sample or monitor, at reasonable times, for the purposes of assuring Order compliance or as otherwise authorized by the CWA or the Water Code, any substances or parameters at any location. (33 U.S.C. section 1318(a)(4)(b); 40 CFR section 122.41(i)(4); Water Code, sections 13267, 13383.)

G. Bypass

1. Definitions
 - a. "Bypass" means the intentional diversion of waste streams from any portion of a treatment facility. (40 CFR section 122.41(m)(1)(i).)
 - b. "Severe property damage" means substantial physical damage to property, damage to the treatment facilities, which causes them to become inoperable, or substantial and permanent loss of natural resources that can reasonably be expected to occur in the absence of a bypass. Severe property damage does not mean economic loss caused by delays in production. (40 CFR section 122.41(m)(1)(ii).)
2. Bypass not exceeding limitations. The Discharger may allow any bypass to occur which does not cause exceedances of effluent limitations, but only if it is for essential maintenance to assure efficient operation. These bypasses are not subject to the provisions listed in Standard Provisions – Permit Compliance I.G.3, I.G.4, and I.G.5 below. (40 CFR section 122.41(m)(2).)
3. Prohibition of bypass. Bypass is prohibited, and the San Diego Water Board may take enforcement action against a Discharger for bypass, unless (40 CFR section 122.41(m)(4)(i)):

- a. Bypass was unavoidable to prevent loss of life, personal injury, or severe property damage (40 CFR section 122.41(m)(4)(i)(A));
 - b. There were no feasible alternatives to the bypass, such as the use of auxiliary treatment facilities, retention of untreated wastes, or maintenance during normal periods of equipment downtime. This condition is not satisfied if adequate back-up equipment should have been installed in the exercise of reasonable engineering judgment to prevent a bypass that occurred during normal periods of equipment downtime or preventive maintenance (40 CFR section 122.41(m)(4)(i)(B)); and
 - c. The Discharger submitted notice to the San Diego Water Board as required under Standard Provisions – Permit Compliance I.G.5 below. (40 CFR section 122.41(m)(4)(i)(C).)
4. The San Diego Water Board may approve an anticipated bypass, after considering its adverse effects, if the San Diego Water Board determines that it will meet the three conditions listed in Standard Provisions – Permit Compliance I.G.3 above. (40 CFR section 122.41(m)(4)(ii).)
5. Notice
 - a. Anticipated bypass. If the Discharger knows in advance of the need for a bypass, it shall submit prior notice, if possible at least 10 days before the date of the bypass. The notice shall be sent to the San Diego Water Board. (40 CFR section 122.41(m)(3)(i).)
 - b. Unanticipated bypass. The Discharger shall submit a notice of an unanticipated bypass as required in Standard Provisions - Reporting V.E below (24-hour notice). The notice shall be sent to the San Diego Water Board. (40 CFR section 122.41(m)(3)(ii).)

H. Upset

Upset means an exceptional incident in which there is unintentional and temporary noncompliance with technology-based permit effluent limitations because of factors beyond the reasonable control of the Discharger. An upset does not include noncompliance to the extent caused by operational error, improperly designed treatment facilities, inadequate treatment facilities, lack of preventive maintenance, or careless or improper operation.
(40 CFR section 122.41(n)(1).)

1. Effect of an upset. An upset constitutes an affirmative defense to an action brought for noncompliance with such technology-based permit effluent limitations if the requirements of Standard Provisions – Permit Compliance I.H.2 below are met. No determination made during administrative review of claims that noncompliance was caused by upset, and before an action for noncompliance, is final administrative action subject to judicial review. (40 CFR section 122.41(n)(2).)

2. Conditions necessary for a demonstration of upset. A Discharger who wishes to establish the affirmative defense of upset shall demonstrate, through properly signed, contemporaneous operating logs or other relevant evidence that (40 CFR section 122.41(n)(3)):
 - a. An upset occurred and that the Discharger can identify the cause(s) of the upset (40 CFR section 122.41(n)(3)(i));
 - b. The permitted facility was, at the time, being properly operated (40 CFR section 122.41(n)(3)(ii));
 - c. The Discharger submitted notice of the upset as required in Standard Provisions – Reporting V.E.2.b below (24-hour notice) (40 CFR section 122.41(n)(3)(iii)); and
 - d. The Discharger complied with any remedial measures required under Standard Provisions – Permit Compliance I.C above. (40 CFR section 122.41(n)(3)(iv).)
3. Burden of proof. In any enforcement proceeding, the Discharger seeking to establish the occurrence of an upset has the burden of proof. (40 CFR section 122.41(n)(4).)

II. STANDARD PROVISIONS – PERMIT ACTION

A. General

This Order may be modified, revoked and reissued, or terminated for cause. The filing of a request by the Discharger for modification, revocation and reissuance, or termination, or a notification of planned changes or anticipated noncompliance does not stay any Order condition. (40 CFR section 122.41(f).)

B. Duty to Reapply

If the Discharger wishes to continue an activity regulated by this Order after the expiration date of this Order, the Discharger must apply for and obtain a new permit. (40 CFR section 122.41(b).)

C. Transfers

This Order is not transferable to any person except after notice to the San Diego Water Board. The San Diego Water Board may require modification or revocation and reissuance of this Order to change the name of the Discharger and incorporate such other requirements as may be necessary under the CWA and the Water Code. (40 CFR sections 122.41(l)(3), 122.61.)

III. STANDARD PROVISIONS – MONITORING

- A. Samples and measurements taken for the purpose of monitoring shall be representative of the monitored activity. (40 CFR section 122.41(j)(1).)
- B. Monitoring must be conducted according to test procedures approved under 40 CFR part 136 for the analyses of pollutants unless another method is required under 40 CFR chapter 1, subchapter N. Monitoring must be conducted according to sufficiently

sensitive test methods approved under 40 CFR part 136 for the analysis of pollutants or pollutant parameters or as required under 40 CFR chapter 1, subchapter N. For the purposes of this paragraph, a method is sufficiently sensitive when:

1. The method Minimum Level (ML) is at or below the level of the most stringent effluent limitation established in the permit for the measured pollutant or pollutant parameter, and either the method ML is at or below the level of the most stringent applicable water quality criterion for the measured pollutant or pollutant parameter or the method ML is above the applicable water quality criterion but the amount of the pollutant or pollutant parameter in the facility's discharge is high enough that the method detects and quantifies the level of the pollutant or pollutant parameter in the discharge; or
2. The method has the lowest ML of the analytical methods approved under 40 CFR part 136 or required under 40 CFR chapter 1, subchapter N for the measured pollutant or pollutant parameter.

In the case of pollutants or pollutant parameters for which there are no approved methods under 40 CFR part 136 or otherwise required under 40 CFR chapter 1, subchapter N, monitoring must be conducted according to a test procedure specified in this Order for such pollutants or pollutant parameters. (40 CFR sections 122.21(e)(3), 122.41(j)(4), 122.44(i)(1)(iv).)

IV. STANDARD PROVISIONS – RECORDS

- A.** The Discharger shall retain records of all monitoring information, including all calibration and maintenance records and all original strip chart recordings for continuous monitoring instrumentation, copies of all reports required by this Order, and records of all data used to complete the application for this Order, for a period of at least three (3) years from the date of the sample, measurement, report or application. This period may be extended by request of the San Diego Water Board Executive Officer at any time. (40 CFR section 122.41(j)(2).)
- B.** Records of monitoring information shall include:
 1. The date, exact place, and time of sampling or measurements (40 CFR section 122.41(j)(3)(i));
 2. The individual(s) who performed the sampling or measurements (40 CFR section 122.41(j)(3)(ii));
 3. The date(s) analyses were performed (40 CFR section 122.41(j)(3)(iii));
 4. The individual(s) who performed the analyses (40 CFR section 122.41(j)(3)(iv));
 5. The analytical techniques or methods used (40 CFR section 122.41(j)(3)(v)); and
 6. The results of such analyses. (40 CFR section 122.41(j)(3)(vi).)
- C.** Claims of confidentiality for the following information will be denied (40 CFR section 122.7(b)):
 1. The name and address of any permit applicant or Discharger (40 CFR section 122.7(b)(1)); and

2. Permit applications and attachments, permits and effluent data.
(40 CFR section 122.7(b)(2).)

V. STANDARD PROVISIONS – REPORTING

A. Duty to Provide Information

The Discharger shall furnish to the San Diego Water Board, State Water Board, or USEPA within a reasonable time, any information which the San Diego Water Board, State Water Board, or USEPA may request to determine whether cause exists for modifying, revoking and reissuing, or terminating this Order or to determine compliance with this Order. Upon request, the Discharger shall also furnish to the San Diego Water Board, State Water Board, or USEPA copies of records required to be kept by this Order. (40 CFR section 122.41(h); Water Code, sections 13267, 13383.)

B. Signatory and Certification Requirements

1. All applications, reports, or information submitted to the San Diego Water Board, State Water Board, and/or USEPA shall be signed and certified in accordance with Standard Provisions – Reporting V.B.2, V.B.3, V.B.4, V.B.5, and V.B.6 below. (40 CFR section 122.41(k).)
2. All permit applications shall be signed by either a principal executive officer or ranking elected official. For purposes of this provision, a principal executive officer of a federal agency includes: (i) the chief executive officer of the agency, or (ii) a senior executive officer having responsibility for the overall operations of a principal geographic unit of the agency (e.g., Regional Administrators of USEPA). (40 CFR section 122.22(a)(3).)
3. All reports required by this Order and other information requested by the San Diego Water Board, State Water Board, or USEPA shall be signed by a person described in Standard Provisions – Reporting V.B.2 above, or by a duly authorized representative of that person. A person is a duly authorized representative only if:
 - a. The authorization is made in writing by a person described in Standard Provisions – Reporting V.B.2 above (40 CFR section 122.22(b)(1));
 - b. The authorization specifies either an individual or a position having responsibility for the overall operation of the regulated facility or activity such as the position of plant manager, operator of a well or a well field, superintendent, position of equivalent responsibility, or an individual or position having overall responsibility for environmental matters for the company. (A duly authorized representative may thus be either a named individual or any individual occupying a named position.) (40 CFR section 122.22(b)(2)); and
 - c. The written authorization is submitted to the San Diego Water Board and State Water Board. (40 CFR section 122.22(b)(3).)
4. If an authorization under Standard Provisions – Reporting V.B.3 above is no longer accurate because a different individual or position has responsibility for the

overall operation of the facility, a new authorization satisfying the requirements of Standard Provisions – Reporting V.B.3 above must be submitted to the San Diego Water Board and State Water Board prior to or together with any reports, information, or applications, to be signed by an authorized representative. (40 CFR section 122.22(c).)

5. Any person signing a document under Standard Provisions – Reporting V.B.2 or V.B.3 above shall make the following certification:

“I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.” (40 CFR section 122.22(d).)

6. Any person providing the electronic signature for documents described in Standard Provisions – V.B.1, V.B.2, or V.B.3 that are submitted electronically shall meet all relevant requirements of Standard Provisions – Reporting V.B, and shall ensure that all relevant requirements of 40 CFR part 3 (Cross-Media Electronic Reporting) and 40 CFR part 127 (NPDES Electronic Reporting Requirements) are met for that submission. (40 C.F.R section 122.22(e).)

C. Monitoring Reports

1. Monitoring results shall be reported at the intervals specified in the Monitoring and Reporting Program (Attachment E) in this Order. (40 CFR section 122.41(l)(4).)
2. Monitoring results must be reported on a Discharge Monitoring Report (DMR) form or forms provided or specified by the San Diego Water Board or State Water Board. As of December 21, 2016, all reports and forms must be submitted electronically to the initial recipient defined in Standard Provisions – Reporting V.J and comply with 40 CFR part 3, 40 CFR section 122.22, and 40 CFR part 127. (40 CFR section 122.41(l)(4)(i).)
3. If the Discharger monitors any pollutant more frequently than required by this Order using test procedures approved under 40 CFR part 136, or another method required for an industry-specific waste stream under 40 CFR chapter 1, subchapter N, the results of such monitoring shall be included in the calculation and reporting of the data submitted in the DMR or reporting form specified by the San Diego Water Board or State Water Board. (40 CFR section 122.41(l)(4)(ii).)
4. Calculations for all limitations, which require averaging of measurements, shall utilize an arithmetic mean unless otherwise specified in this Order. (40 CFR section 122.41(l)(4)(iii).)

D. Compliance Schedules

Reports of compliance or noncompliance with, or any progress reports on, interim and final requirements contained in any compliance schedule of this Order, shall be submitted no later than 14 days following each schedule date. (40 CFR section 122.41(l)(5).)

E. Twenty-Four Hour Reporting

1. The Discharger shall report any noncompliance which may endanger health or the environment. Any information shall be provided orally within 24 hours from the time the Discharger becomes aware of the circumstances. A report shall also be provided within five (5) days of the time the Discharger becomes aware of the circumstances. The report shall contain a description of the noncompliance and its cause; the period of noncompliance, including exact dates and times, and if the noncompliance has not been corrected, the anticipated time it is expected to continue; and steps taken or planned to reduce, eliminate, and prevent reoccurrence of the noncompliance.

For noncompliance events related to combined sewer overflows, sanitary sewer overflows, or bypass events, these reports must include the data described above (with the exception of time of discovery) as well as the type of event (i.e., combined sewer overflow, sanitary sewer overflow, or bypass event), type of overflow structure (e.g., manhole, combined sewer overflow outfall), discharge volume untreated by the treatment works treating domestic sewage, types of human health and environmental impacts of the event, and whether the noncompliance was related to wet weather.

As of December 21, 2020, all reports related to combined sewer overflows, sanitary sewer overflows, or bypass events: must be submitted to the San Diego Water Board and must be submitted electronically to the initial recipient defined in Standard Provisions – Reporting V.J. The reports shall comply with 40 CFR part 3, 40 CFR section 122.22, and 40 CFR part 127. The San Diego Water Board may also require the Discharger to electronically submit reports not related to combined sewer overflows, sanitary sewer overflows, or bypass events under this section. (40 CFR section 122.41(l)(6)(i).)

2. The following shall be included as information that must be reported within 24 hours:
 - a. Any unanticipated bypass that exceeds any effluent limitation in this Order. (40 CFR section 122.41(l)(6)(ii)(A).)
 - b. Any upset that exceeds any effluent limitation in this Order. (40 CFR section 122.41(l)(6)(ii)(B).)
3. The San Diego Water Board may waive the above required written report on a case-by-case basis if an oral report has been received within 24 hours. (40 CFR section 122.41(l)(6)(ii)(B).)

F. Planned Changes

The Discharger shall give notice to the San Diego Water Board as soon as possible of any planned physical alterations or additions to the permitted facility. Notice is required under this provision only when (40 CFR section 122.41(l)(1)):

1. The alteration or addition to a permitted facility may meet one of the criteria for determining whether a facility is a new source in section 122.29(b) (40 CFR section 122.41(l)(1)(i)); or
2. The alteration or addition could significantly change the nature or increase the quantity of pollutants discharged. This notification applies to pollutants that are not subject to effluent limitations in this Order. (40 CFR section 122.41(l)(1)(ii).)

The alteration or addition could significantly change the nature or increase the quantity of pollutants discharged. This notification applies to pollutants that are subject neither to effluent limitations in this Order nor to notification requirements under section 122.42(a)(1) (see Additional Provisions—Notification Levels VII.A.1). (40 CFR section 122.41(l)(1)(ii).)

G. Anticipated Noncompliance

The Discharger shall give advance notice to the San Diego Water Board of any planned changes in the permitted facility or activity that may result in noncompliance with this Order's requirements. (40 CFR section 122.41(l)(2).)

H. Other Noncompliance

The Discharger shall report all instances of noncompliance not reported under Standard Provisions – Reporting V.C, V.D, and V.E above at the time monitoring reports are submitted. The reports shall contain the information listed in Standard Provision – Reporting V.E above. For noncompliance events related to combined sewer overflows, sanitary sewer overflows, or bypass events, these reports shall contain the information described in Standard Provision – Reporting V.E and the applicable required data in appendix A to 40 CFR part 127. The San Diego Water Board may also require the Discharger to electronically submit reports not related to combined sewer overflows, sanitary sewer overflows, or bypass events under this section. (40 CFR section 122.41(l)(7).)

I. Other Information

When the Discharger becomes aware that it failed to submit any relevant facts in a permit application, or submitted incorrect information in a permit application or in any report to the San Diego Water Board, State Water Board, or USEPA, the Discharger shall promptly submit such facts or information. (40 CFR section 122.41(l)(8).)

J. Initial Recipient for Electronic Reporting Data

The owner, operator, or the duly authorized representative is required to electronically submit NPDES information specified in appendix A to 40 CFR part 127 to the initial recipient defined in 40 CFR section 127.2(b). USEPA will identify and publish the list of initial recipients on its website and in the Federal Register, by State and by NPDES data group [see 40 CFR section 127.2(c)]. USEPA will update and maintain this

listing.
(40 CFR section 122.41(l)(9).)

VI. STANDARD PROVISIONS – ENFORCEMENT

The San Diego Water Board is authorized to enforce the terms of this permit under several provisions of the Water Code, including, but not limited to, sections 13268, 13385, 13386, and 13387.

VII. ADDITIONAL PROVISIONS – NOTIFICATION LEVELS

Publicly-Owned Treatment Works (POTWs) - All POTWs shall provide adequate notice to the San Diego Water Board of the following (40 CFR section 122.42(b)):

- A.** Any new introduction of pollutants into the POTW from an indirect discharger that would be subject to sections 301 or 306 of the CWA if it were directly discharging those pollutants (40 CFR section 122.42(b)(1)); and
- B.** Any substantial change in the volume or character of pollutants being introduced into that POTW by a source introducing pollutants into the POTW at the time of adoption of this Order. (40 CFR section 122.42(b)(2).)
- C.** Adequate notice shall include information on the quality and quantity of effluent introduced into the POTW as well as any anticipated impact of the change on the quantity or quality of effluent to be discharged from the POTW. (40 CFR section 122.42(b)(3).)

ATTACHMENT E – MONITORING AND REPORTING PROGRAM

CONTENTS

I.	General Monitoring Provisions.....	E-3
II.	Monitoring Locations.....	E-4
III.	Core Monitoring Requirements.....	E-6
IV.	Receiving Water Monitoring Requirements	E-18
A.	Surf Zone Water Quality Monitoring Requirements	E-19
B.	Nearshore and Offshore Water Quality Monitoring Requirements.....	E-20
C.	Benthic Monitoring Requirements.....	E-23
D.	Fish and Macroinvertebrates Monitoring Requirements	E-28
E.	California Environmental Data Exchange Network	E-32
F.	Receiving Water Status and Trends	E-33
V.	Regional Monitoring Requirements	E-34
A.	Kelp Bed Canopy Monitoring Requirements.....	E-35
B.	Southern California Bight Monitoring Program Participation Requirements.....	E-36
VI.	Special Studies Requirements.....	E-37
A.	Climate Change Action Plan.....	E-37
B.	Plume Tracking Monitoring Program	E-37
VII.	Reporting Requirements.....	E-41
A.	General Monitoring and Reporting Requirements	E-41
B.	Self-Monitoring Reports (SMRs).....	E-41
C.	Discharge Monitoring Reports (DMRs).....	E-44
D.	Other Reports	E-44

TABLES

Table E-1.	Monitoring Station Locations	E-5
Table E-2.	Influent Monitoring at Monitoring Location M-INF.....	E-7
Table E-3.	Effluent Monitoring at Monitoring Location M-001 ¹	E-7
Table E-4.	Effluent Monitoring at Monitoring Location M-002 ¹	E-8
Table E-5.	Effluent Monitoring at Monitoring Location M-003 ¹	E-11
Table E-6.	Surf Zone Water Quality Monitoring ¹	E-20
Table E-7.	Nearshore and Offshore Water Quality Monitoring Requirements ¹	E-21
Table E-8.	List of Parameters to Characterize Sediment Contamination Exposure and Effect. E-25	
Table E-9.	List of Parameters to Characterize Fish Tissue.....	E-31
Table E-10.	Monitoring Periods and Reporting Schedule	E-41
Table E-11.	Other Reports.....	E-44

ATTACHMENT E – MONITORING AND REPORTING PROGRAM (MRP)

Section 308 of the federal Clean Water Act (CWA) and sections 122.41(h), (j)-(l), 122.44(i), and 122.48 of title 40 of the Code of Federal Regulations (40 CFR) require that all National Pollutant Discharge Elimination System (NPDES) permits specify monitoring and reporting requirements. California Water Code (Water Code) sections 13267 and 13383 also authorize the California Regional Water Quality Control Board, San Diego Region (San Diego Water Board) to establish monitoring, inspection, entry, reporting, and recordkeeping requirements. Pursuant to this authority, this MRP establishes conditions for the Fallbrook Public Utility District (Discharger) to conduct routine or episodic self-monitoring of the discharges regulated under this Order at specified effluent, and receiving water monitoring locations. The MRP requires the Discharger to report the results to the San Diego Water Board with information necessary to evaluate discharge characteristics and compliance status.

The purpose of the MRP is to determine and ensure compliance with effluent limitations and other requirements established in this Order, assess treatment efficiency, characterize effluents, and characterize the receiving water and the effects of the discharge on the receiving water. The MRP also specifies requirements concerning the proper use, maintenance, and installation of monitoring equipment and methods, and the monitoring type intervals and frequency necessary to yield data that are representative of the activities and discharges regulated under this Order.

Each monitoring section contains an introductory paragraph summarizing why the monitoring is needed and the key management questions the monitoring is designed to answer. In developing the list of key management questions, the San Diego Water Board considered four basic types of information for each question:

- (1) Management Information Need – Why does the San Diego Water Board need to know the answer?
- (2) Monitoring Criteria – What monitoring will be conducted for deriving an answer to the question?
- (3) Expected Product – How should the answer be expressed and reported?
- (4) Possible Management Actions – What actions will be potentially influenced by the answer?

The framework for this monitoring program has three components that comprise a range of spatial and temporal scales: 1. core monitoring, 2. regional monitoring, and 3. special studies.

1. Core monitoring consists of the basic site-specific monitoring necessary to measure compliance with individual effluent limits and/or impacts to receiving water quality. Core monitoring is typically conducted in the immediate vicinity of the discharge by examining local scale spatial effects.
2. Regional monitoring provides information necessary to make assessments over large areas and serves to evaluate cumulative effects of all anthropogenic inputs. Regional monitoring data also assists in the interpretation of core monitoring studies. In the event that a regional monitoring effort takes place during the permit cycle in which the MRP does not specifically address regional monitoring, the San Diego Water Board may

allow relief from aspects of core monitoring components in order to encourage participation pursuant to section V of this MRP.

3. Special studies are directed monitoring efforts designed in response to specific management or research questions identified through either core or regional monitoring programs. Often, they are used to help understand core or regional monitoring results, where a specific environmental process is not well understood, or to address unique issues of local importance.

I. GENERAL MONITORING PROVISIONS

- A. Samples and measurements taken as required herein shall be representative of the volume and nature of the monitored discharge. All samples shall be taken at the monitoring points specified in section II, Table E-1 and, unless otherwise specified, before the monitored flow joins or is diluted by any other waste stream, body of water, or substance. Monitoring points shall not be changed without notification to and the approval of the San Diego Water Board.
- B. All monitoring instruments and devices used by the Discharger to fulfill the prescribed monitoring program shall be properly maintained and calibrated as necessary to ensure their continued accuracy. Appropriate flow measurement devices and methods consistent with accepted scientific practices shall be selected and used to ensure the accuracy and reliability of measurements of the volume of monitored discharges. The flow measurement devices shall be installed, calibrated at least once per year (i.e., no more than 12 months between calibrations) or more frequently, and maintained to ensure that the accuracy of the measurement is consistent with the accepted capability of that type of device. The flow measurement devices selected shall be capable of measuring flows with a maximum deviation of less than ± 5 percent from true discharge rates throughout the range of expected discharge volumes.
- C. Monitoring must be conducted according to U.S. Environmental Protection Agency (USEPA) test procedures approved at 40 CFR part 136, *Guidelines Establishing Test Procedures for the Analysis of Pollutants Under the CWA* as amended, or an alternative test procedure (ATP) approved by USEPA, or by the San Diego Water Board when there are no methods specified for a pollutant at 40 CFR part 136.
- D. Data produced and reports submitted pursuant to this Order shall be generated by a laboratory accredited by the State of California Environmental Laboratory Accreditation Program (ELAP). The laboratory must hold a valid certificate of accreditation for the analytical test method specified in 40 CFR 136, an ATP approved by USEPA, or by the San Diego Water Board when there are no methods specified for a pollutant at 40 CFR part 136. The laboratory must include quality assurance/quality control data in all data reports required by this Order and submit electronic data as required by the San Diego Water Board. Data generated using field tests is exempt pursuant to California Water Code Section 13176. Additional information on ELAP can be accessed at:

http://www.waterboards.ca.gov/drinking_water/certlic/labs/index.shtmlhttps://www.waterboards.ca.gov/drinking_water/certlic/labs/index.shtml.

- E. Records of monitoring information shall include information required under Standard Provision, Attachment D, section IV of this Order.
- F. The Discharger shall have, and implement, an acceptable written quality assurance (QA) plan for laboratory analyses. Duplicate chemical analyses must be conducted on a minimum of 10 percent of the samples or at least one sample per month, whichever is greater. A similar frequency shall be maintained for analyzing spiked samples. The Discharger should have a success rate equal or greater than 80 percent.
- G. When requested by USEPA or the San Diego Water Board, the Discharger will participate in the NPDES Discharge Monitoring Report QA (DMR-QA) performance study. If the DMR-QA is not requested, the Discharger shall submit the most recent Water Pollution Performance Evaluation Study. The Discharger shall ensure that the results of the DMR-QA Study or the most recent Water Pollution Performance Evaluation Study are submitted annually by December 31 to the State Water Resources Control Board at the following address:

State Water Resources Control Board Quality Assurance Program Officer
Office of Information Management and Analysis
State Water Resources Control Board
1001 I Street, Sacramento, CA 95814
- H. Analysis for toxic pollutants, including chronic toxicity, with effluent limitations or performance goals based on water quality objectives and criteria of the *Water Quality Control Plan for the San Diego Basin* (Basin Plan) and the *Water Quality Control Plan for Ocean Waters of California, California Ocean Plan* (Ocean Plan) shall be conducted in accordance with procedures described in the Ocean Plan and restated in this MRP.
- I. The Discharger shall ensure that analytical procedures used to evaluate compliance with effluent limitations or performance goals established in this Order use minimum levels (ML) no greater than the applicable effluent limitations or performance goals and are consistent with the requirements of 40 CFR part 136, or otherwise approved by USEPA and authorized by the San Diego Water Board. If no authorized ML value is below the effluent limitation, then the method must achieve an ML no greater than the lowest ML value provided in 40 CFR part 136.

II. MONITORING LOCATIONS

The Discharger shall establish the following monitoring locations to demonstrate compliance with the effluent limitations, discharge specifications, and other requirements in this Order. The North latitude and West longitude information contained in Table E-1 below are approximate for administrative purposes.

Table E-1. Monitoring Station Locations

Discharge Point Name	Monitoring Location Name	Monitoring Location Description
--	M-INF	At a location where all influent flows to the Fallbrook Water Reclamation Plant (Fallbrook WRP) are accounted for in monitoring events; upstream of any in-plant return flows; and where representative samples of influent can be collected.
--	M-001	Downstream of any in-plant return flows at the Fallbrook WRP where representative samples of effluent treated solely at the Fallbrook WRP can be collected, prior to commingling with other discharges contributing to the Oceanside Ocean Outfall (OOO).
001	M-002	At a location downstream of M-001 where representative samples of effluent treated solely at the Fallbrook WRP and brine waste from the Santa Margarita GTP can be collected, prior to commingling with other discharges contributing to the OOO. Current location is near the terminus of the Fallbrook Land Outfall and prior to connecting to the La Salina Wastewater Treatment Plant Land Outfall, which is discharges to the OOO.
--	M-003	At a location where representative samples of brine waste from the Santa Margarita Groundwater Treatment Plant (Santa Margarita GTP) can be collected, prior to commingling with other discharges contributing to the Oceanside Ocean Outfall (OOO).
SURF ZONE STATIONS		
--	S1	Surf zone, 5,500 feet south of the outfall (approximately: 33°09'56"N 117°21'36"W).
--	S2	Surf zone, 2,500 feet south of the outfall (approximately: 33°10'21"N 117°21'56"W).
--	S3	Surf zone; at the outfall (approximately: 33°10'40"N 117°22'14"W).
--	S4	Surf zone, 2,000 feet north of the outfall (approximately: 33°10'56"N 117°22'28"W).
--	S5	Surf zone, 5,800 feet north of the outfall (approximately: 33°11'27"N 117°22'54"W).
--	S6	To be determined by the San Diego Water Board based on the results of the Plume Tracking Monitoring Program
--	S7	To be determined by the San Diego Water Board based on the results of the Plume Tracking Monitoring Program
NEARSHORE STATIONS		
--	N1	Opposite S1, at the 30-foot depth contour, mean lower low water (MLLW) (approximately: 33°09'45"N 117°21'55"W).
--	N2	Opposite S2, at the 30-foot depth contour, MLLW (approximately: 33°10'08"N 117°22'15"W).
--	N3	Opposite S3, at the 30-foot depth contour, MLLW (approximately: 33°10'27"N 117°22'32"W).
--	N4	Opposite S4, at the 30-foot depth contour, MLLW (approximately: 33°10'44"N 117°22'47"W).
--	N5	Opposite S5, at the 30-foot depth contour, MLLW (approximately: 33°11'14"N 117°23'15"W).
--	N6	To be determined by the San Diego Water Board based on the results of the Plume Tracking Monitoring Program
--	N7	To be determined by the San Diego Water Board based on the results of the Plume Tracking Monitoring Program
OFFSHORE STATIONS		
--	A1	Northwest corner of a 1,000 ft x 1,000 ft square having one side parallel to shore and the intersection of its diagonals at the seaward end of the outfall (approximately: 33°09'47"N 117°23'37"W).

Discharge Point Name	Monitoring Location Name	Monitoring Location Description
--	A2	Northeast corner of a 1,000 ft x 1,000 ft square having one side parallel to shore and the intersection of its diagonals at the seaward end of the outfall (approximately: 33°09'53"N 117°23'28"W).
--	A3	Southeast corner of a 1,000 ft x 1,000 ft square having one side parallel to shore and the intersection of its diagonals at the seaward end of the outfall (approximately: 33°09'45"N 117°23'21"W).
--	A4	Southwest corner of a 1,000 ft x 1,000 ft square having one side parallel to shore and the intersection of its diagonals at the seaward end of the outfall (approximately: 33°09'39"N 117°23'30"W).
--	A5	At the seaward end of the outfall (approximately: 33°09'46"N 117°23'29"W).
--	B1	One mile downcoast from the outfall, and over the same depth contour as Station A5 (approximately: 33°09'13"N 117°22'41"W).
--	B2	One mile upcoast from the outfall and over the same depth contour as Station A5 (approximately: 33°10'16"N 117°24'20"W).
TRAWL STATIONS		
--	T3	Begin trawl slightly downcoast of station A5, first trawl to be away from station heading downcoast along the 100-foot depth (MLLW) contour
--	T4	Begin trawl slightly downcoast of station B1, first trawl to be away from station heading downcoast along the 100-foot depth (MLLW) contour
--	T5	Begin trawl slightly upcoast of station B2, first trawl to be away from station heading upcoast along the 100-foot depth (MLLW) contour.
RIG FISHING STATIONS		
--	RF1	8,000 feet northeast of the outfall (approximately: 33°08'52"N 117°22'22"W).
--	RF2	At the seaward end of outfall (approximately: 33°09'46"N 117°23'29"W).
--	RF3	8,000 feet southeast of the outfall (approximately: 33°10'59"N 117°24'09"W).

III. Core Monitoring Requirements

A. Influent Monitoring Requirements

Influent monitoring is the collection and analysis of samples or measurements of wastewater prior to the treatment processes. Influent monitoring of a wastewater stream prior to entering the treatment plant is necessary to address the following question:

- (1) What is the frequency of unexpected non-residential pollutants (or pollutants loads) which can cause or contribute to an upset in the wastewater process?
- (2) Is the influent inhibiting or disrupting the plant, its treatment processes or operations, or its sludge processes, use, or disposal?
- (3) Is the Fallbrook WRP complying with permit conditions including, but not limited to, carbonaceous biochemical oxygen demand 5-day @ 20°C (CBOD₅) and total suspended solids (TSS) percent removal limitations?

The Discharger shall monitor the influent at Monitoring Location M-INF as follows:

Table E-2. Influent Monitoring at Monitoring Location M-INF

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Flow	million gallons per day (MGD)	Recorder/Totalizer	Continuous	--
CBOD ₅	milligram per liter (mg/L)	24-hr Composite	1/Week	1
TSS	mg/L	24-hr Composite	1/Week	1

¹ As required under 40 CFR part 136.

B. Effluent Monitoring Requirements

Effluent monitoring is the collection and analysis of samples or measurements of effluents, after all treatment processes, to determine and quantify contaminants and to demonstrate compliance with applicable effluent limitations, standards, and other requirements of this Order.

Effluent monitoring is necessary to address the following questions:

- (1) Does the effluent comply with permit effluent limitations, performance goals, and other requirements of this Order, thereby ensuring that water quality standards are achieved in the receiving water?
 - (2) What is the mass of constituents that are discharged daily, monthly, or annually?
 - (3) Is the effluent concentration or mass changing over time?
 - (4) Are the Facilities being properly operated and maintained to ensure compliance with the conditions of this Order?
1. The Discharger shall monitor the effluent at Monitoring Location M-001 as follows:

Table E-3. Effluent Monitoring at Monitoring Location M-001¹

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Flow	MGD	Recorder/Totalizer	Continuous	--
CBOD ₅	mg/L	24-hr Composite	1/Day ^{2,3,4}	5
TSS	mg/L	24-hr Composite	1/Day ^{2,3,4}	5
pH	standard units	Grab	1/Day ²	5
Oil and Grease	mg/L	Grab	1/Month ³	5
Settleable Solids	milliliter per liter (ml/L)	Grab	1/Day ²	5
Turbidity	nephelometric turbidity unit (NTU)	24-hr Composite	1/Week	5

¹ See Attachment A for definitions of abbreviations and a glossary of common terms used in this Order.

² Applies 5 days per week, except 7 days per week for at least 1 week during July or August of each year.

- ³ The Discharger shall calculate and report the mass emission rate (MER) of the constituent for each sample taken. The MER shall be calculated in accordance with section VII.I.4 of this Order.
- ⁴ The Discharger shall calculate the monthly average percent removal for CBOD₅ and TSS.
- ⁵ As required under 40 CFR part 136.

2. The Discharger shall monitor the effluent at Monitoring Location M-002 (Discharge Point No. 1) as follows:

Table E-4. Effluent Monitoring at Monitoring Location M-002¹

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Flow	MGD	Recorder/Totalizer	Continuous	--
Temperature	°C	Grab	1/Week	2
Dissolved Oxygen	mg/L	Grab	1/Week	2
Specific Electrical Conductivity (@ 25°C)	micromhos per centimeter (µmhos/cm)	Grab	1/Month	2
Fecal Coliform	<u>colony forming units per 100 milliliters</u> (CFU/100 mL) ³	Grab	1/Quarter ⁴³	2
Enterococci	CFU/100 mL ³	Grab	1/Quarter ⁴³	2
Nitrogen, Total	mg/L	24-hr Composite	1/Quarter	2
Phosphorus, Total (as P)	mg/L	24-hr Composite	1/Quarter	2
TABLE 3 PARAMETERS FOR PROTECTION OF MARINE AQUATIC LIFE				
Arsenic, Total Recoverable	microgram per liter (µg/L)	24-hr Composite	2/Year ^{4,55,6}	2
Cadmium, Total Recoverable	µg/L	24-hr Composite	2/Year ^{5,64,5}	2
Chromium (VI), Total Recoverable ⁷	µg/L	24-hr Composite	2/Year ^{5,64,5}	2
Copper, Total Recoverable	µg/L	24-hr Composite	2/Year ^{4,55,6}	2
Lead, Total Recoverable	µg/L	24-hr Composite	2/Year ^{5,64,5}	2
Mercury, Total Recoverable	µg/L	24-hr Composite	2/Year ^{5,64,5}	2
Nickel, Total Recoverable	µg/L	24-hr Composite	2/Year ^{5,64,5}	2
Selenium, Total Recoverable	µg/L	24-hr Composite	2/Year ^{5,64,5}	2
Silver, Total Recoverable	µg/L	24-hr Composite	2/Year ^{5,64,5}	2
Zinc, Total Recoverable	µg/L	24-hr Composite	2/Year ^{5,64,5}	2
Cyanide, Total	µg/L	24-hr Composite	2/Year ^{5,64,5}	2,87
Total Residual Chlorine	µg/L	Grab	1/Week ^{54,98}	2
Ammonia Nitrogen, Total (as N)	mg/L	24-hr Composite	1/Month ^{5,64,5}	2
Chronic Toxicity	<u>Toxicity Units Chronic (TUC) "Pass"/"Fail" (Test of Significant Toxicity)</u> ⁹	24-hr Composite	1/Quarter ¹⁰ 2/Year	4410
Phenolic Compounds (nonchlorinated) ¹	µg/L	24-hr Composite	2/Year ^{5,64,5}	2

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Phenolic Compounds (chlorinated) ¹	µg/L	24-hr Composite	2/Year ^{5,6,5}	2
Endosulfan ¹	µg/L	24-hr Composite	2/Year ^{5,6,5}	2
Endrin	µg/L	24-hr Composite	2/Year ^{5,6,5}	2
HCH ¹	µg/L	24-hr Composite	2/Year ^{5,6,5}	2
Radioactivity (alpha and beta particles)	picocuries per liter (pCi/L)	24-hr Composite	2/Year ^{5,6,5}	2

TABLE 3 PARAMETERS FOR PROTECTION OF HUMAN HEALTH – NONCARCINOGENS

Acrolein	µg/L	Grab	2/Year ^{5,6,5}	2
Antimony, Total Recoverable	µg/L	24-hr Composite	2/Year ^{5,6,5}	2
Bis (2-chloroethoxy) Methane	µg/L	24-hr Composite	2/Year ^{5,6,5}	2
Bis (2-chloroisopropyl) Ether	µg/L	24-hr Composite	2/Year ^{5,6,5}	2
Chlorobenzene	µg/L	Grab	2/Year ^{5,6,5}	2
Chromium (III), Total Recoverable ⁷	µg/L	24-hr Composite	2/Year ^{5,6,5}	2
Di-n-butyl Phthalate	µg/L	24-hr Composite	2/Year ^{5,6,5}	2
Dichlorobenzenes ¹	µg/L	Grab	2/Year ^{5,6,5}	2
Diethyl Phthalate	µg/L	24-hr Composite	2/Year ^{5,6,5}	2
Dimethyl Phthalate	µg/L	24-hr Composite	2/Year ^{5,6,5}	2
4,6-dinitro-2-methylphenol	µg/L	24-hr Composite	2/Year ^{5,6,5}	2
2,4-dinitrophenol	µg/L	24-hr Composite	2/Year ^{5,6,5}	2
Ethylbenzene	µg/L	Grab	2/Year ^{5,6,5}	2
Fluoranthene	µg/L	24-hr Composite	2/Year ^{5,6,5}	2
Hexachlorocyclopentadiene	µg/L	24-hr Composite	2/Year ^{5,6,5}	2
Nitrobenzene	µg/L	24-hr Composite	2/Year ^{5,6,5}	2
Thallium, Total Recoverable	µg/L	24-hr Composite	2/Year ^{5,6,5}	2
Toluene	µg/L	Grab	2/Year ^{5,6,5}	2
Tributyltin	µg/L	24-hr Composite	2/Year ^{5,6,5}	2
1,1,1-trichloroethane	µg/L	Grab	2/Year ^{5,6,5}	2

TABLE 3 PARAMETERS FOR PROTECTION OF HUMAN HEALTH – CARCINOGENS

Acrylonitrile	µg/L	Grab	2/Year ^{5,6,5}	2
Aldrin	µg/L	24-hr Composite	2/Year ^{5,6,5}	2
Benzene	µg/L	Grab	2/Year ^{5,6,5}	2
Benzidine	µg/L	24-hr Composite	2/Year ^{5,6,5}	2
Beryllium, Total Recoverable	µg/L	24-hr Composite	2/Year ^{5,6,5}	2
Bis (2-chloroethyl) Ether	µg/L	24-hr Composite	2/Year ^{5,6,5}	2
Bis (2-ethylhexyl) Phthalate	µg/L	24-hr Composite	2/Year ^{5,6,5}	2
Carbon Tetrachloride	µg/L	Grab	2/Year ^{5,6,5}	2
Chlordane ¹	µg/L	24-hr Composite	2/Year ^{5,6,5}	2
Chlorodibromomethane	µg/L	Grab	2/Year ^{5,6,5}	2
Chloroform	µg/L	Grab	2/Year ^{5,6,5}	2
Dichlorodiphenyltrichloroethane (DDT) ¹	µg/L	24-hr Composite	2/Year ^{5,6,5}	2

FALLBROOK PUBLIC UTILITY DISTRICT
FALLBROOK WATER RECLAMATION PLANT
SANTA MARGARITA GROUNDWATER TREATMENT PLANT

Attachment 1 to Order No. R9-2021-0204
ORDER NO. R9-2019-0169
Amended by Order No. R9-2020-0193 and R9-2021-0204
NPDES NO. CA0108031

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
1,4-dichlorobenzene	µg/L	Grab	2/Year ^{5,64,5}	2
3,3'-dichlorobenzidine	µg/L	24-hr Composite	2/Year ^{5,64,5}	2
1,2-dichloroethane	µg/L	Grab	2/Year ^{5,64,5}	2
1,1-dichloroethylene	µg/L	Grab	2/Year ^{5,64,5}	2
Dichlorobromomethane	µg/L	Grab	2/Year ^{5,64,5}	2
Dichloromethane	µg/L	Grab	2/Year ^{5,64,5}	2
1,3-dichloropropene	µg/L	Grab	2/Year ^{5,64,5}	2
Dieldrin	µg/L	24-hr Composite	2/Year ^{5,64,5}	2
2,4-dinitrotoluene	µg/L	24-hr Composite	2/Year ^{5,64,5}	2
1,2-diphenylhydrazine	µg/L	24-hr Composite	2/Year ^{5,64,5}	2
Halomethanes ¹	µg/L	Grab	2/Year ^{5,64,5}	2
Heptachlor	µg/L	24-hr Composite	1/Month ^{5,64,5}	2
Heptachlor Epoxide	µg/L	24-hr Composite	1/Month ^{5,64,5}	2
Hexachlorobenzene	µg/L	24-hr Composite	2/Year ^{5,64,5}	2
Hexachlorobutadiene	µg/L	24-hr Composite	2/Year ^{5,64,5}	2
Hexachloroethane	µg/L	24-hr Composite	2/Year ^{5,64,5}	2
Isophorone	µg/L	24-hr Composite	2/Year ^{5,64,5}	2
N-nitrosodimethylamine	µg/L	24-hr Composite	2/Year ^{5,64,5}	2
N-nitrosodi-N-propylamine	µg/L	24-hr Composite	2/Year ^{5,64,5}	2
N-nitrosodiphenylamine	µg/L	24-hr Composite	2/Year ^{5,64,5}	2
Polynuclear Aromatic Hydrocarbons (PAHs) ¹	µg/L	24-hr Composite	2/Year ^{5,64,5}	2
Polychlorinated Biphenyls (PCBs) ¹	µg/L	24-hr Composite	2/Year ^{5,64,5}	2
TCDD equivalents ¹	µg/L	24-hr Composite	1/Quarter ^{5,64,5}	2
1,1,2,2-tetrachloroethane	µg/L	Grab	2/Year ^{5,64,5}	2
Tetrachloroethylene	µg/L	Grab	2/Year ^{5,64,5}	2
Toxaphene	µg/L	24-hr Composite	2/Year ^{5,64,5}	2
Trichloroethylene	µg/L	Grab	2/Year ^{5,64,5}	2
1,1,2-trichloroethane	µg/L	Grab	2/Year ^{5,64,5}	2
2,4,6-trichlorophenol	µg/L	24-hr Composite	2/Year ^{5,64,5}	2
Vinyl Chloride	µg/L	Grab	2/Year ^{5,64,5}	2
Parameters Relevant to Ocean Acidification and Hypoxia				
<u>Ammonium¹¹</u>	<u>mg/L</u>	<u>24-hr Composite</u>	<u>1/Month^{5,12}</u>	<u>2</u>
<u>Nitrogen, Total¹³</u>	<u>mg/L</u>	<u>24-hr Composite</u>	<u>1/Month^{5,12}</u>	<u>2</u>
<u>Nitrogen, Total Organic¹⁴</u>	<u>mg/L</u>	<u>24-hr Composite</u>	<u>1/Month^{5,12}</u>	<u>2</u>
<u>Nitrate</u>	<u>mg/L</u>	<u>24-hr Composite</u>	<u>1/Month^{5,12}</u>	<u>2</u>
<u>Nitrite</u>	<u>mg/L</u>	<u>24-hr Composite</u>	<u>1/Month^{5,12}</u>	<u>2</u>
<u>Phosphorus, Total (as P)</u>	<u>mg/L</u>	<u>24-hr Composite</u>	<u>1/Month^{5,12}</u>	<u>2</u>
<u>Phosphate</u>	<u>mg/L</u>	<u>24-hr Composite</u>	<u>1/Month^{5,12}</u>	<u>2</u>
<u>Carbon, Total Organic</u>	<u>mg/L</u>	<u>24-hr Composite</u>	<u>1/Month^{5,12}</u>	<u>2</u>
<u>Carbon, Dissolved Inorganic¹⁵</u>	<u>mg/L</u>	<u>24-hr Composite</u>	<u>1/Month^{5,12}</u>	<u>2</u>
<u>Iron, Dissolved</u>	<u>mg/L</u>	<u>24-hr Composite</u>	<u>1/Month^{5,12}</u>	<u>2</u>

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Alkalinity	mg/L CaCO ₃	24-hr Composite	1/Month ¹²	²

¹ See Attachment A for definitions of abbreviations and a glossary of common terms used in this Order.

² As required under 40 CFR part 136.

²³ ~~The Discharger may report in most probable number (MPN) in lieu of CFU.~~

³⁴ ~~The Discharger shall monitor the effluent on the same day as the receiving water monitoring for the same parameter.~~

⁴⁵ ~~The Discharger shall calculate and report the MER of the constituent for each sample taken. The MER shall be calculated in accordance with section VII.I.4 of this Order.~~

⁵⁶ ~~The minimum frequency of monitoring for this constituent is automatically increased to twice the minimum frequency specified, if any analysis for this constituent yields a result higher than the applicable effluent limitation or performance goal specified in this Order. The increased minimum frequency of monitoring shall remain in effect until the results of a minimum of four consecutive analyses for this constituent are below all applicable effluent limitations or performance goals specified in this Order.~~

⁶⁷ ~~The Discharger may, at their option, meet this performance goal as a total chromium performance goal and monitor for total recoverable chromium in lieu of total recoverable chromium (III) or total recoverable chromium (VI).~~

⁷⁸ ~~If a Discharger can demonstrate to the satisfaction of the San Diego Water Board (subject to USEPA approval) that an analytical method is available to reliably distinguish between strongly and weakly complexed cyanide, performance goals may be evaluated with the combined measurement of free cyanide, simple alkali metals cyanides, and weakly complexed organometallic cyanide complexes. In order for the analytical method to be acceptable, the recovery of free cyanide from metal complexes must be comparable to that achieved by the approved method in 40 CFR part 136, as revised May 14, 1999.~~

⁸⁹ ~~Monitoring of total chlorine residual is not required on days when none of the treatment units that are subject to this Order use chlorine for disinfection. If only one sample is collected for total chlorine residual analysis on a particular day, that sample must be collected at the time when the concentration of total chlorine residual in the discharge would be expected to be greatest. The times of chlorine discharges on the days that samples are collected, and the time at which samples are collected, shall be reported.~~

⁹ ~~To determine if the performance goal was met, chronic toxicity results shall be reported as "Pass" or "Fail". For monitoring purpose only, chronic toxicity results shall also include "Percent Effect".~~

⁴⁰ ~~If the effluent is in compliance with the chronic toxicity performance goal contained in Section IV.A.2, Table 8 of this Order for ten consecutive routine monitoring events, the Discharger may submit a request to the San Diego Water Board to decrease the minimum sampling frequency for chronic toxicity from quarterly to semiannually. If during the reduced frequency the effluent is not in compliance with the chronic toxicity performance goal, the frequency is automatically increased back to once per quarter.~~

¹⁰ ~~As specified in section VII.L of this Order and section III.C of this MRP (Attachment E).~~

¹¹ ~~Ammonium may be determined by conversion from ammonia-nitrogen.~~

¹² ~~Monitoring may be delayed until January 2023 to allow for budgeting and proper laboratory staffing. The minimum sampling frequency shall be once per month for a period of one year. After one year of monthly sampling, the minimum sampling frequency may automatically be reduced to once per quarter.~~

¹³ ~~Total nitrogen is the sum of total organic nitrogen, ammonia nitrogen, nitrate nitrogen, and nitrite nitrogen.~~

¹⁴ ~~Total organic nitrogen can be derived by subtracting ammonia nitrogen from total kjeldahl nitrogen.~~

⁴⁴¹⁵ ~~Dissolved inorganic carbon may be estimated using results of pH and alkalinity.~~

3. The Discharger shall monitor the effluent at Monitoring Location M-003 as follows:

Table E-5. Effluent Monitoring at Monitoring Location M-003¹

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Flow	MGD	Recorder/Totalizer	Continuous	--
TSS	mg/L	Grab	1/Day ^{2,3}	4
pH	standard units	Grab	1/Day ²	4
Oil and Grease	mg/L	Grab	1/Month ³	4

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Settleable Solids	milliliter per liter (ml/L)	Grab	1/Day ²	4
Turbidity	nephelometric turbidity unit (NTU)	Grab	1/Week	4

- ¹ See Attachment A for definitions of abbreviations and a glossary of common terms used in this Order.
- ² Applies 5 days per week, except 7 days per week for at least 1 week during July or August of each year.
- ³ The Discharger shall calculate and report the mass emission rate (MER) of the constituent for each sample taken. The MER shall be calculated in accordance with section VII.I.4 of this Order.
- ⁴ As required under 40 CFR part 136.

C. Whole Effluent Toxicity Testing Requirements

The WET refers to the overall aggregate toxic effect of an effluent measured directly by an aquatic toxicity test(s). The control of WET is one approach this Order uses to control the discharge of toxic pollutants. WET tests evaluate the 1) aggregate toxic effects of all chemicals in the effluent including additive, synergistic, or antagonistic toxicity effects; 2) the toxicity effects of unmeasured chemicals in the effluent; and 3) variability in bioavailability of the chemicals in the effluent.

Monitoring to assess the overall toxicity of the effluent is required to answer the following questions:

- (1) ~~Does the effluent comply with~~ Is the effluent toxicity less than or equal to performance goals for toxicity thereby ensuring that water quality standards are achieved in the receiving water?
- (2) If the effluent ~~does not comply with~~ toxicity is not less than or equal to performance goals for toxicity, are unmeasured pollutants causing risk to aquatic life?
- (3) If the effluent ~~does not comply with~~ toxicity is not less than or equal to performance goals for toxicity, are pollutants in combinations causing risk to aquatic life?

1. Discharge In-stream Waste Concentration (IWC) for Chronic Toxicity

The chronic IWC is calculated by dividing 100 percent by the dilution ratio. The chronic toxicity IWC is 1.15 percent effluent.

2. Sample Volume and Holding Time

The total sample volume shall be determined by the specific toxicity test method used. Sufficient sample volume of the effluent shall be collected to perform the required toxicity test. Sufficient sample volume shall also be collected during accelerated monitoring for subsequent Toxicity Identification Evaluation (TIE) studies, if necessary, at each sampling event. All toxicity tests shall be conducted as soon as possible following sample collection. No more than 36 hours shall elapse before the conclusion of sample collection and test initiation.

3. Chronic Marine Species and Test Methods

If effluent samples are collected from outfalls discharging to receiving waters with salinity >one parts per thousand (ppt), the Discharger shall conduct the following chronic toxicity tests on effluent samples, at the Discharge IWC (1.15 percent effluent), in accordance with species and test methods in *Short-Term Methods for Estimating the Chronic Toxicity of Effluent and Receiving Waters to West Coast Marine Estuarine Organisms* (EPA/600/R-95/136, 1995). Artificial sea salts or hypersaline brine shall be used to increase sample salinity if needed. In no case shall these species be substituted with another test species unless written authorization from the San Diego Water Board is received.

- a. A static renewal toxicity test with the topsmelt, *Atherinops affinis* (Larval Survival and Growth Test Method 1006.01). If laboratory-held cultures of the topsmelt, *Atherinops affinis*, are not available for testing, then the Discharger shall conduct a static renewal toxicity test with the inland silverside, *Menidia beryllina* (Larval Survival and Growth Test Method 1006.01), found in the third edition of *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Marine and Estuarine Organisms* (EPA-821-R-02-014, 2002; Table IA, 40 CFR part 136). Additional species may be used by the Discharger if approved by the San Diego Water Board.
- b. A static non-renewal toxicity test with the purple sea urchin, *Strongylocentrotus purpuratus*/sand dollar, *Dendraster excentricus* (Fertilization Test Method 1008.0 or Larval Development Test Method); or a static non-renewal toxicity test with the red abalone, *Haliotis rufescens* (Larval Shell Development Test Method).
- c. A static non-renewal toxicity test with the giant kelp, *Macrocystis pyrifera* (Germination and Growth Test Method 1009.0).

4. Species Sensitivity Screening

Species sensitivity screening shall be conducted during this Order's first required sample collection, or within 24 months of the most recent screening, whichever is later.

For each suite during the species sensitivity screening, the Discharger shall collect a single effluent sample to initiate and concurrently conduct three toxicity tests using the fish, an invertebrate, and the alga species previously referenced. This sample shall also be analyzed for the parameters required on a monthly frequency for the discharge, during that given month. As allowed under the test method for the *Atherinops affinis*, a second and third sample shall be collected for use as test solution renewal water as the seven-day toxicity test progresses. The species exhibiting the highest Toxic Units Chronic (TUC) If the result of all three species is "Pass," then the species that exhibits the highest "Percent Effect" at the discharge IWC is considered the most sensitive species for that suite. If only one species fails, then that species is considered the most sensitive species for that suite. Likewise, if two or more species result in "Fail," then the species that

~~exhibits the highest “Percent Effect”~~ at the discharge IWC is considered the most sensitive species for that suite.

If the first suite of rescreening tests demonstrates that the same species is the most sensitive, then the rescreening does not need to include more than one suite of tests. If a different species is the most sensitive or if there is ambiguity, then the Discharger shall proceed with suites of screening tests for a minimum of three, but not to exceed five suites.

Species sensitivity rescreening is required every 24 months. The Discharger shall rescreen with the marine vertebrate species, a marine invertebrate species, and the alga species previously referenced, and continue to monitor with the most sensitive species.

The species used during routine monitoring shall be the most sensitive species from the most recent species sensitivity screening.

During the calendar month, toxicity tests used to determine the most sensitive test species shall be reported as effluent ~~compliance~~-monitoring results ~~for to~~ determine if the results are less than or equal to the chronic toxicity performance goals.

5. Quality Assurance (QA) and Additional Requirements

The QA measures, instructions, and other recommendations and requirements are found in the test methods manual previously referenced. Additional requirements are specified below.

- a. The discharge is subject to a performance goal for chronic toxicity based on TUc using the No Observed Effect Concentration (NOEC; also referred to as the No Observed Effect Level or NOEL) approach described in Short Term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to West Coast Marine and Estuarine Organisms, EPA-821/600/R-95/136. TUc is calculated by the following equation: “TUc = 100 / NOEC”. ~~determination of “Pass” or “Fail” from a chronic toxicity test using the Test of Significant Toxicity (TST) statistical t test approach described in National Pollutant Discharge Elimination System Test of Significant Toxicity Implementation Document (EPA 833-R-10-003, 2010), Appendix A, Figure A-1 and Table A-1 and Appendix B, Table B-1. The null hypothesis (Ho) for the TST statistical approach is: Mean discharge IWC response $\leq 0.75 \times$ Mean control response. A test result that rejects this null hypothesis is reported as “Pass”. A test result that does not reject this null hypothesis is reported as “Fail”. This is a t-test (formally Student’s t-test), a statistical analysis comparing two sets of replicate observations—in the case of WET, only two test concentrations (i.e., a control and IWC). The purpose of this statistical test is to determine if the means of the two sets of observations are different (i.e., if the IWC or receiving water concentration differs from the control (the test result is “Pass” or “Fail”). The Welch’s t-test employed by the TST statistical approach is an adaptation of Student’s t-test and is used with two samples having unequal variances. The relative “Percent Effect” at the~~

~~discharge IWC is defined and reported as: $((\text{Mean control response} - \text{Mean discharge IWC response}) \div \text{Mean control response}) \times 100$.~~

- b. If the effluent toxicity test does not meet all test acceptability criteria (TAC) specified in the referenced test method, *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to West Coast Marine and Estuarine Organisms* (EPA/600/R-95/136, 1995), the test should be declared invalid, then the Discharger must resample and re-test within 14 days of test termination.
- c. Dilution water and control water, including brine controls, shall be 1-mircrometer-filtered uncontaminated natural seawater, hypersaline brine prepared using uncontaminated natural seawater, or laboratory water prepared and used as specified in the test methods manual. Dilution water and control water, including brine controls, shall be uncontaminated natural water, as specified in the test methods manual. If dilution water and control water is different from test organism culture water, then a second control using culture water shall also be used.
- d. ~~Monthly r~~Reference toxicant testing ~~is sufficient if shall be conducted~~ in accordance with *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to West Coast Marine and Estuarine Organisms* (EPA/600/R-95/136, 1995). All reference toxicant test results should be reviewed and reported using the effects concentration at 25 percent (EC25).
- e. The Discharger shall perform toxicity tests on final effluent samples. Chlorine and ammonia shall not be removed from the effluent sample prior to toxicity testing, unless explicitly authorized under this section of this MRP and the rationale is explained in the Fact Sheet (Attachment F).

6. Preparation of an Initial Investigation Toxicity Reduction Evaluation (TRE) Work Plan

The Discharger shall prepare and submit a copy of the Discharger's Initial Investigation TRE Work Plan to the San Diego Water Board for approval within one year of the effective date of this Order. If the San Diego Water Board does not disapprove the work plan within 60 days, the work plan shall become effective. The Discharger shall use USEPA manual EPA/833B-99/002 (municipal), or most current version, as guidance. The TRE Work Plan shall describe the steps that the Discharger intends to follow if toxicity is detected, and shall include, at a minimum:

- a. A description of the investigation and evaluation techniques that will be used to identify potential causes and sources of toxicity, effluent variability, and treatment system efficiency;
- b. A description of the Discharger's methods of maximizing in-house treatment efficiency and good housekeeping practices, and a list of all chemicals used in the operation of the Facilities; and,

- c. If a TIE is necessary, an indication of the person who would conduct the TIEs (i.e., an in-house expert or an outside contractor).

7. Accelerated Monitoring Schedule for Maximum Daily Single Result: "Fail"

The maximum daily single result of a ~~"Fail,"~~ greater than or equal to 88 TUC shall be used to determine if accelerated testing needs to be conducted.

Once the Discharger becomes aware that the maximum daily single result is ~~"Fail,"~~ greater than or equal to 88 TUC, the Discharger shall notify the San Diego Water Board and implement an accelerated monitoring schedule within five calendar days of the receipt of the result. However, if the sample is contracted out to a commercial laboratory, the Discharger shall ensure that the San Diego Water Board is notified, and the first of ~~four-six~~ accelerated monitoring tests is initiated within seven calendar days of the Discharger becoming aware of the result. The accelerated monitoring schedule shall consist of ~~four-six~~ toxicity tests (including the discharge IWC), conducted at approximately two-week intervals, over an ~~eight-twelve~~-week period; in preparation for the TRE process and associated reporting, these results shall also be reported using the EC25. If each of the accelerated toxicity tests results in ~~"Pass,"~~ less than 88 TUC, the Discharger shall return to routine monitoring for the next monitoring period. If one of the accelerated toxicity tests results in ~~"Fail,"~~ greater than or equal to 88 TUC, the Discharger shall immediately implement the TRE Process conditions set forth below. During accelerated monitoring schedules, ~~TST results ("Pass" or "Fail")~~ for chronic toxicity tests shall be used to determine effluent compliance for the chronic toxicity ~~MDEL~~ performance goal.

8. TRE Process

During the TRE Process, minimum effluent monitoring shall resume and ~~TST~~ results ~~("Pass" or "Fail")~~ for chronic toxicity tests reported in TUC shall be used to determine effluent compliance for the chronic toxicity performance goals.

- a. Preparation and Implementation of Detailed TRE Work Plan. The Discharger shall immediately initiate a TRE using, according to the type of treatment facility, USEPA manual *Toxicity Reduction Evaluation Guidance for Municipal Wastewater Treatment Plants* (EPA/833/B-99/002, 1999) and, within 15 days of receiving validated results, submit to the San Diego Water Board a Detailed TRE Work Plan, which shall follow the Initial Investigation TRE Work Plan revised as appropriate for this toxicity event. The TRE Work Plan shall include the following information, and comply with additional conditions set by the San Diego Water Board:
 - i. Further actions by the Discharger to investigate, identify, and correct the causes of toxicity;
 - ii. Actions the Discharger will take to mitigate the effects of the discharge and prevent the recurrence of toxicity; and
 - iii. A schedule for these actions, progress reports, and the final report.

- b. TIE Implementation. The Discharger may initiate a TIE as part of a TRE to identify the causes of toxicity using the same species and test method and, as guidance, USEPA manuals: *Methods for Aquatic Toxicity Identification Evaluations: Phase I Toxicity Characterization Procedures* (EPA/600/6-91/003, 1991); *Methods for Aquatic Toxicity Identification Evaluations, Phase II Toxicity Identification Procedures for Samples Exhibiting Acute and Chronic Toxicity* (EPA/600/R-92/080, 1993); *Methods for Aquatic Toxicity Identification Evaluations, Phase III Toxicity Confirmation Procedures for Samples Exhibiting Acute and Chronic Toxicity* (EPA/600/R-92/081, 1993); *Toxicity Identification Evaluation: Characterization of Chronically Toxic Effluents, Phase I* (EPA/600/6-91/005, 1991); and *Marine Toxicity Identification Evaluation (TIE): Phase I Guidance Document* (EPA/600/R-96-054, 1996). The TIE should be conducted on the species demonstrating the most sensitive toxicity response.
- c. Many recommended TRE elements parallel required or recommended efforts for source control, pollution prevention, and storm water control programs. Whenever possible, TRE efforts should be coordinated with such efforts. As toxic substances are identified or characterized, the Discharger shall continue the TRE by determining the sources and evaluating alternative strategies for reducing or eliminating the substances from the discharge. All reasonable steps shall be taken to reduce toxicity to levels consistent with toxicity evaluation parameters.
- d. The Discharger shall continue to conduct the minimum effluent monitoring while the TRE and/or TIE process is taking place. Additional accelerated monitoring and TRE Work Plans are not required once a TRE is begun.
- e. The San Diego Water Board recognizes that toxicity may be episodic and identification of causes and reduction of sources of toxicity may not be successful in all cases. Upon approval from the San Diego Water Board, the TRE may be ended at any stage if routine monitoring finds there is no longer toxicity.
- f. The San Diego Water Board may consider the results of any TRE/TIE studies in an enforcement action.

9. Reporting

The Self-Monitoring Report (SMR) shall include a full laboratory report for each toxicity test. This report shall be prepared using the format and content of the test methods manual chapter called Report Preparation¹, and shall include:

- a. The valid toxicity test results ~~for using the TST statistical~~NOEC approach, reported ~~as "Pass" or "Fail" and "Percent Effect" in TUC~~ at the chronic toxicity IWC for the discharge. All toxicity test results (whether identified as valid or

¹ Section 10 of *Short-Term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to the West Coast Marine and Estuarine Organisms*, August 1995, EPA/600/R-95-136, https://cfpub.epa.gov/si/si_public_file_download.cfm?p_download_id=524691

otherwise) conducted during the calendar month shall be reported on the SMR due date specified in Table E-10.

- b. Summary water quality measurements for each toxicity test (e.g., pH, dissolved oxygen, temperature, conductivity, hardness, salinity, chlorine, ammonia).
- c. The statistical ~~method used to calculate the endpoints. analysis used in National Pollutant Discharge Elimination System Test of Significant Toxicity Implementation Document (EPA 833-R-10-003, 2010) Appendix A, Figure A-1 and Table A-1, and Appendix B, Table B-1.~~
- d. TRE/TIE results. The San Diego Water Board shall be notified no later than 30 days from completion of each aspect of TRE/TIE analyses. Prior to the completion of the final TRE/TIE report, the Discharger shall provide status updates in the monthly SMRs, indicating which TRE/TIE steps are underway, which steps have been completed, and the estimated time to completion of the final TRE/TIE report. The final TRE/TIE report shall be submitted to the San Diego Water Board within 30 days of report completion.
- e. Statistical program ~~(e.g., TST calculator, CETIS, etc.)~~ output results, including graphical plots, for each toxicity test.
- f. Graphical plots clearly showing the laboratory's performance for the reference toxicant for the previous 20 tests and the laboratory's performance for the control mean, control standard deviation, and control coefficient of variation for the previous 12-month period.
- g. Any additional quality assurance/quality control (QA/QC) documentation or any additional chronic toxicity-related information, upon written request from the San Diego Water Board.

D. Land Discharge Monitoring Requirements – Not Applicable

E. Recycling Monitoring Requirements – Not Applicable

IV. RECEIVING WATER MONITORING REQUIREMENTS

The receiving water monitoring requirements set forth below are designed to measure the effects of the OOO discharge on the receiving ocean waters, including effects on coastal water quality, seafloor sediments, and marine life. The overall receiving water monitoring program is intended to answer the following questions:

- (1) Does the receiving water meet water quality standards?
- (2) Are the receiving water conditions getting better or worse over time?
- (3) What is the relative contribution of the Facilities discharge to pollution in the receiving water?
- (4) What are the effects of the discharge on the receiving waters?

Receiving water in the vicinity of the OOO shall be conducted as specified below. This program is intended to document conditions, within the zone of initial dilution (ZID), within

the waste field where initial dilution is completed, at reference stations, and at other areas beyond the ZID where discharge impacts might be reasonably expected. Station location, sampling, sample preservation, and analyses, when not specified, shall be by methods approved by the San Diego Water Board. The monitoring program may be modified by the San Diego Water Board at any time. The Discharger may also submit a list of proposed changes with supporting rationale to these monitoring requirements that it considers to be appropriate to the San Diego Water Board for approval.

The receiving water and sediment monitoring program for the OOO may be conducted either individually or jointly with other dischargers to the OOO.

All receiving water monitoring shall be conducted in accordance with restrictions and requirements established by the State of California Department of Fish and Wildlife and this Order. During monitoring events, sample stations shall be located using a land-based microwave positioning system or a satellite positioning system such as global positioning system (GPS).

In the event that the Discharger is unable to obtain a sample from a monitoring station(s) due to safety, legal, or other reasons, collection of samples at such station(s) can be omitted. The visual observations listed in footnote no. 1 to Table E-6 below shall still be recorded and reported in the monthly SMR to the San Diego Water Board for these stations at the time of the sample collection. If practicable, an effort should be made to return to the sampling station that was omitted and collect the sample during safer conditions within the same reporting period. In the event that a monitoring location is omitted, the Discharger shall submit a statement to the San Diego Water Board containing, at a minimum, the following information:

- The monitoring station(s) that was omitted;
- The date the monitoring station was omitted; and
- A description of the circumstances for omitting the collection of data at the monitoring station.

A. Surf Zone Water Quality Monitoring Requirements

As ocean surface waves come closer to shore they break, forming the foamy, bubbly surface called surf. The region of breaking waves defines the surf zone.

Monitoring of the surf zone is intended to answer the following questions:

- (1) Does the effluent cause or contribute to an exceedance of the water quality standards in the receiving water?
- (2) Does the effluent reach water contact zones or commercial shellfish beds?
- (3) Are densities of bacteria in water contact areas below levels protective of public health?

All surf zone stations listed in Table E-1 (S1 through S7) shall be monitored as follows:

Table E-6. Surf Zone Water Quality Monitoring¹

Parameter	Units	Sample Type	Minimum Sampling Frequency
Visual Observations	--	Visual	²
Temperature	°C	Grab	1/Week
Total Coliform	CFU/100 ml ³	Grab	3/Month ^{4,3}
Fecal Coliform	CFU/100 ml ³	Grab	5/Month 1/Week ^{3,4,5,6}
Enterococci	CFU/100 ml ³	Grab	5/Month 1/Week ^{3,4,5}

¹ See Attachment A for definitions of abbreviations and a glossary of common terms used in this Order.

² Visual observations of the surface water conditions at the designated receiving water stations shall be conducted in such a manner as to enable the observer to describe and report the presence, if any, of floatables of sewage origin. Observations of wind (direction and speed), weather (cloudy, sunny, or rainy), direction of current, tidal conditions (high or low), water color, discoloration, oil and grease, turbidity, and odor shall be recorded. These observations shall be taken whenever a sample is collected. Visual observations shall also be conducted for repeat sampling.

^{2,3} ~~The Discharger may report in MPN in lieu of CFU; however, the receiving water limitations for these parameters will be the same regardless of the reporting units.~~

^{3,4} Sampling shall be spaced equally throughout the month to the extent possible.

⁵ Samples for fecal coliform and enterococci shall be collected on the same day fecal coliform and enterococci are sampled at monitoring location M-~~004002~~.

^{4,6} ~~The Discharger shall ensure a minimum of five samples are collected and analyzed within a rolling 30-day period, which may require more than one sample per week depending on the sampling schedule.~~

B. Nearshore and Offshore Water Quality Monitoring Requirements

The nearshore for the purposes of monitoring and assessment is considered to extend from the shoreline to a distance of 1,000 feet from the shoreline or the 30-foot depth contour, whichever is further from the shoreline. Offshore for the purposes of monitoring and assessment is considered to extend beyond the nearshore zone.

Nearshore and offshore monitoring is necessary to answer the following questions:

- (1) Is natural light significantly reduced at any point outside the ZID as a result of the discharge?
- (2) Does the discharge cause a discoloration of the ocean surface?
- (3) Does the discharge of oxygen demanding waste cause the dissolved oxygen concentration to be depressed at any time more than 10 percent from that which occurs naturally outside the ZID?
- (4) Does the discharge of waste cause the pH to change at any time more than 0.2 standard units from that which occurs naturally outside the ZID?
- (5) Does the discharge of waste cause the salinity to become elevated in the receiving water?
- ~~(6) Do nutrients cause objectionable aquatic growth or degrade indigenous biota?~~
- ~~(7)~~(6) Is the wastewater plume encroaching upon receiving water areas used for swimming, surfing, diving, and shellfish harvesting?
- ~~(8)~~(7) What is the fate of the discharge plume?

~~(9)~~(8) Is fecal indicator bacteria present outside the zone of initial dilution? If so, is the bacteria human source?

1. All nearshore and offshore stations listed in Table E-1 (N1 through N7, A1 through A5, B1, and B2) shall be monitored as follows:

Table E-7. Nearshore and Offshore Water Quality Monitoring Requirements¹

Parameter	Units	Sample Type	Minimum Sampling Frequency
Visual Observations	--	Visual ²	1/Quarter
Fecal Coliform	CFU /100 ml ³	Grab ^{3,4,5}	1/Quarter
<i>Enterococci</i>	CFU/100 ml ³	Grab ^{3,4,5}	1/Quarter
Human Marker HF183	Number of copies (molecules)/100 mL	Grab ^{6,5}	1/Quarter
Nitrogen, Total	mg/L	Grab³	1/Quarter
Phosphorus, Total (as-P)	mg/L	Grab³	1/Quarter
Temperature and Depth	°C, meters	7 ⁶	1/Quarter
Dissolved Oxygen	mg/L	7 ⁶	1/Quarter
Light Transmittance	percent	7 ⁶	1/Quarter
pH	standard units	7 ⁶	1/Quarter
Salinity	ppt	7 ⁶	1/Quarter
Colored Dissolved Organic Matter (CDOM)	ppb	7	1/Quarter
Spectrophotometric pH⁸	standard units	Grab	1/Quarter
Alkalinity, Total⁸	mg/L CaCO₃	Grab	1/Quarter

¹ See Attachment A for definitions of abbreviations and a glossary of common terms used in this Order.

² Visual observations of the surface water conditions at the designated receiving water stations shall be conducted in such a manner as to enable the observer to describe and report the presence, if any, of floatables of sewage origin. Observations of wind (direction and speed), weather (cloudy, sunny, or rainy), direction of current, tidal conditions (high or low), water color, oil and grease, turbidity, and odor shall be recorded. The proximity of recreational and commercial vessels to monitoring locations shall also be recorded. These observations shall be taken whenever a sample is collected.

³ The Discharger may report in MPN in lieu of CFU; however, the receiving water limitations for these parameters will be the same regardless of the reporting units.

³⁴ At the surface for nearshore monitoring locations N1 through N7 and surface and mid-depth for offshore monitoring locations A1 through A5, B1, and B2.

⁴⁵ Samples for fecal coliform and enterococci shall be collected on the same day fecal coliform and enterococci are sampled at monitoring location M-004.

⁵⁶ Samples shall be collected at the surface and mid-depth at offshore monitoring locations A1-A5, B1 and B2 and analyzed in accordance with section IV.B.2 of this MRP.

⁷ For offshore monitoring locations A1-A5, B1 and B2, temperature, depth, dissolved oxygen, light transmittance, pH, CDOM, and salinity profile data shall be measured throughout the entire water column using a CTD profiler during the quarterly sampling events. Depth profile measurements shall be obtained using multiple sensors to measure parameters through the entire water column (from the surface to as close to the bottom as practicable). For nearshore monitoring locations N1 through N7, temperature, depth, dissolved oxygen, light transmittance, pH, CDOM, and salinity shall be measured throughout the entire water column by a CTD profiler or at the surface by grab samples.

⁸ Monitoring for alkalinity and spectrophotometric pH is recommended and requested, but not required. If monitoring is conducted, it is recommended that the monitoring be consistent with the methods described in section IV.B.3 of this MRP.

2. **Human ~~Marker-Associated~~ HF183 ~~Genetic Marker~~ Monitoring Requirements**

The human associated~~Human Marker~~ HF183 genetic marker (HF183) monitoring is required to confirm the presence of human fecal material when the single sample maximum receiving water limitation for fecal coliform is exceeded. Analysis of the Human Marker HF183 sample is not required if the result for fecal coliform is below receiving water limitations.

- a. **Sample Collection.** The Discharger shall collect samples for the ~~Human Marker~~ HF183 concurrently with samples collected for fecal coliform at the offshore monitoring locations A1 through A5, B1, and B2, and in accordance with EPA method 1696, or an alternative method proposed by the Discharger with comparable accuracy, unless the alternative method is not accepted by the San Diego Water Board. Samples shall be filtered through a membrane filter as soon as possible, but no later than 6 hours after sample collection. Following filtration, the membrane filter shall be stored at -80 °C for later analysis.
- b. **Sample Analysis.** If a result for fecal coliform exceeds the single sample maximum receiving water limitation of 400 CFU per 100 mL (section V.A.1.a.i.(b) of this Order), the Discharger shall analyze the ~~Human Marker~~ HF183 sample that was collected concurrently with the fecal coliform sample that exceeded the receiving water limitation. Samples shall be analyzed in accordance with EPA method 1696, the droplet digital polymerase chain reaction (ddPCR) method developed by the Southern California Coastal Waters Research Project (SCCWRP), or an alternative method proposed by the Discharger with comparable accuracy, unless the alternative method is not accepted by the San Diego Water Board. If the Discharger proposes to use the ddPCR method, the Discharger shall submit a QA/QC procedure for acceptance by the San Diego Water Board. The Discharger shall follow all quality control and quality assurance procedures outlined in the method or as approved by the San Diego Water Board. If the results for fecal coliform are below receiving water limitations, the discharger may discard the ~~Human Marker~~ HF183 sample.
- c. **Sample Results (if applicable).** Sample results for ~~Human Marker~~ HF183 shall be submitted with the monthly self-monitoring report to the California Integrated Water Quality System (CIWQS) as an attachment in PDF format.

3. **Total Alkalinity and Spectrophotometric pH Monitoring Requirements**

Monitoring for total alkalinity and spectrophotometric pH is recommended and requested, but not required. The San Diego Water Board recommends using the results for pH measured by spectrophotometric analysis to calibrate the pH results measured by the CTD profiler and to calculate aragonite saturation state.

- a. **Monitoring Locations and Sample Collection.** If monitoring for spectrophotometric pH and total alkalinity is conducted, the San Diego Water Board recommends and requests that the monitoring be consistent with the methods described in *An evaluation of potentiometric pH sensors in coastal*

monitoring applications (McLaughlin et al. 2017). Grab samples for spectrophotometric pH and alkalinity should be collected at the surface, thermocline, and bottom at offshore monitoring locations A1 and B2. It is recommended that a duplicate sample at the bottom at offshore monitoring location A1 is collected during each sampling event. For stations A1 and B2, grab samples should be collected by Niskin bottles attached to the same device as the CTD used to collect profile samples. Water from the Niskin bottles should be transferred to 500 mL Pyrex bottles and overfilled by a minimum of 250 mL leaving approximately 1% headspace. All bottles should be fixed with approximately 120 µL of saturated mercury (II) chloride solution and stored at room temperature until analysis.

a.b. Sample Analysis. If monitoring is conducted, the San Diego Water Board recommends and requests the grab samples for pH be measured using the spectrophotometric technique described in *An automated system for spectrophotometric seawater pH measurements* (Carter et al. 2013), estimating pH at 25 degrees Celsius on the total hydrogen ion scale using m-cresol purple dye indicator and pH calibration equations based on *Purification and characterization of meta-cresol purple for spectrophotometric seawater pH measurements* (Liu et al. 2011). It is recommended that grab samples for total alkalinity be measured by a two-stage, potentiometric, and open-celled titration using coulometrically analyzed hydrochloric acid as described in *Reference material for oceanic CO2 analysis: A method for the certification of total alkalinity* (Dickson et al. 2003). The San Diego Water Board recommends and requests the Discharger use the spectrophotometric pH and total alkalinity results to calibrate and adjust the pH samples collected quarterly by the CTD and to calculate the aragonite saturation state. If results are not available for submittal with the quarterly SMR, the Discharger may submit the results by email to SanDiego@waterboards.ca.gov.

C. Benthic Monitoring Requirements

Seafloor sediments integrate constituents that are discharged to the ocean. Most particles that come from the OOO discharge, and any associated contaminants, will eventually settle to the seafloor where they are incorporated into the existing sediments. Sediments can accumulate these particles over the years until the point where sediment quality is degraded and beneficial uses are impaired.

Benthic organisms are strongly affected by sediment contaminant exposure because these organisms often live in continual direct contact with sediment/pore water, and many species ingest significant quantities of sediment as a source of nutrition. Because the benthos are dependent on their surroundings, they serve as a biological indicator that reflects the overall conditions of the aquatic environment.

The assessment of sediment quality with respect to sediment chemistry, sediment toxicity, and benthic community condition is necessary to answer the following questions:

- (1) Is the dissolved sulfide concentration of waters in sediments significantly increased above that present under natural conditions?
- (2) Is the concentration of substances set forth in Table 3 of the Ocean Plan for protection of marine aquatic life in marine sediments at levels which would degrade the benthic community?
- (3) Is the concentration of organic pollutants in marine sediments at levels that would degrade the benthic community?
- (4) Are benthic communities degraded as a result of the discharge?
- (5) Is the sediment quality changing over time?

The assessment of sediment quality to evaluate potential effects of the OOO discharge and compliance with narrative water quality standards specified in the Ocean Plan consist of the measurement and integration of three lines of evidence: 1) physical and chemical properties of seafloor sediments, 2) seafloor sediment toxicity to assess bioavailability and toxicity of sediment contaminants, and 3) ecological status of the biological communities (benthos) that live in or on the seafloor sediments. The Discharger may request to fulfill the benthic monitoring requirements through a regional monitoring program described in section V.B of this MRP.

1. Sediment Assessment for Physical and Chemical Properties

- a. **Sediment Sampling Stations and Monitoring Frequency.** The sediment monitoring program is designed to assess spatial and temporal trends at the offshore stations and to assess benthic habitat condition in terms of physical and chemical composition (e.g., grain-size distribution, sediment chemistry). Sediment samples for assessment of sediment chemistry shall be collected once during the permit term at each of the offshore stations listed in Table E-1 (A1 through A5, B1, and B2). Results from sediment monitoring are due no later than 180 days prior to the expiration date of this Order.
- b. **Sediment Sample Collection Methods.** Sediment samples shall be taken using a 0.1-square meter modified Van Veen grab sampler. Samples for grain-size and chemical analyses shall be taken from the top two centimeters of the surface sediment. Sediment samples for physical and chemical properties shall be taken concurrently with and adjacent to (as much as possible) the sediment samples for toxicity and benthic community condition. Bulk sediment chemical analysis shall include at a minimum the set of constituents listed in Table E-8.
- c. **Sediment Chemistry Test Methods.** Sediment chemistry is the measurement of the concentration of chemicals of concern in sediments. The chemistry line of evidence is used to assess the potential overall exposure risk to benthic organisms from pollutants in surficial sediments. Chemical analysis of sediment shall be conducted using USEPA-approved methods, methods developed by the National Oceanic and Atmospheric Administration's (NOAA's) *National Status and Trends for Marine Environmental Quality*, or methods developed in conjunction with the

Southern California Bight Regional Monitoring Program. For chemical analysis of sediment, samples shall be reported on a dry weight basis.

Sediment monitoring for physical and chemical properties shall be conducted at the offshore stations listed in Table E-1 (A1 through A5, B1, and B2) for the parameters identified in Table E-8:

Table E-8. List of Parameters to Characterize Sediment Contamination Exposure and Effect

Parameter	Units	Type of Sample	Minimum Sampling Frequency
Sediment Grain Size	micrometer (µm)	Grab	1/Permit Term
Total Organic Carbon	Percent	Grab	1/Permit Term
Total Nitrogen	Percent	Grab	1/Permit Term
Acid Volatile Sulfides	milligram/kilogram (mg/kg)	Grab	1/Permit Term
<u>Dissolved Sulfide</u>	<u>mg/kg</u>	<u>Grab</u>	<u>1/Permit Term</u>
Aluminum	mg/kg	Grab	1/Permit Term
Antimony	mg/kg	Grab	1/Permit Term
Arsenic	mg/kg	Grab	1/Permit Term
Cadmium	mg/kg	Grab	1/Permit Term
Total Chromium	mg/kg	Grab	1/Permit Term
Copper	mg/kg	Grab	1/Permit Term
Iron	mg/kg	Grab	1/Permit Term
Lead	mg/kg	Grab	1/Permit Term
Manganese	mg/kg	Grab	1/Permit Term
Mercury	mg/kg	Grab	1/Permit Term
Nickel	mg/kg	Grab	1/Permit Term
Selenium	mg/kg	Grab	1/Permit Term
Silver	mg/kg	Grab	1/Permit Term
Tin	mg/kg	Grab	1/Permit Term
Zinc	mg/kg	Grab	1/Permit Term
PCBs	nanogram/kilogram (ng/kg)	Grab	1/Permit Term
2,4-DDD	ng/kg	Grab	1/Permit Term
4,4-DDD	ng/kg	Grab	1/Permit Term
2,4-DDE	ng/kg	Grab	1/Permit Term
4,4-DDE	ng/kg	Grab	1/Permit Term
2,4-DDT	ng/kg	Grab	1/Permit Term
4,4-DDT	ng/kg	Grab	1/Permit Term
Aldrin	ng/kg	Grab	1/Permit Term
Alpha-Chlordane	ng/kg	Grab	1/Permit Term
Dieldrin	ng/kg	Grab	1/Permit Term
Endosulfan	ng/kg	Grab	1/Permit Term
Endrin	ng/kg	Grab	1/Permit Term
Gamma-BHC	ng/kg	Grab	1/Permit Term
Heptachlor	ng/kg	Grab	1/Permit Term
Heptachlor Epoxide	ng/kg	Grab	1/Permit Term
Hexachlorobenzene	ng/kg	Grab	1/Permit Term
Mirex	ng/kg	Grab	1/Permit Term
Trans-Nonachlor	ng/kg	Grab	1/Permit Term
Acenaphthene	microgram/kilogram (µg/kg)	Grab	1/Permit Term
Acenaphthylene	µg/kg	Grab	1/Permit Term
Anthracene	µg/kg	Grab	1/Permit Term

Parameter	Units	Type of Sample	Minimum Sampling Frequency
Benzo(a)anthracene	µg/kg	Grab	1/Permit Term
Benzo(eb)fluoranthene	µg/kg	Grab	1/Permit Term
Benzo(k)fluoranthene	µg/kg	Grab	1/Permit Term
Benzo(ghi)pyrene	µg/kg	Grab	1/Permit Term
Benzo(a)pyrene	µg/kg	Grab	1/Permit Term
Benzo(e)pyrene	µg/kg	Grab	1/Permit Term
Biphenyl	µg/kg	Grab	1/Permit Term
Chrysene	µg/kg	Grab	1/Permit Term
Dibenzo(ah)anthracene	µg/kg	Grab	1/Permit Term
Fluoranthene	µg/kg	Grab	1/Permit Term
Fluorene	µg/kg	Grab	1/Permit Term
Ideno(123cd)pyrene	µg/kg	Grab	1/Permit Term
Naphthalene	µg/kg	Grab	1/Permit Term
1-Methylnaphthalene	µg/kg	Grab	1/Permit Term
2-Methylnaphthalene	µg/kg	Grab	1/Permit Term
2,6-Dimethylnaphthalene	µg/kg	Grab	1/Permit Term
2,3,5-Trimethylnaphthalene	µg/kg	Grab	1/Permit Term
Perylene	µg/kg	Grab	1/Permit Term
Phenanthrene	µg/kg	Grab	1/Permit Term
1-Methylphenanthrene	µg/kg	Grab	1/Permit Term
Pyrene	µg/kg	Grab	1/Permit Term

2. Sediment Toxicity

- a. **Toxicity Sampling Stations and Frequency.** Sediment toxicity is a measure of the response of invertebrates exposed to surficial sediments under controlled laboratory conditions. The sediment toxicity line of evidence is used to assess both pollutant-related biological effects and exposure. Sediment samples for assessment of toxicity shall be monitored once during the permit term at each of the offshore stations listed in Table E-1 (A1 through A5, B1, and B2). Results from sediment toxicity sampling are due no later than 180 days prior to the expiration date of this Order.
- b. **Sediment Toxicity Collection Methods.** Sediment samples shall be taken using a 0.1-square meter modified Van Veen grab sampler. Samples for toxicity analyses shall be taken from the top two centimeters of the surface sediment. Sediment samples for toxicity shall be taken concurrently with and adjacent to (as much as possible) the sediment samples for physical and chemical properties, and benthic community condition.
- c. **Sediment Toxicity Test Methods.** Sediment toxicity tests shall utilize alternative amphipod species (*Eohaustorius estuaries*, *Leptocheirus plumulosus*, *Rhepoxynius abronius*) and be conducted in accordance with EPA 600/R-94/0925 (USEPA, 1994), *Methods for Assessing the Toxicity of Sediment-associated Contaminants with Estuarine and Marine Amphipods*, and the Southern California Bight Project sediment toxicity testing guidelines (Bight'13 Toxicology Committee, 2013). Response criteria shall include mortality, emergence from sediment during exposure, and ability to rebury in

clean sediment at the end of the 10-day exposure period. ~~Results shall be reported as "Pass"/"Fail" and percent response.~~

- d. **Data Analysis.** Analysis of sediment toxicity shall include a calculation of the mean control normalized response.

3. Benthic Community Condition

- a. **Benthic Community Sampling Stations and Frequency.** Sediment samples for assessment of benthic community structure shall be collected once during the permit term at each of the offshore stations listed in Table E-1 (A1 through A5, B1, and B2). One sample per station shall be collected for analysis of benthic community structure. Results from benthic community sampling are due no later than 180 days prior to the expiration date of this Order.
- b. **Benthic Community Sample Collection Methods.** Benthic community samples shall be collected using the guidance specified in the most recent field manual developed for the Southern California Bight Regional Monitoring Program. The benthic samples shall be collected using a 0.1-square meter modified Van Veen grab sampler. These grab samples shall be taken concurrently with and adjacent to (as much as possible) samples collected for sediment physical and chemical properties, and toxicity. The samples shall be sieved using a 1.0-millimeter mesh screen. The benthic organisms retained on the sieve shall be fixed in 10 percent buffered formalin and transferred to at least 70 percent ethanol within two to seven days of storage. All benthic invertebrates in the screened sample shall be identified to the lowest possible taxon and; enumerated (counted); ~~measured, and, where feasible, assessed for reproductive condition.~~
- c. **Benthic Community Analysis.** Analysis of benthic community structure shall include determination of the number of species, number of individuals per species, and total numerical abundance present. The following parameters or metrics shall be calculated for each 0.1-square meter grab sample and summarized by station, as appropriate:
 - i. Number of species;
 - ii. Total numerical abundance;
 - iii. Benthic Response Index (BRI);
 - iv. Swartz's 75 percent dominance index;
 - v. Shannon-Weiner's diversity index (H); and
 - vi. Pielou evenness index (J).

4. Benthic Monitoring Work Plan

- a. **Benthic Monitoring Work Plan.** The Discharger shall submit to the San Diego Water Board within 180 days after the effective date of this Order, a Benthic Monitoring Work Plan to implement the sediment monitoring

program. The Benthic Monitoring Work Plan is not required if the Discharger is fulfilling the benthic monitoring requirements contained in Attachment E section IV.C.1 through IV.C.3 by participating in a regional monitoring program, as described in Attachment E section V.B. If required, the Benthic Monitoring Work Plan shall include the following elements:

- i. A Quality Assurance Project Plan (QAPP) describing the project objectives and organization, functional activities, and quality assurance/quality control protocols;
 - ii. Protocols for sediment sample collection and processing;
 - iii. Proposed methods for analyzing sediment data and integrating the three lines of evidence (i.e., sediment physical and chemical properties, sediment toxicity, and benthic community condition); and
 - iv. Schedule for completion of sample collection and submission of the results.
- b. **Benthic Monitoring Work Plan Implementation.** The Discharger shall implement the Benthic Monitoring Work Plan sixty (60) days after submission of the Benthic Monitoring Work Plan, unless otherwise directed in writing by the San Diego Water Board. The Discharger shall notify the San Diego Water Board of the intent to initiate the proposed actions included in the Benthic Monitoring Work Plan; and comply with any conditions set by the San Diego Water Board.

D. Fish and Macroinvertebrates Monitoring Requirements

Many pollutants discharged into receiving waters have the potential to bioaccumulate and persist in tissues of aquatic organisms, including marine fishes. Chemical pollutants that bioaccumulate tend to biomagnify as they pass through the aquatic food chain. Therefore, fish monitoring data is required to assess the human health risks for individuals who may consume fish and to assess trends of contaminants levels in fish tissue over time.

Aquatic benthic invertebrates are excellent indicators of ecosystem health because they are ubiquitous, abundant, diverse, and typically sedentary. The growth, survival, and reproduction of many species of aquatic invertebrates are all sensitive to changes in environmental health, making analysis of assemblage structure a good ecosystem monitoring tool.

Fish and invertebrate monitoring is necessary to answer the following questions:

- (1) Does the concentration of pollutants in fish, shellfish, or other marine organisms used for human consumption bioaccumulate to levels that are harmful to human health?
- (2) Does the concentration of pollutants in marine life bioaccumulate to levels that degrade marine communities?
- (3) Are the concentrations of pollutants in fish and other marine organisms changing over time?

- (4) Is the health of fish changing over time?
- (5) Are the populations of selected species of fish and invertebrates changing over time?

The Discharger may request to fulfill the fish and macroinvertebrate monitoring requirements through a regional monitoring program described in section V.B of this MRP.

1. Fish and Invertebrate Trawls

- a. **Demersal Fish and Invertebrate Trawl Frequency and Locations.** Fish and invertebrate trawls shall be conducted to assess the structure of demersal fish and megabenthic invertebrate communities, while the presence of priority pollutants in fish shall be analyzed from species captured using both trawling and rig fishing techniques. Single community trawls for fish and invertebrates shall be conducted once per permit term at the three biological transect (trawl) stations T3, T4, and T5 listed in Table E-1. These trawl stations represent one area 50 feet downcoast of Discharge Point No. 001 (station T3), one area 1 mile downcoast of Discharge Point No. 001 (station T4), and one area 1 mile upcoast of Discharge Point No. 001 (station T5). The Discharger shall notify the San Diego Water Board of the intent to initiate the fish and invertebrate trawls, no later than 30 days before the survey. Results from the fish and invertebrate trawls are due no later than 180 days prior to the expiration date of this Order.
- b. **Fish and Invertebrate Trawl Method.** Trawls shall be conducted using a Marinovich 7.62-meter (25-foot) head rope otter trawl, using the guidance specified in the most recent field manual developed for the Southern California Bight Regional Monitoring Program. Captured organisms shall be identified at all stations.

In order to minimize negative impacts that may occur due to unsuccessful trawling efforts associated with unusual environmental conditions, the requirement to conduct trawls during any given period may be postponed or waived at the discretion of the Executive Officer of the San Diego Water Board upon receipt of written justification provided by the Discharger. Examples of such unusual events include the presence of large populations of red tuna crabs (*Pleuroncodes planipes*) associated with El Niño and the occurrence of large squid egg masses that prevent hauling in the trawl nets.

- c. **Fish and Invertebrate Community Structure Analysis.** All demersal fishes and megabenthic invertebrates collected by trawls shall be identified by species if possible. For fish, community structure analysis shall consist of determining the standard length and total wet weight, total number of individuals per species, the total numerical abundance of all fish, species richness, species diversity (H'), and multivariate pattern analyses (e.g., ordination and classification analyses). The presence of any physical abnormalities or disease symptoms (e.g., fin erosion, external lesions, and tumors) or external parasites shall also be recorded. For invertebrates,

community structure shall be summarized as the total number of individuals per species, the total numerical abundance of all invertebrates, species richness, and species diversity (H').

- d. **Fish Liver Tissue Chemical Analysis.** Chemical analyses of fish liver tissues shall be performed once per permit term on target species collected at or near the trawl stations. The three trawl stations are classified into three zones for the purpose of collecting sufficient numbers of fish for tissue analyses. Trawl Zone 1 represents the nearfield zone, defined as the area within a 0.5-km radius of trawl station T3; Trawl Zone 2 is considered the down coast zone, defined as the area within a 1-km radius of trawl station T4; and Trawl Zone 3 is considered the up coast zone, and is defined as the area centered within a 1-km radius of trawl station T5.

Liver tissues shall be analyzed during each survey from fishes collected in each of the above three trawl zones. No more than a maximum of five 10-minute (bottom time) trawls shall be required per zone in order to acquire sufficient numbers of fish for composite samples; these trawls may occur anywhere within a defined zone. If sufficient numbers of trawl zone target species cannot be, or are unlikely to be, captured by trawling, fish for tissue analysis from these areas may be collected using alternative methods such as those described below under Rig Fishing in section IV.D.2.b of this MRP (e.g., hook and line, baited lines). Three replicate composite samples shall be prepared from each trawl zone, with each composite consisting of tissues from at least three individual fish of the same species. These liver tissues shall be analyzed for the constituents listed in the Table E-9.

- e. **Fish Targeted for Chemical Analysis.** The species of fish targeted for tissue analysis from the trawl sites shall be primarily flatfish including, but not limited to, Pacific sanddab (*Citharichthys sordidus*), longfin sanddab (*Citharichthys xanthostigma*), bigmouth sole (*Hippoglossina stomata*), California tonguefish (*Symphurus atricaudus*) and hornyhead turbot (*Pleuronichthys verticalis*). If sufficient numbers of these primary flatfish species are not present in a zone, secondary target species such as the longspine combfish (*Zaniolepis latipinnis*), California scorpionfish (*Scorpaena guttata*), California lizardfish (*Synodus lucioceps*) may be collected as necessary.

2. Rig Fishing

- a. **Rig Fishing Frequency.** Fish muscle tissues shall be analyzed once during the permit term from fishes collected in each of the three rig fishing zones described below in order to monitor the uptake of pollutants in selected species. The Discharger shall notify the San Diego Water Board of the intent to initiate sample collection, no later than 30 days before sampling. Results from the fish muscle tissue analyses are due no later than 180 days prior to the expiration date of this Order.

- b. **Rig Fishing Method and Location.** The fish shall be collected by hook and line or by setting baited lines from within zones surrounding rig fishing stations RF1, RF2, and RF3 listed in Table E-1. Rig Fishing Zone 2 is the nearfield area centered within a 0.5-km radius of station RF2; Rig Fishing Zone 1 represents the northern farfield area centered within 1-km radius of station RF1; and Rig Fishing Zone 3 represents the southern farfield area centered within a 1-km radius of station RF3. There are no depth requirements for these three rig fishing zones with regards to the collection of fishes for tissue analysis. Fish samples shall be identified to species, with number of individuals per species, standard length and wet weight recorded. Physical abnormalities or disease symptoms (e.g., fin rot, lesions, and tumors) and external parasites shall be recorded and itemized.
- c. **Rig Fishing Targeted Species.** The species of fish targeted for tissue analysis from the rig fishing stations shall be representative of those caught by recreational and/or commercial fishery activities in the region. The species targeted for muscle tissue analysis shall be primarily rockfish (*Sebastes spp.*), excluding species restricted by the California Department of Fish and Wildlife. If sufficient numbers of rockfish are not present or cannot be caught in a particular zone, secondary target species (e.g., scorpionfish and lingcod) may be collected and analyzed as necessary.
- d. **Rig Fishing Collection.** Three replicate composite samples of the target species shall be obtained from each zone, with each composite consisting of a minimum of three individual fish. Muscle tissue shall be chemically analyzed for the same set of constituents as trawl-caught fish specified in Table E-9.

Table E-9. List of Parameters to Characterize Fish Tissue

Parameter	Units	Type of Sample	Minimum Sampling Frequency
Total Lipids	mg/kgpercent	Composite	1/Permit Term
Aluminum	mg/kg	Composite	1/Permit Term
Antimony	mg/kg	Composite	1/Permit Term
Arsenic	mg/kg	Composite	1/Permit Term
Cadmium	mg/kg	Composite	1/Permit Term
Chromium	mg/kg	Composite	1/Permit Term
Copper	mg/kg	Composite	1/Permit Term
Iron	mg/kg	Composite	1/Permit Term
Lead	mg/kg	Composite	1/Permit Term
Manganese	mg/kg	Composite	1/Permit Term
Mercury	mg/kg	Composite	1/Permit Term
Nickel	mg/kg	Composite	1/Permit Term
Selenium	mg/kg	Composite	1/Permit Term
Silver	mg/kg	Composite	1/Permit Term
Tin	mg/kg	Composite	1/Permit Term
Zinc	mg/kg	Composite	1/Permit Term
PCBs	ng/kg	Composite	1/Permit Term
2,4-DDD	ng/kg	Composite	1/Permit Term
4,4-DDD	ng/kg	Composite	1/Permit Term

Parameter	Units	Type of Sample	Minimum Sampling Frequency
2,4-DDE	ng/kg	Composite	1/Permit Term
4,4-DDE	ng/kg	Composite	1/Permit Term
2,4-DDT	ng/kg	Composite	1/Permit Term
4,4-DDT	ng/kg	Composite	1/Permit Term
Aldrin	ng/kg	Composite	1/Permit Term
Alpha-Chlordane	ng/kg	Composite	1/Permit Term
Dieldrin	ng/kg	Composite	1/Permit Term
Endosulfan	ng/kg	Composite	1/Permit Term
Endrin	ng/kg	Composite	1/Permit Term
Gamma-BHC	ng/kg	Composite	1/Permit Term
Heptachlor	ng/kg	Composite	1/Permit Term
Heptachlor Epoxide	ng/kg	Composite	1/Permit Term
Hexachlorobenzene	ng/kg	Composite	1/Permit Term
Mirex	ng/kg	Composite	1/Permit Term
Trans-Nonachlor	ng/kg	Composite	1/Permit Term
Acenaphthene	µg/kg	Composite	1/Permit Term
Acenaphthylene	µg/kg	Composite	1/Permit Term
Anthracene	µg/kg	Composite	1/Permit Term
Benzo(a)anthracene	µg/kg	Composite	1/Permit Term
Benzo(b)fluoranthene	µg/kg	Composite	1/Permit Term
Benzo(k)fluoranthene	µg/kg	Composite	1/Permit Term
Benzo(ghi)pyrene	µg/kg	Composite	1/Permit Term
Benzo(a)pyrene	µg/kg	Composite	1/Permit Term
Benzo(e)pyrene	µg/kg	Composite	1/Permit Term
Biphenyl	µg/kg	Composite	1/Permit Term
Chrysene	µg/kg	Composite	1/Permit Term
Dibenzo(ah)anthracene	µg/kg	Composite	1/Permit Term
Fluoranthene	µg/kg	Composite	1/Permit Term
Fluorene	µg/kg	Composite	1/Permit Term
Ideno(123cd)pyrene	µg/kg	Composite	1/Permit Term
Naphthalene	µg/kg	Composite	1/Permit Term
1-Methylnaphthalene	µg/kg	Composite	1/Permit Term
2-Methylnaphthalene	µg/kg	Composite	1/Permit Term
2,6-Dimethylnaphthalene	µg/kg	Composite	1/Permit Term
2,3,5-Trimethylnaphthalene	µg/kg	Composite	1/Permit Term
Perylene	µg/kg	Composite	1/Permit Term
Phenanthrene	µg/kg	Composite	1/Permit Term
1-Methylphenanthrene	µg/kg	Composite	1/Permit Term
Pyrene	µg/kg	Composite	1/Permit Term

E. California Environmental Data Exchange Network

In addition to submitting self-monitoring reports (SMRs), the Discharger shall also ensure that all the receiving water monitoring results are submitted to the California Environmental Data Exchange Network (CEDEN) or an equivalent database that is linked to CEDEN no later than 90 days after ~~the completion of receiving the~~ monitoring results. Any data not accepted by CEDEN is not required to be submitted. If the receiving water monitoring is conducted jointly with other dischargers to the OOO, the Discharger shall coordinate the submittal of the receiving water monitoring

results with other agencies discharging through the OOO to ensure data is not duplicated in CEDEN. A statement certifying that all applicable monitoring results have been timely uploaded into CEDEN shall be submitted annually by March 1 of each year. Only monitoring results from the following requirements shall be reported in CEDEN:

1. Shoreline, nearshore, and offshore water quality (sections IV.A and IV.B of this MRP);
2. Sediment assessment for physical and chemistry properties (section IV.C.1 of this MRP);
3. Sediment toxicity (section IV.C.2 of this MRP);
4. Benthic community condition (section IV.C.3 of this MRP);
5. Demersal fish and macroinvertebrate trawls (section IV.D.1 of this MRP); and
6. Rig fishing (section IV.D.2 of this MRP).

F. Receiving Water Status and Trends

1. **Receiving Water Monitoring Report.** The Discharger shall submit a receiving water monitoring report, prepared individually or jointly with other discharges to the OOO, once no later than 180 days prior to the expiration date of this Order. The receiving water monitoring report shall cover the following requirements:
 - a. Shoreline, nearshore, and offshore water quality (sections IV.A and IV.B of this MRP);
 - b. Sediment assessment for physical and chemistry properties (section IV.C.1 of this MRP);
 - c. Sediment toxicity (section IV.C.2 of this MRP);
 - d. Benthic community condition (section IV.C.3 of this MRP);
 - e. Demersal fish and macroinvertebrate trawls (section IV.D.1 of this MRP);
 - f. Rig fishing (section IV.D.2 of this MRP); and
 - g. Plume tracking (section VI.B of this MRP).
2. The report shall include, as a minimum, the following information:
 - a. A description of climatic and receiving water characteristics at the time of sampling (weather observations, floating debris, discoloration, wind speed and direction, swell or wave action, time of sampling, tide height, etc.);
 - b. A description of sampling stations, including, if such information is available, differences unique to each station (e.g., station location, sediment grain size, distribution of bottom sediments, rocks, shell litter, calcareous worm tubes, etc.);
 - c. A description of the sample collection and preservation procedures used in the survey;

- d. A description of the specific method used for laboratory analysis;
 - e. An in-depth discussion, evaluation (e.g., detailed statistical analyses), interpretation and tabulation of the data including interpretations and conclusions as to whether applicable receiving water limitations in this Order have been attained at each station;
 - f. A narrative summary of general observations, including any abnormal conditions; and
 - g. An in-depth discussion addressing the questions proposed in each section of the Receiving Water Monitoring Requirements of this MRP.
3. **State of the Ocean Report.** The Discharger, either individually or collectively with other discharges to the OOO, shall present an oral report to the San Diego Water Board summarizing the conclusions of the receiving water monitoring report. The State of the Ocean Report shall be given once no later than 180 days prior to the expiration date of this Order. If an oral report cannot be scheduled for a San Diego Water Board meeting, the San Diego Water Board may approve submission of a written State of the Ocean Report. The State of the Ocean Report shall include, at minimum, the following elements:
- a. Description of the monitoring effort completed;
 - b. The status and trends of receiving water quality conditions; and
 - c. Plans for future monitoring efforts.

V. REGIONAL MONITORING REQUIREMENTS

Regional ocean water monitoring provides information about the sources, fates, and effects of anthropogenic contaminants in the coastal marine environment necessary to make assessments over large areas. The large-scale assessments provided by regional monitoring describe and evaluate cumulative effects of all anthropogenic inputs and enable better decision-making regarding protection of beneficial uses of ocean waters. Regional monitoring data assists in the interpretation of core monitoring studies by providing a more accurate and complete characterization of reference conditions and natural variability. Regional monitoring also leads to methods standardization and improved quality control through inter-calibration exercise. The coalitions implementing regional monitoring enable sharing of technical resources, trained personnel, and associated costs. Focusing these resources on regional issues and developing a broader understanding of pollutants effects in ocean waters enables the development of more rapid and effective response strategies. Based on all of these considerations the San Diego Water Board supports regional approaches to monitoring ocean waters.

The Discharger shall, as directed by the San Diego Water Board, participate with other regulated entities, other interested parties, and the San Diego Water Board in development and implementation of new and improved monitoring and assessment programs for ocean waters in the San Diego Region and discharges to those waters. These programs shall be developed and implemented so as to answer the following questions:

- (1) What are the status and trends of conditions in ocean waters in the San Diego Region with regard to beneficial uses? For example:
 - i. Are fish and shellfish safe to eat?
 - ii. Is water quality safe for swimming?
 - iii. Are ecosystems healthy?
- (2) What are the primary stressors causing or contributing to conditions of concern?
- (3) What are the major sources of the stressors causing or contributing to conditions of concern?
- (4) Are the actions taken to address such stressors and sources effective (i.e., environmental outcomes)?

Development and implementation of new and improved monitoring and assessment programs for ocean waters will be guided by the following:

1. The Ocean Plan;
2. San Diego Water Board Resolution No. R9-2012-0069, *Resolution in Support of A Regional Monitoring Framework*;
3. San Diego Water Board staff report entitled *A Framework for Monitoring and Assessment in the San Diego Region*; and
4. Other guidance materials, as appropriate.

A. Kelp Bed Canopy Monitoring Requirements

Kelp consists of a number of species of brown algae. Along the central and southern California coast, giant kelp (*Macrocystis pyrifera*) is the largest species colonizing rocky, and in some cases sandy, subtidal habitats. Giant kelp is an important component of coastal and island communities in southern California, providing food and habitat for numerous animals. Monitoring of the kelp beds is necessary to answer the following questions:

- (1) What is the maximum areal extent of the coastal kelp bed canopies each year?
- (2) What is the variability of the coastal kelp bed canopy over time?
- (3) Are coastal kelp beds disappearing? If yes, what are factors that could contribute to the disappearance?
- (4) Are new coastal kelp beds forming?

The Discharger shall participate with other Southern California ocean dischargers in an ongoing regional survey of coastal kelp beds in the Southern California Bight. The intent of these surveys is to provide an indication of the health of these kelp beds, recognizing that the extent of kelp bed canopies may change due to a variety of influences.

Kelp beds shall be monitored by means of vertical aerial infrared photography to determine the maximum areal extent of the canopies of coastal kelp beds each year. Surveys shall be conducted as close as possible to when kelp bed canopies are at

their greatest extent during the year. The entire San Diego Region coastline, from the international boundary to the San Diego Region/Santa Ana Region boundary shall be photographed on the same day.

Annually by October 1, the Discharger shall submit to the San Diego Water Board a report which summarizes the data, analyses, assessment, and images produced by the surveys. The report is a joint collaboration among multiple ocean dischargers in the Southern California (e.g., Regional 9 Kelp Survey Consortium member agencies). In addition to the kelp bed canopies, the images shall show onshore reference points, locations of all ocean outfalls and diffusers, artificial reefs, areas of known hard-bottom substrate (i.e., rocky reefs), and depth contours at intervals of 30-feet mean lower low water (MLLW). The report shall also be made available in a user-friendly format on a website that is readily available to the public.

The surveys shall be conducted on a “continuous improvement” basis, as needed improvements shall be made in monitoring, analysis, assessment, and/or documentation. For example, these could include:

1. More sophisticated analysis of patterns, correlations, and cycles that may be related to the extent of kelp bed canopies; or
2. Projects to improve understanding of influences on kelp beds or of how the extent of the canopies of various kelp beds has changed since the early 20th century.

B. Southern California Bight Monitoring Program Participation Requirements

The Discharger is required to participate in the Southern California Bight Regional Monitoring Program coordinated by the Southern California Coastal Water Research Project (SCCWRP), or any other coordinator named by the San Diego Water Board, pursuant to Water Code sections 13267 and 13383, and 40 CFR section 122.48. The intent of the Southern California Bight Regional Monitoring Program is to maximize the efforts of all monitoring partners using a more cost-effective monitoring design and to best utilize the pooled scientific resources of the Southern California Bight.

During these coordinated sampling efforts, a portion of the Discharger’s receiving water sampling and analytical effort, as defined in section IV of this MRP, may be reallocated to provide a regional assessment of the impact of the discharge of wastewater to the Southern California Bight. In that event, the San Diego Water Board shall notify the Discharger in writing that a portion of the requirements to perform the receiving water sampling and analytical effort defined in section IV of this MRP is suspended for the duration of the reallocation. Anticipated modifications to the monitoring program will be coordinated so as to provide a more comprehensive picture of the ecological and statistical significance of monitoring results and to determine cumulative impacts of various pollution sources. The level of resources in terms of sampling and analytical effort redirected from the receiving water monitoring program required under section IV of this MRP shall approximately equal the level of resources provided to implement the regional monitoring and assessment program, unless the San Diego Water Board and the Discharger agree otherwise. The specific scope and duration of the receiving water monitoring program reallocation and

redirection shall be determined in writing by the San Diego Water Board, in consultation with the Discharger.

VI. SPECIAL STUDIES REQUIREMENTS

A. Climate Change Action Plan

The Discharger shall prepare and submit a Climate Change Action Plan (CCAP) within three years of the effective date of this Order. The Discharger may make use of existing climate-change-related plans to comply with this requirement. Changing climate conditions may fundamentally alter the way wastewater facilities are designed and operated. Climate change research indicates the overarching driver of change is increased atmospheric carbon dioxide (CO₂) from human activity. The increased CO₂ emissions trigger changes to climatic patterns, which increase the intensity of sea level rise and coastal storm surges (Δ Sea Level), lead to more erratic rainfall and local weather patterns (Δ Weather Patterns), trigger a gradual warming of freshwater and ocean temperatures (Δ Water Temperature), and trigger changes to ocean water chemistry (Δ Water pH). The CCAP shall identify projected regional impacts on the Facilities and operations due to climate change if current trends continue. The CCAP shall also identify steps being taken or planned to address greenhouse gas emissions attributable to wastewater treatment plants, solids handling, and effluent discharge processes. The CCAP shall also identify steps being taken or planned to address flooding, wildfire, and sea level rise risks; volatile rain period impacts (both dry and wet weather); challenges in accommodating high and low wastewater flows; impacts on process design parameters due to higher biochemical oxygen demand, ammonia (as N), and TSS influent concentrations; impacts on wastewater treatment operations and quality; the potential need to adjust NPDES permit conditions and the Discharger's pollution control program; the financing needed to pay for planned actions; schedules to update the CCAP as more information on climate change and its effect become more available; and any other factors as appropriate.

B. Plume Tracking Monitoring Program

Plume tracking is a single study or ongoing program designed to assess dispersion and fate of the wastewater plume discharged from the OOO. The Plume Tracking Monitoring Program (PTMP) may be conducted either alone or in collaboration with other agencies (e.g., city of Oceanside; Marine Corps Base, Camp Pendleton; as well as agencies that discharge through the Encina and San Elijo ocean outfalls). The PTMP shall be designed to address, at minimum, the following questions:

- (1) Are the current monitoring locations and methods adequate to determine whether the wastewater plume is encroaching on water recreational areas, including, but not limited to, areas used for swimming, scuba diving, surfing, and fishing? If not, what monitoring locations and/or methods are more appropriate?
- (2) How does the brine discharge from the Discharger's Santa Margarita Groundwater Treatment Plant, City of Oceanside's Mission Basin Groundwater Purification Facility; Genentech, Inc.; the Marine Corps Base, Camp Pendleton's Advanced Water Treatment Plant at Haybarn Canyon; and

future brine discharges (along with increased recycled water use and decreased outfall discharge flows) affect the dynamics of the wastewater plume and initial dilution?

- (3) Does the wastewater plume have the potential to interact with wastewater plumes from other ocean outfalls or other sources of pollution, such as storm water?
- (4) What is the fate of the wastewater plume in typical and atypical oceanographic conditions, and when and under what conditions is the wastewater plume no longer distinguishable from ambient receiving water?
- (5) What parameters are most useful for assessing the presence of the wastewater plume?
- (6) What is the variability in the degree of initial dilution that occurs under typical and atypical oceanographic conditions?
- (7) At what distance is the wastewater plume no longer distinguishable from the ambient receiving water?

The Discharger shall develop and submit the following:

1. **Conceptual Site Model.** The Discharger shall construct a conceptual site model (CSM) based on available site-specific data such as the effluent quality at OOO, occurrence of waste constituents in the receiving water and sediment, and direction and magnitude of surface and subsurface currents. The CSM shall include three-dimensional spatial extent and temporal variability of the waste constituents in the receiving water; and the location and exposure points of actual and potential receptors (humans and marine life).

The CSM shall be refined and updated as data becomes available. The initial CSM shall include a discussion of the level of uncertainty of conclusions, outline data gaps in the initial CSM, and describe the additional work needed to complete the CSM. Updates to the CSM shall be included in the plume tracking semiannual progress reports required in section VI.B.4 of this MRP. The initial CSM shall be submitted to the San Diego Water Board with the work plan described in section VI.B.2 of this MRP.

2. **Plume Tracking Monitoring Program (PTMP) Work Plan.** The Discharger shall prepare and submit to the San Diego Water Board within 180 days of the effective date of this Order, a PTMP Work Plan designed to guide the collection of information to produce the PTMP described in section VI.B.3 of this MRP. The PTMP Work Plan shall include, but is not limited to, the following:
 - a. Conceptual Site Model;
 - b. Evaluation of available technologies that may assist with the PTMP;
 - c. Specific questions that will be answered with the PTMP;
 - d. A description of the approach to be taken for conducting a feasibility analysis for:

- i. Installation and operation by the Discharger of a permanent, real-time oceanographic mooring system located near the terminal diffuser structure of the OOO;
 - ii. Utilizing advanced oceanographic sampling technologies such as an autonomous underwater vehicle (AUV) or remotely operated towed vehicle (ROTV) either alone or in conjunction with an OOO real-time mooring; and
 - iii. An alternative method identified during the evaluation of available technologies that may assist with the PTMP.
- e. How the work on the PTMP will be divided amongst the participating agencies; and
 - f. Schedule for completion of all activities and submission of a final PTMP as described in section VI.B.3 below.

The Discharger shall present the submitted PTMP Work Plan at the next available San Diego Water Board meeting for the Board to consider the PTMP Work Plan for approval. The Discharger shall implement the PTMP Work Plan to develop the final PTMP as directed by the San Diego Water Board sixty (60) days after Board approval of the PTMP Work Plan. The Discharger shall notify the San Diego Water Board of the intent to initiate the proposed actions included in the PTMP Work Plan; and comply with any conditions set by the San Diego Water Board.

- 3. **Plume Tracking Monitoring Program (PTMP).** The Discharger shall, in consultation with the San Diego Water Board, prepare and submit a final PTMP, prepared in accordance with the approved PTMP Work Plan, to evaluate dispersion and fate of the wastewater plume discharged from the OOO. The PTMP shall include, but is not limited to, the following elements:
 - a. A feasibility analysis, including an assessment of advantages, disadvantages, cost, usefulness, and effectiveness of the following:
 - i. Installation and operation by the Discharger of a permanent, real-time oceanographic mooring system located near the terminal diffuser structure of the OOO. The mooring system shall be designed to measure, at minimum, direction and velocity of subsurface currents, and ocean stratification. If determined to be feasible, this element shall also include:
 - (a) A plan for implementation of the OOO real-time mooring system, including data acquisition and processing; and
 - (b) Provision for networking the OOO system to be compatible with similar systems being deployed by other dischargers in the San Diego Region, as well as a third-party system operated by the University of California, San Diego, Scripps Institution of Oceanography in the coastal waters off the City of Del Mar.

- ii. Utilizing advanced oceanographic sampling technologies such as an AUV or ROTV either alone or in conjunction with an OOO real-time mooring system.
- iii. An alternative approach proposed by the Discharger to answer the questions posed above for the PTMP, if applicable.
- b. A detailed description of recommended actions;
- c. A monitoring plan to implement the recommended actions; and
- d. A detailed description and schedule for completion of all activities planned to implement the recommendations in the PTMP, and the schedule for submission of the final results of the PTMP as described in section VI.B.5 of this MRP.

The PTMP shall be submitted in accordance with the timeframe outlined in the approved PTMP Work Plan described in section VI.B.2 of this MRP.

- 4. **PTMP Implementation.** The Discharger shall initiate implementation of the PTMP sixty (60) days after submission in accordance with the schedule contained in the PTMP, unless otherwise directed by the San Diego Water Board. The Discharger shall notify the San Diego Water Board of the intent to initiate the proposed actions included in the PTMP; and comply with any conditions set by the San Diego Water Board. The Discharger shall submit semiannual progress reports on implementation of the PTMP to the San Diego Water Board on February 1 and August 1 of each year.
- 5. **Plume Tracking Monitoring Reports.** The final results of the PTMP shall be submitted with the Receiving Water Monitoring Report and be included in the State of the Ocean report required in section IV.F of this MRP. The Receiving Water Monitoring Report shall include in-depth discussion, evaluation, interpretation, and tabulation of the project data. Report interpretations and conclusions shall include the state of the receiving waters around the OOO and the estimated locations of the OOO plume during the study.
- 6. **Receiving Water Monitoring Recommendations Report.** The Discharger shall submit a Receiving Water Monitoring Recommendations Report that includes recommendations for future shoreline, nearshore, and offshore monitoring locations. At minimum, the Discharger shall propose monitoring locations for monitoring stations S6, S7, N6, and N7. The Discharger shall also evaluate the feasibility of event-based monitoring, including collecting samples during certain oceanographic conditions at locations where the plume is most likely to occur. The Receiving Water Monitoring Recommendations Report shall be submitted to the San Diego Water Board no later than 180 days prior to the expiration date of this Order, unless an extension is granted by the San Diego Water Board.

VII. REPORTING REQUIREMENTS

A. General Monitoring and Reporting Requirements

1. The Discharger shall comply with all Standard Provisions (Attachment D) related to monitoring, reporting, and recordkeeping.
2. The Discharger shall report all instances of noncompliance not reported under sections V.E, V.G, and V.H of the Standard Provisions (Attachment D) at the time monitoring reports are submitted.

B. Self-Monitoring Reports (SMRs)

1. The Discharger shall electronically submit SMRs using the State Water Board's California Integrated Water Quality System (CIWQS) Program website at http://www.waterboards.ca.gov/water_issues/programs/ciwqs/https://www.waterboards.ca.gov/water_issues/programs/ciwqs/. The CIWQS website will provide additional information for SMR submittal in the event there will be a planned or unplanned service interruption for electronic submittal. SMRs must be signed and certified as required by section V of the Standards Provisions (Attachment D). The Discharger shall maintain sufficient staffing and resources to ensure it submits SMRs that are complete and timely. This includes provision for training and supervision of individuals on how to prepare and submit SMRs.
2. The Discharger shall report in the SMR the results for all monitoring specified in this MRP under sections III through IV. The Discharger shall submit SMRs including the results of all required monitoring using USEPA-approved test methods or other test methods specified in this Order. SMRs are to include all new monitoring results obtained since the last SMR was submitted. If the Discharger monitors any pollutant more frequently than required by this Order, the results of this monitoring shall be included in the calculations and reporting of the data submitted in the SMR.
3. Monitoring periods and reporting for all required monitoring shall be completed according to the following schedule:

Table E-10. Monitoring Periods and Reporting Schedule

Sampling Frequency	Monitoring Period Begins On...	Monitoring Period	SMR Due Date
Continuous	Permit effective date	All	First day of second calendar month following month of sampling.
Daily	Permit effective date	(Midnight through 11:59 PM) or any 24-hour period that reasonably represents a calendar day for purposes of sampling.	First day of second calendar month following month of sampling.
Weekly	Sunday following permit effective date or on permit effective date if on a Sunday	Sunday through Saturday	First day of second calendar month following month of sampling.

Sampling Frequency	Monitoring Period Begins On...	Monitoring Period	SMR Due Date
Monthly	First day of calendar month following permit effective date or on permit effective date if that date is first day of the month	1 st day of calendar month through last day of calendar month	First day of second calendar month following month of sampling.
Quarterly	Closest of January 1, April 1, July 1, or October 1 following (or on) permit effective date	January 1 through March 31 April 1 through June 30 July 1 through September 30 October 1 through December 31	May 1 August 1 November 1 February 1
Semiannually	Closest of January 1 or July 1 following (or on) permit effective date	January 1 through June 30 July 1 through December 31	August 1 February 1
Once during the permit term	Permit effective date	Permit term	No later than 180 days prior to the expiration date of this Order

4. **Reporting Protocols.** The Discharger shall report with each sample result the applicable reported Minimum Level (reported ML, also known as the Reporting Level, or RL) and the current Method Detection Limit (MDL), as determined by the procedure in 40 CFR part 136.

The Discharger shall report the results of analytical determinations for the presence of chemical constituents in a sample using the following reporting protocols:

- a. Sample results greater than or equal to the reported ML shall be reported as measured by the laboratory (i.e., the measured chemical concentration in the sample).
- b. Sample results less than the reported ML, but greater than or equal to the laboratory's MDL, shall be reported as "Detected, but Not Quantified," or DNQ. The estimated chemical concentration of the sample shall also be reported.

For the purposes of data collection, the laboratory shall write the estimated chemical concentration next to DNQ. The laboratory may, if such information is available, include numerical estimates of the data quality for the reported result. Numerical estimates of data quality may be percent accuracy ($\pm$ a percentage of the reported value), numerical ranges (low to high), or any other means considered appropriate by the laboratory.

- c. Sample results less than the laboratory's MDL shall be reported as "Not Detected," or ND.
- d. Dischargers are to instruct laboratories to establish calibration standards so that the ML value (or its equivalent if there is differential treatment of samples relative to calibration standards) is the lowest calibration standard. At no time is the Discharger to use analytical data derived from extrapolation beyond the lowest point of the calibration curve.

5. **Compliance Determination.** Compliance with effluent limitations for reportable pollutants shall be determined using sample reporting protocols defined above and in Attachment A of this Order. For purposes of reporting and administrative enforcement by the San Diego Water Board and State Water Board, the Discharger shall be deemed out of compliance with effluent limitations if the concentration of the reportable pollutant in the monitoring sample is greater than the effluent limitation and greater than or equal to the reported ML.
6. **Multiple Sample Data.** When determining compliance with a measure of central tendency (arithmetic mean, geometric mean, median, etc.) of multiple sample analyses and the data set contains one or more reported determinations of DNQ or ND, the Discharger shall compute the median in place of the arithmetic mean in accordance with the following procedure:
 - a. The data set shall be ranked from low to high, ranking the reported ND determinations lowest, DNQ determinations next, followed by quantified values (if any). The order of the individual ND or DNQ determinations is unimportant.
 - b. The median value of the data set shall be determined. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two values around the middle unless one or both of the points are ND or DNQ, in which case the median value shall be the lower of the two data points where DNQ is lower than a value and ND is lower than DNQ.
7. The Discharger shall submit SMRs in accordance with the following requirements:
 - a. The Discharger shall arrange all reported data in a tabular format. The data shall be summarized to clearly illustrate whether the facility is operating in compliance with interim and/or final effluent limitations. The Discharger is not required to duplicate the submittal of data that is entered in a tabular format within CIWQS. When electronic submittal of data is required and CIWQS does not provide for entry into a tabular format within the system, the Discharger shall electronically submit the data in a tabular format as an attachment.
 - b. The Discharger shall attach a cover letter to the SMR. The information contained in the cover letter shall clearly identify violations of the waste discharge requirements; discuss corrective actions taken or planned; and the proposed time schedule for corrective actions. Identified violations must include a description of the requirement that was violated and a description of the violation.
 - c. The Discharger shall add all violations of the waste discharge requirements in CIWQS under the "Violations" tab.

C. Discharge Monitoring Reports (DMRs)

The DMRs are USEPA reporting requirements. The Discharger shall electronically certify and submit DMRs together with SMRs using Electronic Self-Monitoring Reports (eSMR) module eSMR 2.5 or any upgraded version. Electronic DMRs submittal shall be in addition to electronic SMR submittal. Information about electronic DMRs submittal is available at the DMR website at:

http://www.waterboards.ca.gov/water_issues/programs/discharge_monitoring
https://www.waterboards.ca.gov/water_issues/programs/discharge_monitoring.

D. Other Reports

The following reports are required under Special Provisions (sections VI.A and VI.C of this Order), sections I, III, IV, V, and VI of this MRP, and the California Code of Regulations (CCR). The reports shall be submitted to the San Diego Water Board using the State Water Board's CIWQS program website, unless otherwise noted. The reports must be signed and certified as required by section V of the Standards Provisions (Attachment D). The CIWQS website will provide additional information for SMR submittal in the event of a planned or unplanned service interruption for electronic submittal.

Table E-11. Other Reports

Report	Location of requirement	Due Date
ROWD (for reissuance)	Table 3 and Section VI.A.2.b	No later than 180 days before the Order expiration date ¹
<u>Performance Goal Exceedance Report</u>	<u>Section IV.A.2</u>	<u>30 days after the third successive exceedance of a performance goal</u>
Treatment Plant Capacity Report	Section VI.C.5.a	Four years prior to reaching plant design capacity ¹
Industrial Waste Survey	Section VI.C.5.b.i	December 1, 2021
Annual Biosolids Report	Section VI.C.5.c.viii	Annually no later than February 19
DMR-QA Study	Section I.G of this MRP	Annually no later than December 31 ²
Initial Investigation TRE Work Plan	Section III.C.6 of this MRP	Within one year of the effective date of this Order
Sediment Monitoring Work Plan	Section IV.C.4 of this MRP	Within 180 days of the effective date of this Order
CEDEN	Section IV.E of this MRP	Annually no later than March 1
Receiving Water Monitoring Report	Sections IV.F.1 and IV.F.2 of this MRP	No later than 180 days prior to the expiration date of this Order
State of the Ocean Report	Section IV.F.3 of this MRP	No later than 180 days prior to the expiration date of this Order
Kelp Bed Canopy Report	Section V.A of this MRP	Annually no later than October 1

FALLBROOK PUBLIC UTILITY DISTRICT
 FALLBROOK WATER RECLAMATION PLANT
 SANTA MARGARITA GROUNDWATER TREATMENT PLANT

Attachment 1 to Order No. R9-2021-0204
 ORDER NO. R9-2019-0169
 Amended by Order No. R9-2020-0193 and R9-2021-0204
 NPDES NO. CA0108031

Report	Location of requirement	Due Date
CCAP	Section VI.A of this MRP	No later than three years of the effective date of this Order
Plume Tracking Monitoring Program Work Plan	Section VI.B.2 of this MRP	Within 180 days of the effective date of this Order
Plume Tracking Monitoring Program	Section VI.B.3 of this MRP	As specified in the Plume Tracking Monitoring Plan Work Plan
Plume Tracking Semiannual Progress Report	Section VI.B.4 of this MRP	After implementation of PTMP January 1 through June 30: due August 1 July 1 through December 31: due February 1
Receiving Water Monitoring Recommendations Report	Section VI.B.6 of this MRP	No later than 180 days prior to the expiration date of this Order

¹ Submit in person or by mail to the San Diego Water Board office (2375 Northside Drive, Suite 100, San Diego, CA 92108) or by email at SanDiego@waterboards.ca.gov.

² See section I.H. of this MRP for instructions on how to submit the study.

ATTACHMENT F – FACT SHEET

CONTENTS

I.	Permit Information	F-3
II.	Facility Description.....	F-5
	A. Description of Wastewater and Biosolids Treatment and Controls	F-5
	B. Discharge Points and Receiving Waters.....	F-7
	C. Summary of Existing Requirements and Self-Monitoring Report (SMR) Data	F-10
	D. Compliance Summary	F-11
	E. Planned Changes	F-12
III.	Applicable Plans, Policies, and Regulations	F-12
	A. Legal Authorities	F-12
	B. California Environmental Quality Act (CEQA).....	F-13
	C. State and Federal Laws, Regulations, Policies, and Plans	F-13
	D. Impaired Water Bodies on the CWA section 303(d) List.....	F-15
	E. Other Plans, Policies, and Regulations	F-15
IV.	Rationale for Effluent Limitations and Discharge Specifications	F-16
	A. Discharge Prohibitions.....	F-16
	B. Technology-Based Effluent Limitations (TBELs)	F-16
	1. Scope and Authority	F-16
	2. Applicable Technology-Based Effluent Limitations	F-17
	C. Water Quality-Based Effluent Limitations (WQBELs)	F-19
	1. Scope and Authority	F-19
	2. Applicable Beneficial Uses and Water Quality Criteria and Objectives	F-20
	3. Determining the Need for WQBELs	F-21
	4. WQBEL Calculations	F-25
	5. Whole Effluent Toxicity (WET).....	F-30
	D. Final Effluent Limitation Considerations.....	F-34
	1. Anti-Backsliding Requirements	F-34
	2. Antidegradation Policies	F-35
	3. Stringency of Requirements for Individual Pollutants.....	F-35
	E. Interim Effluent Limitations – Not Applicable	F-36
	F. Land Discharge Specifications – Not Applicable	F-36
	G. Recycling Specifications – Not Applicable	F-36
V.	Rationale for Receiving Water Limitations	F-36
VI.	Rationale for Provisions	F-37
	A. Standard Provisions.....	F-37
	B. Special Provisions	F-37
	1. Reopener Provisions	F-37
	2. Special Studies, Technical Reports, and Additional Monitoring Requirements	F-37
	3. Best Management Practices and Pollution Prevention	F-38
	4. Construction, Operation, and Maintenance Specifications	F-38
	5. Special Provisions for Wastewater Facilities	F-38
	6. Other Special Provisions – Not Applicable	F-41

7. Compliance Schedules – Not Applicable	F-41
VII. Rationale for Monitoring and Reporting Requirements	F-41
A. Core Monitoring Requirements	F-41
B. Receiving Water Monitoring.....	F-45
1. Surf Zone Water Quality Monitoring Requirements	F-45
2. Nearshore and Offshore Water Quality Monitoring Requirements.....	F-45
3. Benthic Monitoring Requirements.....	F-48
4. Fish and Macroinvertebrate Monitoring Requirements	F-49
5. <u>Receiving Water Monitoring Reporting Requirements</u>	<u>F-50</u>
6. <u>Groundwater – Not Applicable</u>	<u>F-51</u>
C. Regional Monitoring Requirements.....	F-51
1. Kelp Bed Canopy Monitoring Requirements	F-52
2. Southern California Bight Regional Monitoring Program Participation Requirements	F-52
D. Special Studies Requirements.....	F-53
E. Other Monitoring Requirements	F-54
VIII. Public Participation	F-54
A. Notification of Interested Parties	F-54
B. Written Comments	F-55
C. Public Hearing	F-55
D. Reconsideration of Waste Discharge Requirements	F-55
E. Information and Copying.....	F-56
F. Register of Interested Persons	F-56
G. Additional Information	F-56

TABLES

Table F-1. Information on the Facilities.....	F-3
Table F-2. San Luis Rey Land Outfall.....	F-8
Table F-3. Historic Effluent Limitations and Monitoring Data at M-001 ¹	F-10
Table F-4. Historic Effluent Limitations and Monitoring Data at M-002 ¹	F-11
Table F-5. Basin Plan Beneficial Uses	F-13
Table F-6. Ocean Plan Beneficial Uses	F-14
Table F-7. Summary of TBELs Based on Secondary Treatment Standards.....	F-18
Table F-8. Summary of TBELs on Table 4 of the Ocean Plan.....	F-19
Table F-9. RPA Results Summary ¹	F-22
Table F-10. Pollutants Having Background Concentrations ¹	F-25
Table F-11. Example Parameter Water Quality Objectives ¹	F-26
Table F-12. Summary of WQBELs at Monitoring Location M-002 ¹	F-27
Table F-13. Summary of Performance Goals for Monitoring Location M-002 ¹	F-28

ATTACHMENT F – FACT SHEET

As described in section II.B of this Order, the California Regional Water Quality Control Board, San Diego Region (San Diego Water Board) incorporates this Fact Sheet as findings of the San Diego Water Board supporting the issuance of this Order. This Fact Sheet includes the legal requirements and technical rationale that serve as the basis for the requirements of this Order.

This Order has been prepared under a standardized format to accommodate a broad range of discharge requirements for Dischargers in California. Only those sections or subsections of this Order that are specifically identified as “not applicable” have been determined not to apply to this Discharger. Sections or subsections of this Order not specifically identified as “not applicable” are fully applicable to this Discharger.

I. PERMIT INFORMATION

The following table summarizes administrative information related to the Facilities.

Table F-1. Information on the Facilities

WDID	9 000000115
Discharger	Fallbrook Public Utility District
Name of Facility	Fallbrook Water Reclamation Plant (Fallbrook WRP) and Santa Margarita Groundwater Treatment Plant (Santa Margarita GTP)
Facility Address	1425 S. Alturas Road Fallbrook, CA 92028 San Diego County
Facility Contact, Title and Phone	Owni Toma, Chief Plant Operator, (760) 999-2802
Authorized Person to Sign and Submit Reports	Jack Bebee, General Manager, (760) 728-1125 Owni Toma, Chief Plant Operator, (760) 999-2802
Mailing Address	990 East Mission Road, Fallbrook, CA 92028
Billing Address	Same as mailing address
Type of Facility	Publicly-owned treatment works (POTWs, SIC No. 4952) and desalination facility
Major or Minor Facility	Major
Threat to Water Quality	1
Complexity	A
Pretreatment Program	No
Recycling Requirements	Producer and distributor (regulated under separate waste discharge requirements (WDRs))

Facility Permitted Flow	Combined flow from the Fallbrook WRP and Santa Margarita GTP: May to October: 2.7 million gallons per day (MGD) November to April: 3.6 MGD
Facility Design Flow	Fallbrook WRP: 2.7 MGD (average dry weather flow) 3.6 MGD (maximum wet weather flow) Santa Margarita GTP: Treats a maximum of 800 acre-feet of raw groundwater per month
Watershed	Pacific Ocean
Receiving Water	Pacific Ocean
Receiving Water Type	Ocean waters

- A. The Fallbrook Public Utility District (Discharger) is the owner and operator of the Fallbrook WRP (formerly the Fallbrook Public Utility District Treatment Plant No. 1) and the associated pump stations/land outfalls; together these facilities comprise a POTW. The Discharger has begun construction of a new groundwater treatment plant, the Santa Margarita Groundwater Treatment Plant (Santa Margarita GTP), which will also be owned and operated by the Discharger and will treat groundwater from the Santa Margarita River Basin. The Fallbrook WRP, Santa Margarita GTP, and the associated pump station(s)/land outfall(s) are collectively referred to as the Facilities.

For the purposes of this Order, references to the “discharger” or “permittee” in applicable federal and State laws, regulations, plans, or policy are held to be equivalent to references to the Discharger herein.

- B. The Fallbrook WRP discharges tertiary-treated wastewater to the Pacific Ocean, a water of the United States. The Discharger was previously regulated by Order No. R9-2012-0004 and National Pollutant Discharge Elimination System (NPDES) Permit No. CA0108031 adopted on August 8, 2012 and expired on September 27, 2017. As requested by the Discharger in its report of waste discharge (ROWD), this Order incorporates an additional facility, the Santa Margarita GTP. Attachment B provides a map of the area around the Facilities. Attachment C provides flow schematics of the Facilities.
- C. The Discharger filed a ROWD and submitted an application for reissuance of its WDRs and NPDES permit on March 29, 2017. The application was deemed complete on August 15, 2017. The San Diego Water Board conducted a site visit of the Fallbrook WRP on April 9, 2019, to observe operations and collect additional data to develop permit limitations and requirements for waste discharge. The final design and construction of the Santa Margarita GTP has not been completed; thus, the San Diego Water Board could not conduct a site visit to view the constructed Santa Margarita GTP.

- D. Regulations at title 40 of the Code of Federal Regulations (40 CFR) section 122.46 limit the duration of NPDES permits to a fixed term not to exceed five years. Accordingly, Table 3 of this Order limits the duration of the discharge authorization. However, pursuant to California Code of Regulations (CCR), title 23, section 2235.4, the terms and conditions of an expired permit are automatically continued pending reissuance of the permit if the Discharger complies with all federal NPDES requirements for continuation of expired permits.

II. FACILITY DESCRIPTION

The Fallbrook WRP serves approximately 23,000 residents, treating predominately residential and commercial wastewater. The Discharger reports that no significant industrial users (SIUs) discharge to the Fallbrook WRP.

A. Description of Wastewater and Biosolids Treatment and Controls

1. Fallbrook WRP

All wastewater entering the Fallbrook WRP is subjected to full tertiary treatment consisting of preliminary treatment by mechanical bar screening and aerated grit removal, primary sedimentation and scum removal, biological treatment using activated sludge followed by secondary clarification, tertiary treatment consisting of coagulation followed by single-media anthracite filter, and chlorine disinfection. Dechlorination of chlorinated effluent is accomplished with sodium bisulfite (NaHSO_3).

Screenings from the headworks and solids from grit removal are washed and compacted prior to being hauled to a local landfill for disposal. Sludge from the secondary treatment processes at the Fallbrook WRP is aerobically digested, then dewatered by a centrifuge. Dewatered sludge is fed to a thermal dryer to meet Class A biosolids requirements. Class A biosolids are utilized as fertilizer by local farmers. If the dryer is inoperative, dewatered sludge (Class B biosolids) is either sent to one of twelve drying beds for a minimum of three months, or to hauling containers for disposal. The Discharger contracts with Denali Solid Solutions for transporting Class B biosolids for land application in Yuma, Arizona.

During high flows or emergency operating or maintenance periods, a 1.2-million-gallon (MG) secondary effluent equalization basin is available for temporary storage. When this volume is exceeded, secondary effluent is discharged to the Fallbrook Public Utility District's 16-inch-diameter land outfall (Fallbrook Land Outfall) at rates not exceeding the hydraulic capacity of the Fallbrook Land Outfall of 3.6 MGD. Additionally, the Fallbrook WRP has a 1.0-MG recycled water storage basin.

The Discharger currently provides recycled water within its service area and through the Fallbrook Land Outfall to Caltrans in Oceanside. Recycled water

distributed from the Fallbrook WRP is regulated under separate WDRs, Order No. 91-39¹.

Treated effluent from the Fallbrook WRP not distributed as recycled water in the Discharger's service area is dechlorinated with sodium bisulfate and discharged to the Fallbrook Land Outfall, which conveys the effluent to the Oceanside Ocean Outfall (OOO).

The Fallbrook WRP is sized to treat an average dry weather flow of 2.7 MGD and a maximum wet weather flow of 3.6 MGD. In 2018, the average influent flow to the Fallbrook WRP was 1.43 MGD and the average effluent flow from the Fallbrook WRP to the OOO was 0.64 MGD. From January 1, 2015 to December 31, 2016 FPUD sent an annual average of 79.17 MGD to reuse sites for irrigation and 129.13 MGD to reuse sites for agriculture.

2. Santa Margarita GTP

The Marine Corps Base, Camp Pendleton (MCB Camp Pendleton), U.S. Department of the Interior, Bureau of Reclamation, and the Discharger are constructing and will operate a proposed Conjunctive Use Project. The project will enhance groundwater recharge and recovery capacity within the Lower Santa Margarita River Basin and develop a conjunctive use program that will increase available water supplies for the benefit of Marine Corps Base, Camp Pendleton and the Discharger. The proposed project includes the construction and use of facilities within MCB Camp Pendleton and the Discharger's service area and within the boundaries of the Santa Margarita River watershed. The project will capture additional surface runoff during high streamflow events in the Santa Margarita River that currently flows out to the Pacific Ocean. This surface water will be used to recharge groundwater through existing groundwater percolation ponds and stored or "banked" in groundwater basins during wet years and used to augment water supplies during dry years, reducing reliance on imported water. The proposed project includes a bidirectional pipeline to deliver water to the Discharger and provide MCB Camp Pendleton with an off-base water supply during drier than normal conditions or emergency situations, and the proposed Santa Margarita GTP. Raw groundwater would be pumped from the aquifer and conveyed to MCB Camp Pendleton's water treatment plant (Advanced Water Treatment Plant at Haybarn Canyon) and through the bidirectional pipeline to the Santa Margarita GTP. Raw groundwater delivered to the Santa Margarita GTP is expected to average 3,100 acre-feet per year and is not expected to exceed 800 acre-feet in any given month. However, total volumes of raw water deliveries to FPUD will vary annually, depending upon multiple factors including, but not limited to, precipitation, river surface flows, surface diversions, and environmental considerations. Treated imported water from the San Diego County Water

¹ On May 20, 1991, the San Diego Water Board adopted Order No. 91-39, *Waste Discharge Requirements for Fallbrook Public Utility District Plant Nos. 1 and 2 Reclamation Projects, San Diego County*. Addendums to Order No. 91-39 were adopted on August 10, 1995, February 13, 1997, and August 13, 1997.

Authority can also be delivered through the same bidirectional facility from the Discharger to MCB Camp Pendleton, if and when needed.

The Santa Margarita GTP will be constructed on the same property as the Fallbrook WRP. The FPUD WTP includes an equalization tank, pretreatment oxidation with sodium hypochlorite, iron and manganese filters, sodium bisulfite and anti-scalent to protect the reverse osmosis (RO) units, RO demineralization, neutralization with sodium hydroxide, clearwell, sodium hypochlorite for primary disinfection, ammonia hydroxide to form a chloramine residual in the pipeline. The FPUD WTP will be connected to and controlled by the Discharger's SCADA system.

The iron and manganese filters are expected to remove 97% of the iron and manganese in the system, reducing the effluent iron and manganese concentrations to below 0.3 milligrams/liter (mg/L) and 0.05 mg/L, respectively. Iron and manganese system reject streams will flow to reclaim tanks for solids separation. The liquid will be decanted and returned to the start of the iron and manganese filters for treatment. The remaining solids would be pumped to existing sludge drying beds at the Fallbrook WRP. Decant from the drying beds will be pumped to the Discharger's sewage collection system and the remaining dry solids will be removed for disposal at a landfill.

The feed water for the Santa Margarita GTP is expected to have relatively low salinity based on water quality data from existing production wells in the Lower Santa Margarita River Basin. Therefore, to maximize the RO treatment process efficiency and achieve a product water with a total dissolved solids (TDS) of 500 mg/L, approximately 49% of the effluent from the iron and manganese filters bypasses the sodium bisulfite, anti-scalent, RO units, and sodium hydroxide, and blends with the RO permeate upstream of the clearwell.

The Discharger's ROWD stated that the concentrated flows (i.e., brine waste) from the RO unit would be discharged to the Pacific Ocean through the OOO, with a maximum flow of about 0.65 MGD and a TDS concentration of 6,567 mg/L. In emails dated July 24 and 29, 2019, the Discharger stated that it plans to discharge the brine waste to its Fallbrook WRP, chlorine contact basin wet well, after the reclaimed water pump station/storage, but before the effluent monitoring for the ocean outfall, however this is subject to change. The July 24 email also states that the total flow from the Fallbrook WRP and Santa Margarita GTP to the OOO will remain within the flow limitations contained in Order No. R9-2012-0004. The maximum product water for the Santa Margarita GTP is estimated to be about 8 MGD or 92% of the feed groundwater flow based on cumulative process recoveries in the system.

B. Discharge Points and Receiving Waters

By letter dated July 6, 2016, the Discharger informed the San Diego Water Board that the Discharger and City of Oceanside (City) are engaged in negotiations to transfer ownership and operation of the lower portion of the Fallbrook Public Utility District Land Outfall (FPUD Land Outfall) from the Discharger to the City. In a letter dated

November 17, 2016 from the San Diego Water Board to the Discharger and the City, the San Diego Water Board stated that the City shall ensure that the proposed transfer and any resulting land outfall modifications do not cause capacity issues in the San Luis Rey Land Outfall or Oceanside Ocean Outfall. By December 2019, the transfer was completed. The tertiary-treated wastewater from the Discharger's Fallbrook Water Reclamation Plant discharges to the 16-inch-diameter FPUD Land Outfall. The FPUD Land Outfall discharges to the San Luis Rey Land Outfall at Pala Road, downstream of San Luis Rey Water Reclamation Facility (SLRWRF, start of San Luis Rey Land Outfall) and upstream of the City's Mission Basin Groundwater Purification Facility's (MBGPF's) discharge into the San Luis Rey Land Outfall; Genentech's discharge into the San Luis Rey Land Outfall; and City's La Salina Wastewater Treatment Plant (LSWTP, end of San Luis Rey Land Outfall). The unchlorinated tertiary-treated wastewater from the Fallbrook WRP discharges to the Fallbrook Land Outfall, which conveys the effluent approximately 14 miles from the Fallbrook WRP to the OOO at the City of Oceanside's (City's) La Salina Wastewater Treatment Plant (LSWTP). At the end of the Fallbrook Land Outfall, at the LSWTP, the effluent from the Fallbrook WRP discharges to the LSWTP Land Outfall and commingles with discharges from the LSWTP, the City's San Luis Rey Water Reclamation Facility (SLRWRF); the City's Mission Basin Groundwater Purification Facility (MBGPF); and Genentech, Inc. after the Combined Meter and before the OOO.² The LSWTP Land Outfall discharges to the start of the OOO, just north of the mouth of the Loma Alta Creek and east of South Pacific Street.

Table F-2. San Luis Rey Land Outfall

<u>Land Outfall</u>	<u>Section of Land Outfall</u>	<u>Pipeline Capacity (MGD)</u>	<u>Facilities Discharging to this section (permitted discharge)</u>	<u>Rated Capacity of Upstream Facilities (MGD)</u>
<u>San Luis Rey Land Outfall</u>	<u>From SLRWRF to FPUD Land Outfall Connection</u>	<u>13.5</u>	<u>SLRWRF (13.5 MGD)</u>	<u>15.4</u>
	<u>From FPUD Land Outfall Connection to</u>	<u>13.5</u>	<u>SLRWRF (13.5 MGD) Fallbrook Public Utility District (2.7/3.6 MGD)</u>	<u>17.1</u>

² Wastewater and waste brine from the City is regulated by separate WDRs, Order No. R9-2019-0166, as Amended By Order Nos. R9-2012-0042, R9-2012-0060, and R9-2014-0108, NPDES No. CA0107433, *Waste Discharge Requirements for the City Of Oceanside, San Luis Rey Water Reclamation Facility, La Salina Wastewater Treatment Plant, and Mission Basin Groundwater Purification Facility Discharges to the Pacific Ocean via the Oceanside Ocean Outfall*.

Waste brine from Genentech, Inc. is regulated by separate WDRs, Order No. R9-2019-0168, NPDES No. CA0109193, *Waste Discharge Requirements for Genentech Inc., Discharge to the Pacific Ocean via the Oceanside Ocean Outfall*).

<u>Land Outfall</u>	<u>Section of Land Outfall</u>	<u>Pipeline Capacity (MGD)</u>	<u>Facilities Discharging to this section (permitted discharge)</u>	<u>Rated Capacity of Upstream Facilities (MGD)</u>
	<u>MBGPF Connection</u>			
	<u>From MBGPF Connection to Genentech, Inc. Connection</u>	<u>13.5</u>	<u>SLRWRF (13.5 MGD)</u> <u>Fallbrook Public Utility District (2.7/3.6 MGD)</u> <u>MBGPF (2.0 MGD)</u>	<u>19.1</u>
	<u>From Genentech, Inc. Connection to LSWTP</u>	<u>13.5</u>	<u>SLRWRF (13.5 MGD)</u> <u>Fallbrook Public Utility District (2.7/3.6 MGD)</u> <u>MBGPF (2.0 MGD)</u> <u>Genentech, Inc. (0.155 MGD)</u>	<u>19.255</u>
<u>LSWTP Land Outfall</u>	<u>From LSWTP to OOO</u>	<u>16.6</u>	<u>SLRWRF (13.5 MGD)</u> <u>Fallbrook Public Utility District (2.7/3.6 MGD)</u> <u>MBGPF (2.0 MGD)</u> <u>Genentech, Inc. (0.155 MGD)</u> <u>LSWTP (5.5 MGD)</u>	<u>24.755</u>

The effluent from Marine Corps Base, Camp Pendleton's Southern Regional Tertiary Treatment Plant and Advanced Water Treatment Plant at Haybarn Canyon discharges to the 16-inch-diameter land outfall (Camp Pendleton Land Outfall). The Camp Pendleton Land Outfall discharges to the OOO, on South Pacific Street just north of the mouth of the Loma Alta Creek, where it comeslingles with the effluent from the Fallbrook WRP, LSWTP; SLRWRF; the MBGPF; and the Genentech, Inc. from the LSWTP Land Outfall.³

The City owns and operates the OOO which starts just north of the mouth of the Loma Alta Creek and east of South Pacific Street. The OOO extends southwesterly approximately 8,850 feet offshore to a depth of approximately 100 feet. The OOO is a steel pipe with a 1-inch-thick cement mortar interior lining and 2.75-inch-thick cement mortar outer jacket. In the letter dated May 1, 2014, the City reported that the average effective internal diameter of the OOO, considering the muck and sediment buildup, is 34.3 inches. The OOO terminates with a 230-foot diffuser collinear with the rest of the outfall and extends to a depth of approximately 108 feet. The diffuser has fourteen 5-

³ Wastewater and waste brine from the Southern Regional Tertiary Treatment Plant and the Advanced Water Treatment Plant at Haybarn Canyon is regulated by separate WDRs, Order No. R9-2019-0167, NPDES No. CA0109347, *Waste Discharge Requirements for the Marine Corps Base, Camp Pendleton, Southern Regional Tertiary Treatment Plant and Advanced Water Treatment Plant, Discharge to the Pacific Ocean via the Oceanside Ocean Outfall.*

inch-diameter ports and ten 4-inch-diameter ports. The terminus of the diffuser is located at Latitude 33° 09' 46" North, Longitude 117° 23' 29" W.

As the owner/operator, the City has the ability to control discharges to the OOO. The City has a contract with the Discharger to discharge an average annual flowrate of 2.4 MGD of treated wastewater from the Fallbrook WRP through the OOO, subject to WDRs contained in this Order. The City has a contract with the Marine Corps Base, Camp Pendleton to discharge up to 3.6 MGD of disinfected secondary-treated and tertiary-treated wastewater from the Southern Regional Tertiary Treatment Plant and brine waste from the Advanced Water Treatment Plant at Haybarn Canyon to the Pacific Ocean through the OOO, subject to WDRs contained in Order No. R9-2019-0167 (NPDES Permit No. CA0109347). As of 2008, the City has a contract with Genentech, Inc. to discharge a wastewater flow up to 1.396 MGD to the SLRWRF and to discharge brine flow up to 0.85 MGD to the OOO. In the letter dated May 1, 2014, the City reported that, due to the muck, sediment, and biological growth buildup in the OOO resulting in high friction losses in the OOO, the capacity of the OOO is 41.5 MGD.

C. Summary of Existing Requirements and Self-Monitoring Report (SMR) Data

Effluent limitations contained in Order No. R9-2012-0004 for discharges from Monitoring Location M-001 and from Discharge Point No. 001 (Monitoring Location M-002) and representative monitoring data from the term of Order No. R9-2012-0004 are as follows:

Table F-2.3. Historic Effluent Limitations and Monitoring Data at M-001¹

Parameter	Units	Effluent Limitation			Monitoring Data (From August 2012 to June 2019)		
		Average Monthly	Average Weekly	Instantaneous Maximum	Highest Average Monthly Discharge	Highest Average Weekly Discharge	Highest Instantaneous Maximum Discharge
Carbonaceous Biochemical Oxygen Demand (5-day @ 20°C) (CBOD ₅)	mg/L	25	40	--	7.2	11	--
	lbs/day	560	900	--	86	139	--
	% Removal	85 ³	--	--	96 ³	--	--
Total Suspended Solids	mg/L	30	45	--	13.4	8.2	--
	lbs/day	680	1,000	--	141	154	--
	% Removal	85 ³	--	--	96 ³	--	--
pH	standard units	--	--	6.0 – 9.0 ⁴	--	--	5.91 – 7.98 ⁴
Oil and Grease	mg/L	25	40	75 ²	1.5	1.5	6.9
	lbs/day	560	900	1,700	6.1	6.1	6.1
Settleable Solids	mL/L	1.0	1.5	3.0 ²	0.13	3.6	20
Turbidity	NTU	75	100	225 ²	3.62	14.7	56.2

- ¹ See Attachment A for definitions of abbreviations and a glossary of common terms used in this Order.
² Mass emission rate (MER) effluent limitations are based on the permitted flow rate for the Fallbrook WRP (2.7 MGD).
³ Lowest average monthly percent removal.
⁴ Minimum and maximum value.

Table F-3.4. Historic Effluent Limitations and Monitoring Data at M-002¹

Parameter	Units	Effluent Limitation ^{2,3}				Monitoring Data (From August 2012 to June 2019)			
		Six-Month Median	Average Monthly	Maximum Daily	Instantaneous Maximum	Highest Six-Month Median Discharge	Highest Maximum Average Monthly Discharge	Highest Maximum Daily Discharge	Instantaneous Maximum
Flow (May to October)	MGD	2.7	--	--	--	1.3	--	--	--
Flow (November to April)	MGD	3.6	--	--	--	1.5	--	--	--
Total Residual Chlorine	µg/L	180	--	700	5,300	4,175	--	9,200	9,200
	lbs/day	4.0	--	16	120	27	--	70	70
TCDD Equivalents	µg/L	--	3.4E-07	--	--	--	7.3E-07 ⁴	--	--
	lbs/day	--	7.7E-09	--	--	--	DNQ	--	--

- ¹ See Attachment A for definitions of abbreviations and a glossary of common terms used in this Order.
² Scientific "E" notation is used to express certain values. In scientific "E" notation, the number following the "E" indicates that position of the decimal point in the value. Negative numbers after the "E" indicate that the value is less than 1, and positive numbers after the "E" indicate that the value is greater than 1. In this notation a value of 6.1 E-02 represents 6.1 x 10⁻² or 0.061, 6.1E+02 represents 6.1 x 10² or 610, and 6.1E+00 represents 6.1 x 10⁰ or 6.1.
³ The mass emission rate (MER) limitation, in lbs/day, was calculated based on the following equation: MER (lbs/day) = 8.34 x Q x C, where Q is the permitted flow for the Fallbrook WRP (2.7 MGD) and C is the concentration (mg/L).
⁴ Estimated effluent concentration above the MDL and below the ML. As per compliance determination procedures, when concentrations are reported as DNQ, the mass is also reported as DNQ.

D. Compliance Summary

As of July 2019, the Discharger has reported the following violations of Order No. R9-2012-0004.

- Order No. R9-2012-0004, section IV.A.1.a, Table 7 states that the instantaneous maximum effluent limitation for settleable solids is 3.0 ml/L. On January 30, 2019, the Discharger exceeded this limitation with a reported value of 20 ml/L.
- Order No. R9-2012-0004, section IV.A.1.a, Table 7 states that the average weekly effluent limitation for settleable solids is 1.5 ml/L. On February 2, 2019, the Discharger exceeded this limitation with a reported value of 3.6 ml/L.
- Order No. R9-2012-0004, Attachment E, section III.A.1, Table E-2 states that the CBOD₅ shall be monitored at least once per week at Monitoring Location M-INF. On August 27, 2014, the Discharger failed to properly analyze the CBOD₅ at

Monitoring Location M-INF and thus did not meet the minimum monitoring requirements.

4. Order No. R9-2012-0004, Attachment E, section IV.A.1, Table E-3 states that the CBOD₅ shall be monitored at least five days per week at Monitoring Location M-001. On July 28 and August 8, 2014, the Discharger failed to properly analyze the CBOD₅ effluent samples and thus did not meet the minimum monitoring requirements. On August 19, 2015, the Discharger failed to monitor for CBOD₅ and thus did not meet the minimum monitoring requirements.
5. Order No. R9-2012-0004, Attachment E, section IV.A.1, Table E-3 states that the settleable solids shall be monitored at least five days per week at Monitoring Location M-001. On August 19, 2015, the Discharger failed to monitor for settleable solids and thus did not meet the minimum monitoring requirements.
6. Order No. R9-2012-0004, Attachment E, section IV.A.2, Table E-4 states that the temperature and dissolved oxygen shall be monitored at least once per week at Monitoring Location M-001 or M-002. The Discharger failed to monitor for these two parameters during the weeks of July 16, 2014 and April 26, 2015 and thus did not meet the minimum monitoring requirements.
7. Order No. R9-2012-0004, Attachment E, section IV.A.2, Table E-4 states that the flow shall be monitored continuous using a recorder/totalizer at Monitoring Location M-001 or M-002. On May 8, 9, 11, 12, 13, 14, 15, 16, 17, 19, 20, and 21, 2017, the Discharger failed to monitor the effluent flow using continuous recorder/totalizer and thus did not meet the minimum monitoring requirements.
8. Order No. R9-2012-0004, Attachment E, section IV.A.2, Table E-4 states that the total chlorine residual shall be monitored at least once per week at Monitoring Location M-001 or M-002. The Discharger failed to monitor for this parameter during the week of August 20, 2014 and thus did not meet the minimum monitoring requirements.
9. Order No. R9-2012-0004, Attachment E, section X.B.3, Table E-14 includes the due dates for the required monitoring reports. The monthly monitoring reports are due the first day of the second calendar month following month of sampling. The May 2013 monthly monitoring report was submitted on July 23, 2013, 23 days past the due date in violation of Table E-14.

E. Planned Changes

As noted in section II.A.2 of this Fact Sheet, the Discharger is proposing to construct the Santa Margarita GTP. The Discharger awarded the contract for construction of the Santa Margarita GTP in July of 2019. Construction is expected to take approximately 2 years and operations is expected to begin in July or August of 2021.

III. APPLICABLE PLANS, POLICIES, AND REGULATIONS

The requirements contained in this Order are based on the requirements and authorities described in this section.

A. Legal Authorities

This Order serves as WDRs pursuant to article 4, chapter 4, division 7 of the California Water Code (Water Code) (commencing with section 13260). This Order is also issued pursuant to section 402 of the federal Clean Water Act (CWA) and implementing regulations adopted by the U. S. Environmental Protection Agency (USEPA) and chapter 5.5, division 7 of the Water Code (commencing with section 13370). It shall serve as an NPDES permit authorizing the Discharger to discharge into waters of the United States at the discharge location described in Table 2 subject to the WDRs in this Order.

B. California Environmental Quality Act (CEQA)

Under Water Code section 13389, this action to adopt an NPDES permit is exempt from the provisions of chapter 3 of CEQA, (commencing with section 21100) of division 13 of the Public Resources Code.

C. State and Federal Laws, Regulations, Policies, and Plans

1. **Water Quality Control Plan.** The San Diego Water Board adopted a *Water Quality Control Plan for the San Diego Basin* (Basin Plan) on September 8, 1994 that designates beneficial uses, establishes water quality objectives, and contains implementation programs and policies to achieve those objectives for the Pacific Ocean and other receiving waters addressed through the plan. Subsequent revisions to the Basin Plan have also been adopted by the San Diego Water Board and approved by the State Water Board. Beneficial uses applicable to the Pacific Ocean specified in the Basin Plan are summarized in Table F-4-5 below:

Table F-4-5. Basin Plan Beneficial Uses

Discharge Point	Receiving Water Name	Beneficial Use(s)
001	Pacific Ocean	Industrial service supply (IND); navigation (NAV); water contact recreation (REC-1); non-contact recreation (REC-2); commercial and sport fishing (COMM); preservation of biological habitats of special significance (BIOL); wildlife habitat (WILD); rare, threatened, or endangered species (RARE); marine habitat (MAR); aquaculture (AQUA); migration of aquatic organisms (MIGR); spawning, reproduction, and/or early development (SPWN); and shellfish harvesting (SHELL).

In order to protect the beneficial uses, the Basin Plan establishes water quality objectives and a program of implementation. Requirements of this Order implement the Basin Plan.

2. **California Ocean Plan.** The State Water Board adopted the *Water Quality Control Plan for Ocean Waters of California, California Ocean Plan* (Ocean Plan) in 1972 and amended it in 1978, 1983, 1988, 1990, 1997, 2000, 2005, 2009, 2012, and 2015. The State Water Board adopted the latest amendment on August 7, 2018, the USEPA approved the amendments on March 22, 2019, and it became effective on March 22, 2019. The Ocean Plan is applicable, in its entirety, to point source discharges to the ocean. The Ocean Plan identifies

beneficial uses of ocean waters of the State to be protected as summarized in Table F-5-6 below:

Table F-5-6. Ocean Plan Beneficial Uses

Discharge Point	Receiving Water	Beneficial Uses
001	Pacific Ocean	IND; REC-1; REC-2, including aesthetic enjoyment; NAV; COMM; mariculture; preservation and enhancement of designated Areas of Special Biological Significance (ASBS); rare and endangered species; MAR; fish migration; fish spawning; and SHELL.

In order to protect the beneficial uses, the Ocean Plan establishes water quality objectives and a program of implementation. Requirements of this Order implement the Ocean Plan.

- Alaska Rule.** On March 30, 2000, USEPA revised its regulation that specifies when new and revised State and tribal water quality standards (WQS) become effective for CWA purposes (40 CFR section 131.21, 65 Fed. Reg. 24641 (April 27, 2000)). Under the revised regulation (also known as the Alaska Rule), new and revised standards submitted to USEPA after May 30, 2000, must be approved by USEPA before being used for CWA purposes. The final rule also provides that standards already in effect and submitted to USEPA by May 30, 2000, may be used for CWA purposes, whether or not approved by USEPA.
- Antidegradation Policy.** Federal regulation 40 CFR section 131.12 requires that the State water quality standards include an antidegradation policy consistent with the federal policy. The State Water Board established California's antidegradation policy in State Water Board Resolution 68-16, *Statement of Policy with Respect to Maintaining High Quality of Waters in California*. Resolution 68-16 is deemed to incorporate the federal antidegradation policy where the federal policy applies under federal law. Resolution 68-16 requires that existing water quality be maintained unless degradation is justified based on specific findings. The San Diego Water Board's Basin Plan implements, and incorporates by reference, both the State and federal antidegradation policies. The permitted discharge must be consistent with the antidegradation provision of 40 CFR section 131.12 and State Water Board Resolution 68-16.
- Anti-Backsliding Requirements.** Sections 402(o) and 303(d)(4) of the CWA and federal regulations at 40 CFR section 122.44(l) restrict backsliding in NPDES permits. These Anti-backsliding provisions require that effluent limitations in a reissued permit must be as stringent as those in the previous permit, with some exceptions in which limitations may be relaxed.
- Endangered Species Act Requirements.** This Order does not authorize any act that results in the taking of a threatened or endangered species or any act that is now prohibited, or becomes prohibited in the future, under either the California Endangered Species Act (Fish and Game Code, sections 2050 to 2097) or the federal Endangered Species Act (16 U.S.C.A. sections 1531 to 1544). This Order requires compliance with effluent limitations, receiving water limits, and other

requirements to protect the beneficial uses of waters of the State, including protecting rare and endangered species. The Discharger is responsible for meeting all requirements of the applicable Endangered Species Act.

7. **Sewage Sludge and Biosolids.** This Order does not authorize any act that results in violation of requirements administered by USEPA to implement 40 CFR part 503, *Standards for the Use or Disposal of Sewage Sludge*. These standards regulate the final use or disposal of sewage sludge that is generated during the treatment of domestic sewage in a municipal wastewater treatment facility. The Discharger is responsible for meeting all applicable requirements of 40 CFR part 503 that are under USEPA's enforcement authority.

D. Impaired Water Bodies on the CWA section 303(d) List

In August 2018, USEPA-approved the list of impaired water bodies, prepared by the State Water Board pursuant to CWA section 303(d), which are not expected to meet applicable water quality standards after implementation of technology-based effluent limitations (TBELs) for point sources. The 303(d) list for waters in the Pacific Ocean in the vicinity of the OOO include:

1. 0.3 miles of Pacific Ocean shoreline, San Luis Rey Hydrologic Unit (HU), at San Luis Rey River mouth for indicator bacteria (enterococcus for REC-1 and total coliform for SHELL);
2. 0.3 miles of Pacific Ocean shoreline, San Luis Rey HU, Oceanside Pier for trash; and
3. 0.3 miles of Pacific Ocean shoreline, Loma Alta hydrologic subarea (HSA), at Loma Alta Creek mouth for indicator bacteria and trash

Currently, there is no effective total maximum daily load (TMDL) to address the specific impairments listed above. The San Diego Water Board is addressing the water quality impairments and will develop appropriate regulatory actions for each impairing pollutant in each listed waterbody. These actions may include the adoption of a TMDL.

E. Other Plans, Policies, and Regulations

1. **Secondary Treatment Regulations.** Part 133 of 40 CFR establishes the minimum levels of effluent quality to be achieved by secondary treatment. These limitations, established by the USEPA, are incorporated into this Order, except where more stringent limitations are required by other applicable plans, policies, or regulations.
2. **Storm Water.** Pursuant to Order No 2014-0057-DWQ, NPDES Permit No. CAS000001, *General Permit for Storm Water Discharges Associated with Industrial Activities* (Storm Water Order), sewerage treatment plants are classified (per Occupational Safety and Health Administration) as Standard Industrial Classification (SIC) code 4952 or Sewerage Systems. SIC code 4952 (https://www.waterboards.ca.gov/water_issues/programs/stormwater/sicnum.shtml) falls within the Regulated SIC Code for enrollment under the Storm Water

Order. The eligibility for enrollment under the Order is not based on treatment design flow or capacity of the sewerage treatment plants. It is the industrial activity that is regulated. The Fallbrook WRP has the SIC code 4952 and is enrolled under the Storm Water Order.

IV. RATIONALE FOR EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS

The CWA requires point source dischargers to control the amount of conventional, non-conventional, and toxic pollutants that are discharged into the waters of the United States. The control of pollutants discharged is established through effluent limitations and other requirements in NPDES permits. There are two principal bases for effluent limitations in the Code of Federal Regulations: 40 CFR section 122.44(a) requires that permits include applicable technology-based limitations and standards; and 40 CFR section 122.44(d) requires that permits include water quality-based effluent limitations (WQBELs) to attain and maintain applicable numeric and narrative water quality criteria to protect the beneficial uses of the receiving water.

A. Discharge Prohibitions

This Order retains the discharge prohibitions from Order No. R9-2012-0004 as described below.

1. Order No. R9-2012-0004 contained Discharge Prohibition III.A, which clearly defined what types of discharges are prohibited. This prohibition has been retained in this Order as Discharge Prohibitions III.A and III.B.
2. Order No. R9-2012-0004 contained Discharge Prohibition III.B, which prohibits bypassing of untreated wastes, except as allowed by Order No. R9-2012-0004, Attachment D, federal Standard Provisions I.G or I.H of this Order. This prohibition has been retained as Discharge Prohibition III.D.
3. Order No. R9-2012-0004 contained Discharge Prohibition III.C, which prohibited the discharge from the Fallbrook WRP through the OOO at Discharge Point No. 001 during dry weather months (May to October) in excess of a monthly average effluent flow of 2.7 MGD, and during wet-weather months (November to April) in excess of a monthly average effluent flow of 3.6 MGD. Because the OOO flow prohibitions are now included as effluent limitations, these requirements are not retained in section III of this Order.
4. Order No. R9-2012-0004 contained Discharge Prohibitions III.D and III.E, which required compliance with the discharge prohibitions of the Ocean Plan and Basin Plan, respectively. These prohibitions have been retained in this Order as Discharge Prohibitions III.C and III.D.

B. Technology-Based Effluent Limitations (TBELs)

1. Scope and Authority

Section 301(b) of the CWA and implementing USEPA permit regulations at 40 CFR section 122.44 require that permits include conditions meeting applicable technology-based requirements at a minimum, and any more stringent effluent limitations necessary to meet applicable water quality standards. The discharge

from the Fallbrook WRP authorized by this Order must meet minimum federal technology-based requirements based on Secondary Treatment Standards at 40 CFR part 133. Discharges must also meet TBELs based on Ocean Plan Table 4.

Regulations promulgated in 40 CFR section 125.3(a)(1) require TBELs for municipal dischargers to be placed in NPDES permits based on *Secondary Treatment Standards or Equivalent to Secondary Treatment Standards*.

The CWA established the minimum performance requirements for POTWs [defined in section 304(d)(1)]. Section 301(b)(1)(B) of that Act requires that such treatment works must, as a minimum, meet effluent limitations based on secondary treatment as defined by the USEPA Administrator.

Based on this statutory requirement, USEPA developed secondary treatment regulations, which are specified in 40 CFR part 133. These technology-based regulations apply to all municipal wastewater treatment plants and identify the minimum level of effluent quality attainable by secondary treatment in terms of biochemical oxygen demand (BOD), TSS, and pH.

In compliance with 40 CFR sections 122.45(f)(1) and 423.15, mass-based limitations have also been established in this Order for conventional, nonconventional, and toxic pollutants, with some exceptions. Section 122.45(f)(2) of 40 CFR allows pollutants that are limited in terms of mass to additionally be limited in terms of other units of measurement. This Order includes effluent limitations expressed in terms of mass and concentration. In addition, pursuant to the exceptions to mass-based limitations provided in 40 CFR section 122.45(f)(1), some effluent limitations are not expressed in terms of mass, such as pH and temperature.

Mass-based effluent limitations were calculated using the following equation:
$$\text{lbs/day} = \text{flow (MGD)} \times \text{pollutant concentration (mg/L)} \times 8.34$$

2. Applicable Technology-Based Effluent Limitations

- a. **Federal Regulations.** Part 133 of 40 CFR establishes the minimum weekly and monthly average level of effluent quality attainable by secondary treatment for BOD and TSS. Section 133.102(a)(4) of 40 CFR allows for effluent limitations for CBOD₅ to be applied in lieu of effluent limitations for BOD where BOD may not provide a reliable measure of the oxygen demand of the effluent. USEPA has determined that a 30-day average effluent limitation of 25 mg/L and a 7-day average effluent limitation of 40 mg/L for CBOD₅ are effectively equivalent to the secondary treatment standards for BOD. Consistent with Order No. R9-2012-0004, this Order includes effluent limitations for CBOD₅.

Section 133.102 of 40 CFR, in describing the minimum level of effluent quality attainable by secondary treatment, states that the 30-day average percent removal of CBOD₅ and TSS shall not be less than 85 percent. Consistent with Order No. R9-2012-0004, this Order contains limitations

requiring an average of 85 percent removal of CBOD₅ and TSS over each calendar month.

The secondary treatment regulations at 40 CFR part 133 also require that pH be maintained between 6.0 and 9.0 standard units.

These TBELs based on secondary treatment standards are applicable to the Fallbrook WRP prior to commingling with any other wastewater. Thus, compliance with these effluent limitations must be determined at Monitoring Location M-001, upstream of the location where these wastewaters commingle with other wastewaters.

Section 122.45(d) of 40 CFR require that all permit limitations be expressed, unless impracticable, as average monthly effluent limitations (AMELs) and average weekly effluent limitations (AWELs) for wastewater facilities. TBELs based on secondary treatment standards for CBOD₅, TSS, and pH are summarized in Table F-6-7 below, applying AMELs in lieu of 30-day average and AWELs in lieu of 7-day average.

Table F-6-7. Summary of TBELs Based on Secondary Treatment Standards

Parameter	Units	Effluent Limitations			
		Average Monthly	Average Weekly	Instantaneous Minimum	Instantaneous Maximum
CBOD ₅	mg/L	25	40	--	--
	% Removal	≥85	--	--	--
TSS	mg/L	30	45	--	--
	% Removal	≥85	--	--	--
pH	standard units	--	--	6.0	9.0

- b. **Ocean Plan.** The Ocean Plan is applicable, in its entirety, to point source discharges to the ocean. Therefore, the discharge of wastewater to the Pacific Ocean at Discharge Point No. 001 is subject to the Ocean Plan.

The Ocean Plan establishes water quality objectives, general requirements for management of waste discharged to the ocean, effluent quality requirements for waste discharges, discharge prohibitions, and general provisions. Further, Table 4 of the Ocean Plan establishes TBELs for wastewater facilities and industrial facilities for which effluent limitation guidelines have not been established (including the waste brine from the Santa Margarita GTP). Consistent with Order No. R9-2012-0004, numeric effluent limitations based on Table 4 of the Ocean Plan are being established in this Order for each contributing facility, at Monitoring Location M-001 (Fallbrook WRP) and M-003 (Santa Margarita GTP).

Table A of the Ocean Plan requires dischargers to, as a monthly average, achieve a percent removal of 75 percent for suspended solids from the influent stream before discharging wastewater to the Pacific Ocean, except that the effluent limitation to be met shall not be less than 60 mg/L. Because

the Santa Margarita GTP is not a POTW, an effluent limitation of 60 mg/L is more appropriate and has been established for the waste brine for the Santa Margarita GTP.

Because secondary treatment standards contain effluent limitations for TSS that are more stringent than Table 4 of the Ocean Plan, the more stringent effluent limitations for TSS will be applied to discharges from the Fallbrook WRP.

The TBELs from the Ocean Plan are summarized in Table F-7-8 below:

Table F-7-8. Summary of TBELs on Table 4 of the Ocean Plan

Parameter	Units	Effluent Limitations		
		Average Monthly	Average Weekly	Instantaneous Maximum
Oil and Grease	mg/L	25	40	75
TSS	mg/L	60 ¹	--	--
	% Removal	1	--	--
Settleable Solids	ml/L	1.0	1.5	3.0
Turbidity	NTU	75	100	225
pH	standard units	Within the limits of 6.0 to 9.0 at all times		

¹ Table 4 of the Ocean Plan requires that the Discharger shall, as a monthly average, remove 75 percent of suspended solids from the influent stream before discharging wastewater to the Pacific Ocean, except that the effluent limitation to be met shall not be less than 60 mg/L.

- c. **Effluent Flow.** Order No. R9-2012-0004 contained a discharge prohibition which prohibited the discharge from the Fallbrook WRP through the OOO at Discharge Point No. 001 during dry-weather months (May to October) in excess of a monthly average effluent flow of 2.7 MGD, and during wet-weather months (November to April) in excess of a monthly average effluent flow of 3.6 MGD. As stated in section II.A.2 of this Fact Sheet, the total flow from the Fallbrook WRP and Santa Margarita GTP to the OOO will remain within the flow prohibitions contained in Order No. R9-2012-0004. These flow prohibitions are being carried over as effluent limitations for the total flow from the Discharger to the OOO.

C. Water Quality-Based Effluent Limitations (WQBELs)

1. Scope and Authority

CWA section 301(b) and 40 CFR section 122.44(d) require that permits include limitations more stringent than applicable federal technology-based requirements where necessary to achieve applicable water quality standards.

Section 122.44(d)(1)(i) of 40 CFR requires that permits include effluent limitations for all pollutants that are or may be discharged at levels that have the reasonable potential to cause or contribute to an exceedance of a water quality standard, including numeric and narrative objectives within a standard. Where reasonable potential has been established for a pollutant, but there is no numeric criterion or objective for the pollutant, WQBELs must be established using: (1) USEPA criteria guidance under CWA section 304(a), supplemented where necessary by

other relevant information; (2) an indicator parameter for the pollutant of concern; or (3) a calculated numeric water quality criterion, such as a proposed State criterion or policy interpreting the State's narrative criterion, supplemented with other relevant information, as provided in 40 CFR section 122.44(d)(1)(vi).

The process for determining reasonable potential and calculating WQBELs when necessary is intended to protect the designated uses of the receiving water as specified in the Basin Plan and Ocean Plan, and achieve applicable water quality objectives and criteria contained in other State plans and policies, or any applicable water quality criteria contained in the Ocean Plan.

2. **Applicable Beneficial Uses and Water Quality Criteria and Objectives**

The Basin Plan and Ocean Plan designate beneficial uses, establish water quality objectives, and contain implementation programs and policies to achieve those objectives for all waters.

- a. **Basin Plan.** The beneficial uses specified in the Basin Plan applicable to the Pacific Ocean are summarized in section III.C.1 of this Fact Sheet.

The Basin Plan water quality objective for dissolved oxygen applicable to ocean waters is stated as follows: "The dissolved oxygen concentration in ocean waters shall not at any time be depressed more than 10 percent from that which occurs naturally, as the result of the discharge of oxygen demanding waste materials."

The Basin Plan states, "The pH value shall not be changed at any time more than 0.2 pH units from that which occurs naturally."

- b. **Ocean Plan.** The beneficial uses specified in the Ocean Plan for the Pacific Ocean are summarized in section III.C.2 of this Fact Sheet. The Ocean Plan also includes water quality objectives for the ocean receiving water for bacterial characteristics, physical characteristics, chemical characteristics, biological characteristics, and radioactivity.

Table 3 of the Ocean Plan includes the following water quality objectives for toxic pollutants and whole effluent toxicity:

- i. Six-month median, daily maximum, and instantaneous maximum objectives for 19 chemicals and chemical characteristics, including total chlorine residual, for the protection of marine aquatic life.
- ii. 30-day average objectives for 20 non-carcinogenic chemicals for the protection of human health. These have been applied as AMELs.
- iii. 30-day average objectives for 42 carcinogenic chemicals for the protection of human health. These have been applied as AMELs.
- iv. Daily maximum objectives for acute and chronic toxicity.

3. Determining the Need for WQBELs

The San Diego Water Board evaluated the need for effluent limitations for non-conventional and toxic pollutant parameters, based on water quality objectives in Table 3 of the Ocean Plan. The evaluation was performed in accordance with 40 CFR section 122.44(d) and guidance for statistically determining the “reasonable potential” for a discharged pollutant to exceed an objective, as outlined in the revised *Technical Support Document for Water Quality-based Toxics Control* (TSD; EPA/505/2-90-001, 1991) and the Ocean Plan Reasonable Potential Analysis (RPA) Amendment that was adopted by the State Water Board on April 21, 2005. The statistical approach combines knowledge of effluent variability (as estimated by a coefficient of variation) with the uncertainty due to a limited amount of effluent data to estimate a maximum effluent value at a high level of confidence. This estimated maximum effluent value is based on a lognormal distribution of daily effluent values. Projected receiving water values (based on the estimated maximum effluent value or the reported maximum effluent value and minimum probable initial dilution) can then be compared to the appropriate objective to determine potential for an exceedance of that objective and the need for an effluent limitation. According to the Ocean Plan amendment, the RPA can yield three endpoints: 1) Endpoint 1, an effluent limitation is required and monitoring is required; 2) Endpoint 2, an effluent limitation is not required and the San Diego Water Board may require monitoring; and 3) Endpoint 3, the RPA is inconclusive, monitoring is required, and an existing effluent limitation may be retained or a permit reopener clause may be included to allow inclusion of an effluent limitation if future monitoring warrants the inclusion. Endpoint 3 is typically the result when there are fewer than 16 data points and all are censored data (i.e., below quantitation or method detection levels for an analytical procedure).

The implementation provisions for Table 3 of the Ocean Plan specify that the minimum initial dilution (Dm) is the lowest average initial dilution within any single month of the year. Dilution estimates are to be based on observed waste flow characteristics, observed receiving water density structure, and the assumption that no currents, of sufficient strength to influence the initial dilution process, flow across the discharge structure.

In 2005, the State Water Board had determined the Dm ratio for the OOO to be 87 parts seawater to 1 part wastewater (87:1). This determination was based on a total flow rate of 29.055 MGD from the Facilities; the City of Oceanside; the Marine Corps Base, Camp Pendleton; and Genentech, Inc. In the ROWD, the Discharger did not note any significant changes that would alter the previously determined dilution characteristics. Therefore, the previous Dm of 87 to 1 will be retained in this Order and applied to WQBELs established herein.

Conventional pollutants were not considered as part of the RPA. TBELs for these pollutants are included in this Order as described in section IV.B of this Fact Sheet.

This Order does not include effluent limitations for bacterial indicators for the following reasons:

- The discharge point (Discharge Point No. 001) is located at the terminus of the OOO, located 8,800 feet offshore at a depth of 100 feet.
- The dilution factor is 87.
- The San Diego Water Board is not aware of any shellfish harvesting within the zone of initial dilution of the OOO.
- There are no kelp beds within the zone of initial dilution of the OOO.
- The Discharger chlorinates/dechlorinates the secondary-treated wastewater from the Fallbrook WRP.

Using the RPcalc 2.0 software tool developed by the State Water Board for conducting reasonable potential analyses, the San Diego Water Board has conducted the RPA for the constituents listed in Table F-8-9 below. For constituents that do not display reasonable potential, this Order includes desirable maximum effluent concentrations which were derived using effluent limitation determination procedures described below and are referred to in this Order as “performance goals.” A narrative limit statement to comply with all Ocean Plan objectives requirements is provided for those parameters not displaying reasonable potential. The Discharger is required to monitor for these constituents as stated in the Monitoring and Reporting Program (MRP, Attachment E) of this Order in order to gather data for use in RPAs for future permit reissuances.

Effluent data provided in the Discharger’s monitoring reports for the Fallbrook WRP from September 2012 through July 2017 were used in the RPA. A Dm ratio of 87:1 was considered in this evaluation.

A summary of the RPA results is provided below:

Table F-8-9. RPA Results Summary¹

Parameter	Units	N ²	MEC ^{3,4}	Most Stringent Criteria	Background	RPA Endpoint ⁵
Arsenic, Total Recoverable	µg/L	9	<5	8 ⁶	3 ⁷	3
Cadmium, Total Recoverable	µg/L	9	<2	1 ⁶	0	3
Chromium, Total Recoverable	µg/L	9	3.6	2 ⁶	0	3
Copper, Total Recoverable	µg/L	9	16	3 ⁶	2 ⁷	2
Lead, Total Recoverable	µg/L	9	4.8	2 ⁶	0	3
Mercury, Total Recoverable	µg/L	9	1.1	0.04 ⁶	0.0005 ⁷	3
Nickel, Total Recoverable	µg/L	9	3.3	5 ⁶	0	3
Selenium, Total Recoverable	µg/L	9	11	15 ⁶	0	3
Silver, Total Recoverable	µg/L	9	<5	0.7 ⁶	0.16 ⁷	3
Zinc, Total Recoverable	µg/L	9	68	20 ⁶	8 ⁷	2
Cyanide	µg/L	9	18	1 ⁶	0	3
Total Residual Chlorine	µg/L	1370	9200	2 ⁶	0	1
Ammonia Nitrogen, Total (as N)	µg/L	58	41000	600 ⁶	0	2
Acute Toxicity	TUa	9	0.69	0.3 ⁸	0	3
Chronic Toxicity	TUc	29	25	18	0	2

FALLBROOK PUBLIC UTILITY DISTRICT
 FALLBROOK WATER RECLAMATION PLANT
 SANTA MARGARITA GROUNDWATER TREATMENT PLANT

ORDER NO. R9-2019-0169
 Amended by Order No. R9-2020-0193 and R9-2021-0204
 NPDES NO. CA0108031

Parameter	Units	N ²	MEC ^{3,4}	Most Stringent Criteria	Background	RPA Endpoint ⁵
Phenolic Compounds (non-chlorinated) ¹	µg/L	10	<7.9	30 ⁶	0	3
Chlorinated Phenolics ¹	µg/L	10	<3.0	1 ⁶	0	3
Endosulfan ¹	µg/L	9	0.0085	0.009 ⁶	0	3
Endrin	µg/L	9	<0.002	0.002 ⁶	0	3
HCH (BHC) ¹	µg/L	9	0.04	0.004 ⁶	0	2
Radioactivity	pci/L	9	18	⁹	0	¹³
Acrolein	µg/L	4	<2.5	220 ¹⁰	0	3
Antimony	µg/L	4	<0.006	1,200 ¹⁰	0	3
Bis(2-chloroethoxyl)methane	µg/L	4	<0.097	4.4 ¹⁰	0	3
Bis(2-chloroisopropyl)ether	µg/L	4	<0.097	1,200 ¹⁰	0	3
Chlorobenzene	µg/L	4	<0.25	570 ¹⁰	0	3
Chromium (III), Total Recoverable	µg/L	4	<0.0025	190,000 ¹⁰	0	3
Di-n-butyl phthalate	µg/L	4	<0.079	3,500 ¹⁰	0	3
Dichlorobenzenes ¹	µg/L	4	<0.5	5,100 ¹⁰	0	3
Diethyl phthalate	µg/L	4	0.67	33,000 ¹⁰	0	3
Dimethyl phthalate	µg/L	4	<0.28	820,000 ¹⁰	0	3
4,6-Dinitro-2-methylphenol	µg/L	10	<0.79	220 ¹⁰	0	3
2,4-Dinitrophenol	µg/L	10	<2.2	4.0 ¹⁰	0	3
Ethylbenzene	µg/L	4	<0.25	4,100 ¹⁰	0	3
Fluoranthene	µg/L	4	<0.23	15 ¹⁰	0	3
Hexachlorocyclopentadiene	µg/L	4	<0.26	58 ¹⁰	0	3
Nitrobenzene	µg/L	4	<0.26	4.9 ¹⁰	0	3
Thallium, Total Recoverable	µg/L	5	<0.005	2 ¹⁰	0	3
Toluene	µg/L	4	<0.25	85,000 ¹⁰	0	3
Tributyltin	µg/L	4	<0.013	0.0014 ¹⁰	0	3
1,1,1-Trichloroethane	µg/L	4	<0.25	540,000 ¹⁰	0	3
Acrylonitrile	µg/L	4	<1	0.10 ¹⁰	0	3
Aldrin	µg/L	4	<0.0016	0.000022 ¹⁰	0	3
Benzene	µg/L	4	<0.25	5.9 ¹⁰	0	3
Benzidine	µg/L	4	<2.6	0.000069 ¹⁰	0	3
Beryllium	µg/L	4	<0.001	0.033 ¹⁰	0	3
Bis(2-chloroethyl) ether	µg/L	4	<0.23	0.045 ¹⁰	0	3
Bis(2-ethylhexyl) phthalate	µg/L	6	51	3.5 ¹⁰	0	2
Carbon tetrachloride	µg/L	4	<0.25	0.90 ¹⁰	0	3
Chlordane ¹	µg/L	4	<0.08	0.000023 ¹⁰	0	3
Chlorodibromomethane (Dibromochloromethane) ¹	µg/L	4	1.5	8.6 ¹⁰	0	3
Chloroform	µg/L	4	54	130 ¹⁰	0	2
Dichlorodiphenyltrichloroethane (DDT) ¹	µg/L	4	<0.051	0.00017 ¹⁰	0	3
1,4-Dichlorobenzene	µg/L	4	<0.25	18 ¹⁰	0	3
3,3-Dichlorobenzidine	µg/L	4	<1.3	0.0081 ¹⁰	0	3
1,2-Dichloroethane	µg/L	4	<0.25	28 ¹⁰	0	3
1,1-Dichloroethylene	µg/L	4	<0.25	0.9 ¹⁰	0	3
Dichlorobromomethane	µg/L	6	21	6.2 ¹⁰	0	2
Dichloromethane (Methylene Chloride)	µg/L	4	1	450 ¹⁰	0	3
1,3-Dichloropropene (1,3-Dichloropropylenes)	µg/L	4	<0.22	8.9 ¹⁰	0	3

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ORDER NO. R9-2019-0169
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NPDES NO. CA0108031

Parameter	Units	N ²	MEC ^{3,4}	Most Stringent Criteria	Background	RPA Endpoint ⁵
Dieldrin	µg/L	4	<0.0022	0.00004 ¹⁰	0	3
2,4-Dinitrotoluene	µg/L	4	<0.053	2.6 ¹⁰	0	3
1,2-Diphenylhydrazine	µg/L	4	<0.5	0.16 ¹⁰	0	3
Halomethanes ¹	µg/L	4	0.45	130 ¹⁰	0	3
Heptachlor	µg/L	7	0.11	0.00005 ¹⁰	0	1
Heptachlor Epoxide	µg/L	7	0.027	0.00002 ¹⁰	0	1
Hexachlorobenzene	µg/L	4	<0.26	0.00021 ¹⁰	0	3
Hexachlorobutadiene	µg/L	4	<0.5	14 ¹⁰	0	3
Hexachloroethane	µg/L	4	<0.5	2.5 ¹⁰	0	3
Isophorone	µg/L	4	<0.26	730 ¹⁰	0	3
N-nitrosodimethylamine	µg/L	4	<0.26	7.3 ¹⁰	0	3
N-nitrosodi-N-propylamine	µg/L	4	<0.26	0.38 ¹⁰	0	3
N-nitrosodiphenylamine	µg/L	4	<0.26	2.5 ¹⁰	0	3
Polynuclear Aromatic Hydrocarbons (PAHs) ¹	µg/L	4	<3.8	0.0088 ¹⁰	0	3
Polychlorinated Biphenyls (PCBs) ¹	µg/L	4	<1.9	0.000019 ¹⁰	0	3
TCDD equivalents ¹	µg/L	13	7.3E-07	0.0000000039 ¹⁰	0	3
1,1,2,2-Tetrachloroethane	µg/L	4	<0.25	2.3 ¹⁰	0	3
Tetrachloroethylene (Tetrachloroethene)	µg/L	4	<0.25	2.0 ¹⁰	0	3
Toxaphene	µg/L	4	<0.25	0.00021 ¹⁰	0	3
Trichloroethylene (Trichloroethene)	µg/L	4	<0.25	27 ¹⁰	0	3
1,1,2-Trichloroethane	µg/L	4	<0.25	9.4 ¹⁰	0	3
2,4,6-Trichlorophenol	µg/L	10	<0.26	0.29 ¹⁰	0	3
Vinyl Chloride	µg/L	4	<0.25	36 ¹⁰	0	3

¹ See Attachment A for definitions of abbreviations and a glossary of common terms used in this Order.

² Number of data points available for the RPA.

³ If there is a detected value, the highest reported value is summarized in the table. If there are no detected values, the lowest MDL is summarized in the table.

⁴ Note that the reported MEC does not account for dilution. The RPA does account for dilution; therefore, it is possible for a parameter with an MEC in exceedance of the most stringent criteria not to present a reasonable potential (i.e. Endpoint 2).

⁵ End Point 1 – Reasonable potential determined, limit required, monitoring required.

End Point 2 – Discharger determined not to have reasonable potential, monitoring may be established.

End Point 3 – RPA was inconclusive, carry over previous limitations if applicable, and establish monitoring.

⁶ Based on the Six-Month Median in the Table 3 of the Ocean Plan.

⁷ Background concentrations contained in Table 5 of the Ocean Plan.

⁸ Based on the Daily Maximum in Table 3 of the Ocean Plan.

⁹ Not to exceed limits specified in title 17, division 1, chapter 5, subchapter 4, group 3, article 3, section 30253 of the CCR. Levels of radioactivity that exceed the applicable criteria are not expected in the discharge.

¹⁰ Based on 30-Day Average in Table 3 of the Ocean Plan.

Consistent with 40 CFR sections 303(d)(4) and 402(o), effluent limitations from Order No. R9-2012-0004 will not be retained for constituents for which reasonable potential has not been determined. Instead, performance goals have been assigned for these constituents. Parameters for which Endpoint 2 was concluded are determined not to have reasonable potential, thus it is inappropriate to establish effluent limitations for these parameters.

For parameters for which Endpoint 3 was concluded, reasonable potential was inconclusive. For parameters for which Endpoint 3 was concluded and previous effluent limitations had not been established in Order No. R9-2012-0004, performance goals have been retained. For parameters for which Endpoint 3 was concluded and previous effluent limitations had been established in Order No. R9-2012-0004, effluent limitations have been retained. Endpoint 3 was determined for TCDD equivalents. Effluent limitations for these parameters have been carried over from Order No. R9-2012-0004.

Reasonable potential to cause or contribute to an exceedance of water quality objectives contained within the Ocean Plan (i.e., Endpoint 1) was determined for total chlorine residual, heptachlor, and heptachlor epoxide. Effluent limitations for total chlorine residual have been retained from Order No. R9-2012-0004. Effluent limitations for heptachlor, and heptachlor epoxide (based on the initial dilution of 87:1, as discussed below) have been established in this Order.

The MRP (Attachment E) is designed to obtain additional information for these constituents to determine if reasonable potential exists for these constituents in future permit renewals and/or updates.

4. WQBEL Calculations

- a. From the Table 3 water quality objectives of the Ocean Plan, effluent limitations and performance goals are calculated according to the following equation for all pollutants, except for acute toxicity (if applicable) and radioactivity:

$$C_e = C_o + D_m (C_o - C_s) \text{ where,}$$

C_e = the effluent limitation ($\mu\text{g/L}$)

C_o = the water quality objective to be met at the completion of initial dilution ($\mu\text{g/L}$)

C_s = background seawater concentration

D_m = minimum probable initial dilution expressed as parts seawater per part wastewater

- b. Initial dilution (D_m) has been determined to be 87 to 1 by the San Diego Water Board through the application of USEPA's dilution model, Visual Plumes.
- c. Table 5 of the Ocean Plan establishes background concentrations for some pollutants to be used when determining reasonable potential (represented as " C_s "). In accordance with Table 3 implementing procedures, C_s equals zero for all pollutants not established in Table 5. The background concentrations provided in Table 5 are summarized in Table F-9-10 below:

Table F-9-10. Pollutants Having Background Concentrations¹

Pollutant	Background Seawater Concentration
Arsenic, Total Recoverable	3 $\mu\text{g/L}$
Copper, Total Recoverable	2 $\mu\text{g/L}$
Mercury, Total Recoverable	0.0005 $\mu\text{g/L}$

Silver, Total Recoverable	0.16 µg/L
Zinc, Total Recoverable	8 µg/L

¹ See Attachment A for definitions of abbreviations and a glossary of common terms used in this Order.

- d. As an example, effluent limitations for total chlorine residual are determined as follows:

Water quality objectives from the Ocean Plan for total chlorine residual are:

Table F-~~10-11~~. Example Parameter Water Quality Objectives¹

Parameter	Units	Six-Month Median	Daily Maximum	Instantaneous Maximum
Total Chlorine Residual	µg/L	2	8	60

¹ See Attachment A for definitions of abbreviations and a glossary of common terms used in this Order.

Using the equation, $C_e = C_o + D_m (C_o - C_s)$, effluent limitations/performance goals are calculated as follows.

Total Chlorine Residual:

$$C_e = 2 + 87 (2 - 0) = 176 \text{ (Six-Month Median)}$$

$$C_e = 8 + 87 (8 - 0) = 704 \text{ (Daily Maximum)}$$

$$C_e = 60 + 87 (60 - 0) = 5,280 \text{ (Instantaneous Maximum)}$$

Based on the implementing procedures described above, effluent limitations and performance goals have been calculated for all parameters in Table 3 of the Ocean Plan and incorporated into this Order.

- e. Section 122.45(f)(1) of the 40 CFR requires effluent limitations be expressed in terms of mass, with some exceptions, and 40 CFR section 122.45(f)(2) allows pollutants that are limited in terms of mass to additionally be limited in terms of other units of measurement. However, section III.C.4.j of the Ocean Plan requires that mass limitations be established for all parameters in Table 3 of the Ocean Plan. This Order includes effluent limitations expressed in terms of mass and concentration. In addition, pursuant to the exceptions to mass limitations provided in 40 CFR section 122.45(f)(1), some effluent limitations are not expressed in terms of mass, such as pH and temperature, and when the applicable standards are expressed in terms of concentration (e.g., CTR criteria and MCLs) and mass limitations are not necessary to protect the beneficial uses of the receiving water.

Mass-based effluent limitations were calculated using the following equation:
 lbs/day = permitted flow (MGD) x pollutant concentration (mg/L) x 8.34

- f. Based on the results of the RPA and BPJ, a summary of the WQBELs established in this Order are provided below:

Table F-44.12. Summary of WQBELs at Monitoring Location M-002¹

Parameter	Unit	Effluent Limitations ^{2,3}			
		Six-Month Median	Average Monthly	Maximum Daily	Instantaneous Maximum
OBJECTIVES FOR PROTECTION OF MARINE AQUATIC LIFE					
Total Chlorine Residual ²	µg/L	1.8E+02	--	7.0E+02	5.3E+03
	lbs/day	4.0E+00	--	1.6E+01	1.2E+02
OBJECTIVES FOR PROTECTION OF HUMAN HEALTH - CARCINOGENS					
Heptachlor	µg/L	--	4.4E-03	--	--
	lbs/day	--	9.9E-05	--	--
Heptachlor Epoxide	µg/L	--	1.8E-03	--	--
	lbs/day	--	4.0E-05	--	--
TCDD Equivalents ¹	µg/L	--	3.4E-07	--	--
	lbs/day	--	7.7E-09	--	--

¹ See Attachment A for definitions of abbreviations and a glossary of common terms used in this Order.

² Scientific "E" notation is used to express certain values. In scientific "E" notation, the number following the "E" indicates that position of the decimal point in the value. Negative numbers after the "E" indicate that the value is less than 1, and positive numbers after the "E" indicate that the value is greater than 1. In this notation a value of 6.1 E-02 represents 6.1 x 10⁻² or 0.061, 6.1E+02 represents 6.1 x 10² or 610, and 6.1E+00 represents 6.1 x 10⁰ or 6.1.

³ The mass emission rate (MER) limitation, in lbs/day, was calculated based on the following equation: MER (lbs/day) = 8.34 x Q x C, where Q is the permitted flow for the Facilities (2.7 MGD) and C is the concentration (mg/L).

⁴ ~~As specified in section VII.L of this Order and section III.C of the MRP (Attachment E).~~

⁵ ~~The chronic toxicity effluent limitation is protective of both the numeric acute and chronic toxicity Ocean Plan water quality objectives. The effluent limitation will be implemented using *Short term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to West Coast Marine and Estuarine Organisms* (EPA/600/R-95/136, 1995); current *USEPA guidance in the National Pollutant Discharge Elimination System Test of Significant Toxicity Implementation Document* (EPA 833-R-10-003, June 2010) (https://www3.epa.gov/npdcs/pubs/wet_final_tst_implementation2010.pdf); and USEPA Regions 8, 9, and 10, *Toxicity Training Tool* (January 2010).~~

- g. Constituents that do not have reasonable potential to cause or contribute to an exceedance of water quality objectives have been assigned performance goals. Pursuant to Water Code section 13383, this Order includes monitoring and reporting requirements related to performance goals. Performance goals serve to maintain existing treatment levels and effluent quality and supports state and federal antidegradation policies. Additionally, performance goals provide all interested parties with information regarding the expected level of pollutants in the discharge that should not be exceeded in order to maintain the water quality objectives established in the Ocean Plan and the overall pollutant control performance of the Facility. Performance goals are not limitations or standards for the regulation of the discharge. Effluent concentrations above the performance goals will not be considered as violations of this Order but serve as indicators that the effluent may be causing or contributing to an exceedance of water quality objectives. However, any two consecutive exceedances of the performance goals will trigger an investigation into the cause of the exceedance. If the exceedance persists in three successive monitoring events, the Discharger is required submit a written Performance Goal Exceedance Report to the San Diego

Water Board within 30 days on the nature of the exceedance, the results of the investigation as to the cause of the exceedance, and the corrective actions taken or proposed corrective measures with timetable for implementation, if necessary. Repeated exceedances of performance goals may prompt the San Diego Water Board to reopen and amend this Order to replace performance goals for constituents of concern with effluent limitations, or the San Diego Water Board may coordinate such actions with the next permit reissuance. The Discharger is in violation of this Order if it does not comply with the performance goal investigation and reporting requirements, when required by the terms of this Order. A summary of the performance goals is provided below.

Table F-42.13. Summary of Performance Goals for Monitoring Location M-002¹

Parameter	Units	Performance Goals ²			
		Six-Month Median	Average Monthly	Maximum Daily	Instantaneous Maximum
OBJECTIVES FOR PROTECTION OF MARINE AQUATIC LIFE					
Arsenic, Total Recoverable	µg/L	4.43E+02	--	2.56E+03	6.78E+03
Cadmium, Total Recoverable	µg/L	8.80E+01	--	3.52E+02	8.80E+02
Chromium (VI), Total Recoverable ³	µg/L	1.76E+02	--	7.04E+02	1.76E+03
Copper, Total Recoverable	µg/L	9.00E+01	--	8.82E+02	2.47E+03
Lead, Total Recoverable	µg/L	1.76E+02	--	7.04E+02	1.76E+03
Mercury, Total Recoverable	µg/L	3.48E+00	--	1.40E+01	3.52E+01
Nickel, Total Recoverable	µg/L	4.40E+02	--	1.76E+03	4.40E+03
Selenium, Total Recoverable	µg/L	1.32E+03	--	5.28E+03	1.32E+04
Silver, Total Recoverable	µg/L	4.77E+01	--	2.32E+02	6.02E+02
Zinc, Total Recoverable	µg/L	1.06E+03	--	6.34E+03	1.69E+04
Cyanide, Total	µg/L	8.80E+01	--	3.52E+02	8.80E+02
Ammonia Nitrogen, Total (as N)	µg/L	5.28E+04	--	2.11E+05	5.28E+05
Chronic Toxicity (Test of Significant Toxicity) ^{4,5}	<u>"Pass"/ "Fail" Toxicity Units Chronic (TUc)</u>	--	--	<u>"Pass"8.8E+01</u>	--
Phenolic Compounds (non-chlorinated) ¹	µg/L	2.64E+03	--	1.06E+04	2.64E+04
Chlorinated Phenolics ¹	µg/L	8.80E+01	--	3.52E+02	8.80E+02
Endosulfan ¹	µg/L	7.92E-01	--	1.58E+00	2.38E+00
Endrin	µg/L	1.76E-01	--	3.52E-01	5.28E-01
HCH (BHC) ¹	µg/L	3.52E-01	--	7.04E-01	1.06E+00
Radioactivity	picocuries per liter (pCi/L)	Not to exceed limits specified in title 17, division 1, chapter 5, subchapter 4, group 3, article 3, section 30253 of the CCR, reference to section 30253 is prospective, including future changes to any incorporated provisions of federal law, as the changes take effect.			
OBJECTIVES FOR PROTECTION OF HUMAN HEALTH – NONCARCINOGENS					
Acrolein	µg/L	--	1.94E+04	--	--
Antimony	µg/L	--	1.06E+05	--	--

FALLBROOK PUBLIC UTILITY DISTRICT
 FALLBROOK WATER RECLAMATION PLANT
 SANTA MARGARITA GROUNDWATER TREATMENT PLANT

ORDER NO. R9-2019-0169
 Amended by Order No. R9-2020-0193 and R9-2021-0204
 NPDES NO. CA0108031

Parameter	Units	Performance Goals ²			
		Six-Month Median	Average Monthly	Maximum Daily	Instantaneous Maximum
Bis(2-chloroethoxy) Methane	µg/L	--	3.87E+02	--	--
Bis(2-chloroisopropyl) Ether	µg/L	--	1.06E+05	--	--
Chlorobenzene	µg/L	--	5.02E+04	--	--
Chromium (III), Total Recoverable ³	µg/L	--	1.67E+07	--	--
Di-n-butyl Phthalate	µg/L	--	3.08E+05	--	--
Dichlorobenzenes ¹	µg/L	--	4.49E+05	--	--
Diethyl Phthalate	µg/L	--	2.90E+06	--	--
Dimethyl Phthalate	µg/L	--	7.22E+07	--	--
4,6-dinitro-2-methylphenol	µg/L	--	1.94E+04	--	--
2,4-dinitrophenol	µg/L	--	3.52E+02	--	--
Ethylbenzene	µg/L	--	3.61E+05	--	--
Fluoranthene	µg/L	--	1.32E+03	--	--
Hexachlorocyclopentadiene	µg/L	--	5.10E+03	--	--
Nitrobenzene	µg/L	--	4.31E+02	--	--
Thallium, Total Recoverable	µg/L	--	1.76E+02	--	--
Toluene	µg/L	--	7.48E+06	--	--
Tributyltin	µg/L	--	1.23E-01	--	--
1,1,1-trichloroethane	µg/L	--	4.75E+07	--	--
OBJECTIVES FOR PROTECTION OF HUMAN HEALTH – CARCINOGENS					
Acrylonitrile	µg/L	--	8.80E+00	--	--
Aldrin	µg/L	--	1.94E-03	--	--
Benzene	µg/L	--	5.19E+02	--	--
Benzidine	µg/L	--	6.07E-03	--	--
Bis(2-chloroethyl) Ether	µg/L	--	3.96E+00	--	--
Bis(2-ethylhexyl) Phthalate	µg/L	--	3.1E+02	--	--
Carbon Tetrachloride	µg/L	--	7.92E+01	--	--
Chlordane ¹	µg/L	--	2.02E-03	--	--
Chlorodibromomethane (Dibromochloromethane)	µg/L	--	7.57E+02	--	--
Chloroform	µg/L	--	1.14E+04	--	--
Dichlorodiphenyltrichloroethane (DDT) ¹	µg/L	--	1.50E-02	--	--
1,4-dichlorobenzene	µg/L	--	1.58E+03	--	--
3,3'-dichlorobenzidine	µg/L	--	7.13E-01	--	--
1,2-dichloroethane	µg/L	--	2.46E+03	--	--
1,1-dichloroethylene	µg/L	--	7.92E+01	--	--
Dichlorobromomethane	µg/L	--	5.5E+02	--	--
Dichloromethane (Methylene Chloride)	µg/L	--	3.96E+04	--	--
1,3-dichloropropene (1,3-Dichloropropylenes)	µg/L	--	7.83E+02	--	--
Dieldrin	µg/L	--	3.52E-03	--	--

Parameter	Units	Performance Goals ²			
		Six-Month Median	Average Monthly	Maximum Daily	Instantaneous Maximum
2,4-dinitrotoluene	µg/L	--	2.29E+02	--	--
1,2-diphenylhydrazine	µg/L	--	1.41E+01	--	--
Halomethanes ¹	µg/L	--	1.14E+04	--	--
Hexachlorobutadiene	µg/L	--	1.23E+03	--	--
Hexachloroethane	µg/L	--	2.20E+02	--	--
Isophorone	µg/L	--	6.42E+04	--	--
N-nitrosodimethylamine	µg/L	--	6.42E+02	--	--
N-nitrosodi-N-propylamine	µg/L	--	3.34E+01	--	--
N-nitrosodiphenylamine	µg/L	--	2.20E+02	--	--
PAHs ¹	µg/L	--	7.74E-01	--	--
PCBs ¹	µg/L	--	1.67E-03	--	--
1,1,2,2-tetrachloroethane	µg/L	--	2.02E+02	--	--
Tetrachloroethylene (Tetrachloroethene)	µg/L	--	1.76E+02	--	--
Toxaphene	µg/L	--	1.85E-02	--	--
Trichloroethylene (Trichloroethene)	µg/L	--	2.38E+03	--	--
1,1,2-trichloroethane	µg/L	--	8.27E+02	--	--
2,4,6-trichlorophenol	µg/L	--	2.55E+01	--	--
Vinyl Chloride	µg/L	--	3.17E+03	--	--

¹ See Attachment A for definitions of abbreviations and a glossary of common terms used in this Order.

² Scientific "E" notation is used to express certain values. In scientific "E" notation, the number following the "E" indicates that position of the decimal point in the value. Negative numbers after the "E" indicate that the value is less than 1, and positive numbers after the "E" indicate that the value is greater than 1. In this notation a value of 6.1 E-02 represents 6.1 x 10⁻² or 0.061, 6.1E+02 represents 6.1 x 10² or 610, and 6.1E+00 represents 6.1 x 10⁰ or 6.1.

³ Dischargers may, at their option, apply this performance goal as a total chromium performance goal.

⁴ As specified in section VII.L of this Order and section III.C of the MRP (Attachment E).

³⁵ The chronic toxicity performance goal is protective of both the numeric acute and chronic toxicity Ocean Plan water quality objectives. The performance goal will be implemented using *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to West Coast Marine and Estuarine Organisms* (EPA/600/R-95/136, 1995). Chronic toxicity expressed as Toxic Units Chronic (TUC) = 100/NOEC, where NOEC (No Observed Effect Concentration; also referred to as the No Observed Effect Level or NOEL) is expressed as the maximum percent effluent or receiving water that causes no observable effect on a test organism.

5. Whole Effluent Toxicity (WET)

- The WET testing protects receiving waters from the aggregate toxic effect of a mixture of pollutants in the effluent. Because of the nature of discharges into the wastewater facilities, it is possible that toxic constituents could be present in the Fallbrook WRP effluent or could have synergistic or additive effects.
- For chronic toxicity, Order No. R9-2012-0004 established a performance goal of 88 TUC and quarterly monitoring. During the term of Order No. R9-2012-0004, the reported effluent chronic toxicity was 25 TUC. This Order reestablishes a performance goal and maintains quarterly monitoring for

chronic toxicity. Amending Order No. R9-2021-0204 reduces the monitoring frequency to semiannual. ~~This Order reduces the minimum sampling frequency for chronic toxicity upon request if ten consecutive routine monitoring events are all in compliance with the chronic toxicity effluent limitations.~~

~~For this Order, chronic toxicity in the discharge is evaluated using USEPA's 2010 Test of Significant Toxicity (TST) hypothesis testing approach at the discharge "in-stream" waste concentration (IWC), as described in section VII.L of this Order and section III.C of the MRP (Attachment E). The TST statistical approach is described in the *National Pollutant Discharge Elimination System Test of Significant Toxicity Implementation Document* (EPA 833-R-10-003, 2010), Appendix A, Figure A-1 and Table A-1. The TST null hypothesis shall be "mean discharge IWC response $\leq 0.75 \times$ mean control response." A test that rejects this null hypothesis shall be reported as "Pass." A test that does not reject this null hypothesis shall be reported as "Fail." The chronic toxicity performance goal is expressed as "Pass" for each maximum daily individual result. The Discharger shall also report the "Percent Effect" as part of chronic toxicity result.~~

This Order contains a reopener to require the San Diego Water Board to modify the performance goal for toxicity, if necessary, to make it consistent with any new policy, law, or regulation.

- c. For acute toxicity, Order No. R9-2012-0004 established performance goals and no effluent monitoring. An acute toxicity test is conducted over a short time period and measures mortality. A chronic toxicity test is conducted over a short or a longer period of time and may measure mortality, reproduction, and growth. A chemical at a low concentration could have chronic effects but no acute effects until the chemical was at a higher concentration. Thus, chronic toxicity is a more stringent requirement than acute toxicity. To ensure the aggregated impacts of pollutants present within the Discharger's effluent does not result in the presence of toxicity within the receiving water, this Order maintains the performance goal for chronic toxicity. This Order removes acute toxicity performance goals. Removal of numeric acute toxicity performance goals does not constitute backsliding because chronic toxicity is a more stringent requirement than acute toxicity.
- d. ~~Section III.F of the Ocean Plan provides for more stringent requirements if necessary to protect the designated beneficial uses of ocean waters. Diamond et al. (2013) examined the side-by-side comparison of No Observed Effect Concentration (NOEC) and TST results using California chronic toxicity test data (including data from POTWs) for the West Coast marine methods and test species required under this Order. See Table 1 (method types 1 through 5) on page 1103 in Diamond J., Denton D., Roberts J., Zheng L. 2013. *Evaluation of the Test of Significant Toxicity for Determining the Toxicity of Effluents and Ambient Water Samples*. Environ Toxicol Chem 32:1101-1108. This comparison shows that while the TST and~~

~~NOEC statistical approaches perform similarly most of the time, the TST performs better in identifying toxic and nontoxic samples, a desirable characteristic for chronic toxicity testing conducted under this Order. This examination also signals that the test methods' false positive rate (β no higher than 0.05 at a mean effect of 10%) and false negative rate (α no higher than 0.05 (0.25 for topmelt) at a mean effect of 25%) are indeed low. This highlights that using the TST in this Order – in conjunction with other Ocean Plan requirements (West Coast WET method/test species for monitoring and limiting chronic toxicity, the IWC representing the critical condition for water quality protection, the initial dilution procedure, and a single test for compliance) – provides increased assurance that statistical error rates are more directly addressed and accounted for in decisions regarding chronic toxicity in the discharge. As a result, and in accordance with Ocean Plan section III.F, the San Diego Water Board is exercising its discretion to use the TST statistical approach for this discharge.~~

~~In January 2010, USEPA published a guidance document entitled: *USEPA Regions 8, 9 and 10 Toxicity Training Tool*, which among other things discusses permit limitation expression for chronic toxicity. The document acknowledges that NPDES regulations at 40 CFR section 122.45(d) require that all permit limits be expressed, unless impracticable, as an AWEL and AMEL for POTWs. Following section 5.2.3 of the Technical Support Document (TSD), the use of an AWEL and AMEL are not appropriate for WET. In lieu of an AWEL and AMEL for POTWs, USEPA recommends establishing a maximum daily effluent limitation (MDEL) for toxic pollutants and pollutants in water quality permitting, including WET. This is appropriate for two reasons. The basis for the average weekly and average monthly requirements for POTWs derives from secondary treatment regulations and is not related to the requirement to ensure achievement of water quality standard. Moreover, an average weekly and monthly requirement comprising up to seven and 31 daily samples, respectively, could average out daily peak toxic concentrations for WET and, therefore, the discharge's potential for causing acute and chronic effects would be missed. It is impracticable to use an AWEL and AMEL, because short-term spikes of toxicity levels that would be permissible under the 7-day and 31-day average scheme, respectively, would not be adequately protective of all beneficial uses. The MDEL is the highest allowable value for the discharge measured during a calendar day or 24-hour period representing a calendar day. Thus, this Order includes maximum daily performance goals rather than average monthly or average weekly performance goal.~~

~~Later in June 2010, USEPA published another guidance document titled, *National Pollutant Discharge Elimination System Test of Significant Toxicity Implementation Document* (EPA 833-R-10-003, June 2010), in which they recommend the following: "Permitting authorities should consider adding the TST approach to their implementation procedures for analyzing valid WET data for their current NPDES WET Program." The TST approach is another~~

~~statistical option for analyzing valid WET test data. Use of the TST approach does not result in any changes to USEPA's WET test methods. Section 9.4.1.2 of USEPA's *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to West Coast Marine and Estuarine Organisms* (EPA/600/R-95/136, 1995), current USEPA *Guidance in the National Pollutant Discharge Elimination System Test of Significant Toxicity Implementation Document* (EPA-833-R-10-003, June 2010), recognizes that, "the statistical methods in this manual are not the only possible methods of statistical analysis." The TST approach can be applied to acute (survival) and chronic (sublethal) endpoints and is appropriate to use for both freshwater and marine USEPA WET test methods.~~

~~The USEPA's WET testing program and acute and chronic WET methods rely on the measurement result for a specific test endpoint, not upon achievement of specified concentration-response patterns to determine toxicity. USEPA's WET methods do not require achievement of specified effluent or ambient concentration-response patterns prior to determining that toxicity is present.⁴ Nevertheless, USEPA's acute and chronic WET methods require that effluent and ambient concentration-response patterns generated for multi-concentration acute and chronic toxicity tests be reviewed—as a component of test review following statistical analysis—to ensure that the calculated measurement result for the toxicity test is interpreted appropriately. (EPA-821-R-02-012, section 12.2.6.2; EPA-821-R-02-013, section 10.2.6.2). In 2000, USEPA provided guidance for such reviews to ensure that test endpoints for determining toxicity based on the statistical approaches utilized at the time the guidance was written (NOEC), percent waste giving 50 percent survival of test organisms (lethal concentration 50, LC 50), effects concentration at 25 percent (EC25) were calculated appropriately (EPA-821-B-00-004).~~

~~USEPA designed its 2000 guidance as a standardized step-by-step review process that investigates the causes for ten commonly observed concentration-response patterns and provides for the proper interpretation of the test endpoints derived from these patterns for NOECs, LC 50, and EC25, thereby reducing the number of misclassified test results. The guidance provides one of three determinations based on the review steps: that calculated effect concentrations are reliable and should be reported, that calculated effect concentrations are anomalous and should be explained, or that the test was inconclusive and should be repeated with a newly collected sample. The standardized review of the effluent and receiving water concentration-response patterns provided by USEPA's 2000 guidance decreased discrepancies in data interpretation for NOEC, LC 50, and EC25 test results, thereby lowering the chance that a truly nontoxic sample would be misclassified and reported as toxic.~~

⁴ See, Supplementary Information in support of the Final Rule establishing WET test methods at 67 Fed. Reg. 69952, 69963, Nov. 19, 2002.

~~Appropriate interpretation of the measurement result from USEPA's TST statistical approach ("Pass"/"Fail") for effluent and receiving water samples is, by design, independent from the concentration-response patterns of the toxicity tests for those samples. Therefore, when using the TST statistical approach, application of USEPA's 2000 guidance on effluent and receiving waters concentration-response patterns will not improve the appropriate interpretation of TST results as long as all Test Acceptability Criteria (TAC) and other test review procedures—including those related to quality assurance for effluent and receiving water toxicity tests, reference toxicity tests, and control performance (mean, standard deviation, and coefficient of variation)—described by the WET test methods manual and TST guidance, are followed. The 2000 guidance may be used to identify reliable, anomalous, or inconclusive concentration-response patterns and associated statistical results to the extent that the guidance recommends review of test procedures and laboratory performance already recommended in the WET test methods manual. The guidance does not apply to single-concentration (IWC) and control statistical t-tests and does not apply to the statistical assumptions on which the TST is based. The San Diego Water Board will not consider a concentration-response pattern as sufficient basis to determine that a TST t-test result for a toxicity test is anything other than valid, absent other evidence. In a toxicity laboratory, unexpected concentration-response patterns should not occur with any regular frequency and consistent reports of anomalous or inconclusive concentration-response patterns or test results that are not valid will require an investigation of laboratory practices.~~

~~Any Data Quality Objectives or Standard Operating Procedure used by the toxicity testing laboratory to identify and report valid, invalid, anomalous, or inconclusive effluent or receiving water toxicity test measurement results from the TST statistical approach which include a consideration of concentration-response patterns and/or Percent Minimum Significant Differences (PMSDs) must be submitted for review by the San Diego Water Board, in consultation with USEPA, and the State Water Board's Quality Assurance Officer and Environmental Laboratory Accreditation Program (ELAP) (40 CFR section 122.44(h)). As described in the bioassay laboratory audit directives to the San Jose Creek Water Quality Laboratory from the State Water Board dated August 7, 2014, and from the USEPA dated December 24, 2013, the PMSD criteria only apply to compliance for NOEC and the sublethal endpoints of the NOEC, and therefore are not used to interpret TST results.~~

D. Final Effluent Limitation Considerations

1. Anti-Backsliding Requirements

NPDES permits must conform with Anti-backsliding requirements discussed in section III.C.5 of this Fact Sheet. These Anti-backsliding provisions require effluent limitations in a reissued permit to be as stringent as those in the previous permit, with some exceptions where limitations may be relaxed. This permit

complies with all applicable federal and State Anti-backsliding regulations and the effluent limitations are as stringent as those in Order No. R9-2012-0004.

2. Antidegradation Policies

The WDRs for the Discharger must conform with antidegradation requirements discussed in section III.C.4 of this Fact Sheet. The antidegradation policies require that beneficial uses and the water quality necessary to maintain those beneficial uses in the receiving waters of the discharge shall be maintained and protected, and, if existing water quality is better than the quality required to maintain beneficial uses, the existing water quality shall be maintained and protected unless allowing a lowering of water quality is necessary to accommodate important economic and social development or consistent with maximum benefit to the people of California. When a significant lowering of water quality is allowed by the San Diego Water Board, an antidegradation analysis is required in accordance with the State Water Board's Administrative Procedures Update (July 2, 1990), *Antidegradation Policy Implementation for NPDES Permitting*.

This Order carries over the TBELs in Order No. R9-2012-0004 for the Fallbrook WRP and adds TBELs for the proposed brine waste discharge. No degradation of the receiving water is expected.

This Order adds WQBELs for heptachlor and heptachlor epoxide due to reasonable potential being demonstrated for these constituents.

This Order complies with the antidegradation provision of 40 CFR section 131.12 and State Water Board Resolution No. 68-16.

3. Stringency of Requirements for Individual Pollutants

This Order contains both technology-based and WQBELs for individual pollutants. The TBELs consist of restrictions on CBOD₅, TSS, oil and grease, settleable solids, turbidity, and pH. Restrictions on these pollutants are discussed in section IV.B of this Fact Sheet. This Order's technology-based pollutant restrictions implement the minimum, applicable federal technology-based requirements. These limitations are not more stringent than required by the CWA.

WQBELs have been derived to implement water quality objectives that protect beneficial uses. Both the beneficial uses and the water quality objectives have been approved pursuant to federal law and are the applicable federal water quality standards. The procedures for calculating the individual WQBELs are based on the Ocean Plan, which was approved by USEPA on February 14, 2006 and has since been further amended. All beneficial uses and water quality objectives contained in the Basin Plan were approved under State law and submitted to and approved by USEPA prior to May 30, 2000. Any water quality objectives and beneficial uses submitted to USEPA prior to May 30, 2000, but not approved by USEPA before that date, are nonetheless "applicable water quality standards for purposes of the CWA" pursuant to 40 CFR section 131.21(c)(1).

Collectively, this Order's restrictions on individual pollutants are no more stringent than required to implement the requirements of the CWA.

E. Interim Effluent Limitations – Not Applicable

F. Land Discharge Specifications – Not Applicable

G. Recycling Specifications – Not Applicable

V. RATIONALE FOR RECEIVING WATER LIMITATIONS

Receiving water limitations of this Order are derived from the water quality objectives for ocean waters established by the Basin Plan and the Ocean Plan.

Prior to 2009, the San Diego Water Board interpreted the Bacterial Characteristics Water-contact Standards of the Ocean Plan to apply only in the zone bounded by the shoreline and a distance 1,000 feet from the shoreline or the 30-foot depth contour, whichever is further from the shoreline, and within kelp beds. The Ocean Plan provides that these Bacteriological Standards also apply in designated areas outside this zone used for water contact sports, as determined by the Regional Water Boards (i.e., all waters designated with the contact water recreation (REC-1) beneficial use). These designated areas must be specifically defined in the Basin Plan. Because the San Diego Water Board has designated the ocean waters with the REC-1 beneficial use in the Basin Plan, the Ocean Plan Bacterial Standards apply throughout State of California territorial marine waters in the San Diego Region, which extend from surface to bottom, out to three nautical miles from the shoreline. This interpretation has been confirmed by USEPA.

The Ocean Plan Bacterial Standards for total coliform, fecal coliform, and enterococcus were exceeded approximately 79 times at the offshore receiving water monitoring locations (including the offshore monitoring locations used as a reference) between 2011 and 2019. However, there is currently not enough information to confirm that the discharge from the Discharger was the cause of these exceedances in the receiving waters. Also, the Discharger sends tertiary-treated wastewater from the Fallbrook WRP to the OOO.

This Order includes the new bacterial provisions contained in the 2018 amendment to the Ocean Plan adopted by the State Water Board on August 7, 2018, approved by the USEPA on March 22, 2019, and effective on March 22, 2019.

VI. RATIONALE FOR PROVISIONS

A. Standard Provisions

Standard Provisions, which apply to all NPDES permits in accordance with 40 CFR section 122.41, and additional conditions applicable to specified categories of permits in accordance with 40 CFR section 122.42, are provided in the Standard Provisions (Attachment D).

Sections 122.41(a)(1) and (b) through (n) of 40 CFR establish conditions that apply to all State-issued NPDES permits. These conditions must be incorporated into the permits either expressly or by reference. If incorporated by reference, a specific citation to the regulations must be included in the order. Section 123.25(a)(12) of 40 CFR allows the State to omit or modify conditions to impose more stringent requirements. In accordance with 40 CFR section 123.25, this Order omits federal conditions that address enforcement authority specified in 40 CFR sections 122.41(j)(5) and (k)(2) because the enforcement authority under the Water Code is more stringent. In lieu of these conditions, this Order incorporates by reference Water Code section 13387(e).

B. Special Provisions

1. Reopener Provisions

This Order may be re-opened and modified, revoked and reissued, or terminated for cause in accordance with the provisions of 40 CFR parts 122, 123, 124, and 125. The San Diego Water Board may reopen the permit to modify permit conditions and requirements. Causes for modification include, but are not limited to, revisions to effluent limitations, receiving water requirements, and monitoring and reporting requirements; participation in the Southern California Coastal Water Research Project (SCCWRP) monitoring program or other regional or water body monitoring coalition as determined by the San Diego Water Board; revisions to sludge use or disposal practices; or adoption of new or revised regulations, water quality control plans, or policies by the State Water Board or the San Diego Water Board, including revisions to the Basin Plan or Ocean Plan.

2. Special Studies, Technical Reports, and Additional Monitoring Requirements

a. Spill Prevention and Response Plans

The CWA largely prohibits any discharge of pollutants from point sources to waters of the United States except as authorized under an NPDES permit. In general, any point source discharge of sewage effluent to waters of the United States must comply with technology-based, secondary treatment standards, at a minimum, and any more stringent requirements necessary to meet applicable water quality standards and other requirements. The unpermitted discharge of wastewater to waters of the United States is illegal under the CWA. Further, the Basin Plan prohibits discharges of waste to land, except as authorized by WDRs or the terms described in Water Code section 13264. The Basin Plan also prohibits the unauthorized discharge of

treated or untreated sewage to waters of the State or to a storm water conveyance system. Further, Discharge Prohibitions III.A and III.B of this Order prohibits the discharge of waste from the Facilities to a location other than Discharge Point No. 001.

Sanitary collection and treatment systems experience periodic failures resulting in discharges that may affect waters of the United States and/or State. There are many factors which may affect the likelihood of a spill. To ensure appropriate funding, management, and planning to reduce the likelihood of a spill, and to increase the level of response if a spill does occur, this Order requires the Discharger to maintain and implement Spill Prevention and Response Plans.

b. Spill Reporting Requirements

To determine compliance with Discharge Prohibitions III.A and III.B and provide appropriate notification to the general public for the protection of public health, spill reporting requirements have been established in section VI.C.2.b of this Order.

3. Best Management Practices and Pollution Prevention

The Pollutant Minimization Program is based on the requirements of the section III.C.9 of the Ocean Plan.

4. Construction, Operation, and Maintenance Specifications

- a. This Order carries over provisions from Order No. R9-2012-0004 to ensure that new treatment facilities and expansions of existing treatment facilities are completely constructed and operable prior to initiation of the discharge from the new or expanded facilities.
- b. This Order carries over a provision from Order No. R9-2012-0004 to ensure the waste treatment, containment, and disposal facilities are protected against the impact of storm events.
- c. This Order adds a provision to ensure the waste treatment, containment, and disposal facilities are protected against regional impacts due to climate change (e.g., sea level rise and floods).
- d. This Order adds a provision based on the requirements of 40 CFR section 122.41(e) to ensure the waste treatment, containment, and disposal facilities have adequate power.

5. Special Provisions for Wastewater Facilities

a. Ensuring Adequate Treatment Plant Capacity

Order No. R9-2012-0004 required the Discharger to submit a written report to the Executive Officer within 90 days after the monthly average influent flow rate equals or exceeds 75 percent of the secondary treatment design capacity of the Fallbrook WRP. In lieu of a written report for each time the monthly average influent flow rate equals or exceeds 75 percent of the

secondary treatment design capacity, the requirement has been changed to require the report be submitted four years prior to the time wastewater flows are projected to reach plant capacity, as stated in title 23, section 2232 of the CCR. The revised requirement states:

Four years prior to reaching POTW design capacity, the Discharger shall submit a Treatment Plant Capacity Report to the San Diego Water Board showing how flow volumes will be prevented from exceeding existing capacity or how capacity will be increased. A notification and copy of the report shall be sent to appropriate local elected officials, local permitting agencies, and the press. The required technical report shall be reviewed, approved, and jointly submitted to the San Diego Water Board by all planning and building departments having jurisdiction in the area served by the POTW. Opportunities for public participation and involvement are required during the preparation and development of the technical report. The report shall be accompanied by a statement outlining how interested persons were involved in the preparation of the technical report.

If the San Diego Water Board finds that the technical report indicates adequate steps are not being taken to address the capacity problem, the San Diego Water Board will adopt a time schedule order or other enforcement order. Such action will be preceded by notice and a hearing.

b. **Pretreatment Program**

Because the Fallbrook WRP does not currently receive discharges from industries that are subject to USEPA's pretreatment standards, the Discharger is not currently required to develop and implement an industrial pretreatment program. Consistent with Order No. R9-2012-0004, this Order requires the Discharger to perform an Industrial Waste Survey (IWS) and influent priority pollutant monitoring to determine whether a pretreatment program is required pursuant to 40 CFR part 403.

c. **Sludge (Biosolids) Requirements**

The use and disposal of biosolids within the U.S. is regulated under State and federal laws and regulations, including permitting requirements and technical standards included in 40 CFR part 503. The Discharger is required to comply with the standards and time schedules contained in 40 CFR part 503 for biosolids used or disposed of within the U.S.

Title 27, division 2, subdivision 1, section 20005 of the CCR establishes approved methods for the disposal of collected screenings, residual sludge, biosolids, and other solids removed from liquid wastes. Requirements to ensure the Discharger disposes of solids in compliance with State and federal regulations have been included in this Order.

d. **Collection System**

The State Water Board issued Order No. 2006-0003-DWQ, *Statewide General Waste Discharge Requirements for Sanitary Sewer System*

(Statewide General SSO Order) on May 2, 2006. The State Water Board amended the MRP for the Statewide General SSO Order through Order WQ 2013-0058-EXEC on August 6, 2013. The Statewide General SSO Order requires public agencies that own or operate sanitary sewer systems greater than one mile in length that collect and/or convey untreated or partially treated wastewater to a publicly-owned treatment facility to enroll for coverage and comply with the Statewide General SSO Order. The Statewide General SSO Order requires agencies to develop sanitary sewer management plans (SSMPs) and report all sanitary sewer overflows, among other requirements and prohibitions.

The Statewide General SSO Order contains requirements for operation and maintenance of collection systems and for reporting and mitigating sanitary sewer overflows that are more extensive, and therefore, more stringent than the requirements under federal standard provisions. The Discharger and public agencies that are discharging wastewater into the facility's collection system were required to obtain enrollment for regulation under the Statewide General SSO Order by December 1, 2006.

The San Diego Water Board issued Order No. R9-2007-0005, *Waste Discharge Requirements for Sewage Collection Agencies in the San Diego Region (Regional General SSO Order)*. Order No. R9-2007-0005 is more stringent and prescriptive than the Statewide General SSO Order. Agencies that are enrolled under the Statewide General SSO Order are also required to also comply with the Regional General SSO Order.

e. **Requirements for Receipt of Anaerobically Digestible Material**

Some wastewater facilities choose to accept organic material such as food waste, fats, oils, and grease into their anaerobic digesters for co-digestion to increase production of methane and other biogases for energy production and to prevent such materials from being discharged into the collection system, which could cause sanitary sewer overflows. The California Department of Resources Recycling and Recovery has proposed an exemption from requiring Process Facility/Transfer Station permits where this activity is regulated under WDRs or NPDES permits. The proposed exemption is restricted to anaerobically digestible material that has been prescreened, slurried, and processed/conveyed in a closed system to be co-digested with regular wastewater sludge. The proposed exemption requires that a wastewater facility develop Standard Operating Procedures (SOPs) for the proper handling, processing, tracking, and management of the anaerobically digestible material before it is received by the wastewater facility.

The SOPs are required for wastewater facilities that accept hauled food waste, fats, oil, and grease for injection into anaerobic digesters. The development and implementation of SOPs for management of these materials is intended to allow the California Department of Resources Recycling and Recovery to exempt this activity from separate and redundant

permitting programs. If the wastewater facility does not accept food waste, fats, oil, or grease for resource recovery purposes, it is not required to develop and implement SOPs.

6. Other Special Provisions – Not Applicable

7. Compliance Schedules – Not Applicable

VII. RATIONALE FOR MONITORING AND REPORTING REQUIREMENTS

CWA section 308 and 40 CFR sections 122.41(h), (j)-(l), 122.44(i), and 122.48 require that all NPDES permits specify monitoring and reporting requirements. Water Code sections 13267 and 13383 also authorize the San Diego Water Board to establish monitoring, inspection, entry, reporting, and recordkeeping requirements. The MRP (Attachment E) establishes monitoring, reporting, and recordkeeping requirements that implement federal and State requirements. The following provides the rationale for the monitoring and reporting requirements contained in the MRP for these Facilities. The burdens, including costs, of the MRP (Attachment E) required by this Order bear a reasonable relationship to the need for and benefits to be obtained from the MRP (Attachment E) to ensure compliance the Order, protect beneficial uses, and obtain other benefits as described in this Fact Sheet.

A. Core Monitoring Requirements

1. Influent Monitoring

Influent monitoring is required to assess the performance of treatment facilities and to evaluate compliance with effluent limitations. Influent monitoring frequencies and sample types for flow, CBOD₅, TSS, and oil and grease have been retained from Order No. R9-2012-0004.

Refer to section III.A of the MRP (Attachment E).

2. Effluent Monitoring

Effluent monitoring is required to determine compliance with the conditions of this Order, to identify operational problems, to improve plant performance, and to conduct reasonable potential analyses for subsequent orders. Effluent monitoring also provides information on wastewater characteristics for use in interpreting water quality and biological data. Effluent monitoring requirements have been carried over from Order No. R9-2012-0004, with the following exceptions.

- a. This Order adds monthly monitoring requirements for specific electrical conductivity to Monitoring Location M-002 to evaluate whether the dilution credit established in 2005 is still applicable and appropriate and to re-assess the dilution credit if the brine discharges from the Discharger, City of Oceanside; Genentech, Inc.; the Marine Corps Base, Camp Pendleton; and future brine discharges (along with increased recycled water use and decreased outfall discharge flows) changes effluent quality discharged at Discharge Point No. 001.

- b. This Order adds quarterly monitoring requirements for fecal coliform and enterococci to document the effluent's relationship with the receiving water monitoring data.
- c. This Order adds monthly monitoring requirements for ammonium, total nitrogen, nitrate, nitrite, total organic nitrogen, total phosphorus, phosphate, total organic carbon, dissolved inorganic carbon, dissolved iron, and alkalinity to gather data on the contribution of the discharge to ocean acidification, hypoxia, and harmful algal blooms. After one year of monitoring these parameters, the monitoring frequency for these parameters may be reduced from monthly to quarterly. This monitoring requirement may be delayed until January 2023 to allow for budgeting and proper laboratory staffing. The chemistry data will be used in the ocean acidification and hypoxia model currently under development by the Southern California Coastal Water Research Project (SCCWRP) to improve understanding of how changing seawater chemistry conditions manifest as adverse biological effects in vulnerable marine organisms. Increasing evidence suggests wastewater discharges of organic carbon and nutrients to ocean waters can contribute to ocean acidification and hypoxia. Upon discharge to ocean waters, organic carbon is broken down by bacteria, which consume dissolved oxygen during the decomposition process, triggering hypoxic conditions, increasing carbon dioxide (CO₂) levels and lowering pH. When nutrients such as nitrogen and phosphorus are introduced to ocean waters, they can trigger algae blooms which create more dissolved CO₂ when the algae die. Following death of the algae, the algae decomposed by bacteria further decreases dissolved oxygen levels and increases acidity. ~~quarterly monitoring requirements for total nitrogen and total phosphorus to gather data on the contribution nutrients to the receiving water, which has implications for ocean acidification, hypoxia, and harmful algal blooms.~~
- d. This Order increases the monitoring frequency for heptachlor, and heptachlor epoxide from annually to monthly consistent with monitoring programs in other POTW permits recently adopted by the San Diego Water Board for parameters with reasonable potential. (see section IV.C.3 of this Fact Sheet for more information).
- e. This Order increases the monitoring frequency for TCDD equivalents from semiannually to quarterly consistent with monitoring programs in other POTW permits recently adopted by the San Diego Water Board for parameters with reasonable potential. (see section IV.C.3 of this Fact Sheet for more information).
- f. This Order increases the monitoring frequency for many of the Table 3 Ocean Plan parameters from annually to semiannually consistent with monitoring programs in other POTW permits recently adopted by the San Diego Water Board for parameters without reasonable potential (see section IV.C.3 of this Fact Sheet for more information).

- g. For this Order, the Discharger may apply the performance goal for both chromium (VI) and chromium (III) as a total chromium performance goal. The Ocean Plan allows dischargers to meet the objective for chromium (VI) as a total chromium objective (footnote a, of Table 3 of the Ocean Plan). Total chromium includes both chromium (VI) and chromium (III) and applicable federal regulations in 40 CFR 136 under the CWA do not specify an analytical method for chromium (III)⁵. Thus, this Order allows the Discharger to also meet the objective for chromium (III) as a total chromium objective. If the Discharger only monitors for total chromium to meet the requirements for both chromium (VI) and chromium (III), the total chromium data will be used to determine if reasonable potential exists for both chromium (VI) and chromium (III) in future permit reissuances and/or updates.
- h. This Order adds monitoring requirements for the proposed discharge from the Santa Margarita GTP. The monitoring requirements include flow (recorder/totalizer), TSS (grab), pH (grab), oil and grease (grab), settleable solids (grab), and turbidity (grab). This is consistent with the brine monitoring requirements for Marine Corps Base, Camp Pendleton's Advanced Water Treatment Plant at Haybarn Canyon.
- i. This Order reduces the minimum sampling frequency for chronic toxicity from once per quarter to twice per year. upon request if ten consecutive routine monitoring events are all in compliance with the chronic toxicity effluent limitations.

Refer to section III.B of the MRP (Attachment E).

3. Whole Effluent Toxicity Testing Requirements

This Order contains a chronic toxicity performance goal as described in section IV.C.5 of this Fact Sheet.

Consistent with the requirements of the Ocean Plan, section III.C.6 of the MRP (Attachment E) requires the Discharger to develop an Initial Investigation Toxicity Reduction Evaluation (TRE) Work Plan and submit the Initial Investigation TRE Work Plan within one year of the effective date of this Order. The one-year timeframe was provided instead of the standard 90 days by request of the Discharger. The Initial Investigation TRE Work Plan must describe steps the Discharger intends to follow if the performance goal for chronic toxicity is exceeded.

Section III.C.10 of the Ocean Plan requires a TRE if a discharge consistently exceeds ~~the performancean effluent limitation goal~~ based on a toxicity objective in Table 3 of the Ocean Plan. While this Order does not include an effluent limitation for toxicity, it does include a performance goal for chronic toxicity that is based on the chronic toxicity objective in Table 3 of the Ocean Plan. Exceeding

⁵ In order to obtain a value for chromium (III), two separate methods must be used: one for total chromium determination and one for chromium (VI) determination. The value for chromium (III) is obtained by subtracting the chromium (VI) value from the total chromium value.

the chronic toxicity performance goal is not a violation of the Order; however, exceeding the performance goal is an indication of toxicity in the effluent that poses a threat to aquatic life and could cause a violation of the receiving water limitation in section V.A.4.a of the Order, which requires that marine communities not be degraded as the result of the discharge. Exceeding the chronic toxicity performance goal is also an indication of poor treatment performance or source control measures. A TRE can assist with identifying and correcting the cause of the toxicity in the effluent. Therefore, this Order requires a TRE if the discharge consistently exceeds the chronic toxicity performance goal to ensure the protection of marine communities and appropriate treatment performance and source control measures.

To determine if the discharge consistently exceeds the toxicity performance goal, this Order requires the Discharger to notify the San Diego Water Board and to accelerate toxicity testing if the ~~performance goal~~~~effluent limitation~~ for chronic toxicity is exceeded in any one test. If any of the additional tests demonstrate toxicity, in accordance with section III.C.10 of the Ocean Plan, the Discharger is required to submit a Detailed TRE Work Plan in accordance with its submitted Initial Investigation TRE Work Plan and USEPA guidance⁶ which shall include: further steps taken by the Discharger to investigate, identify, and correct the causes of toxicity; actions the Discharger will take to mitigate the effects of the discharge and prevent the recurrence of toxicity; and a schedule for these actions. The Discharger must also implement a Toxicity Identification Evaluation (TIE), as necessary, based upon the magnitude and persistence of toxicity performance goal exceedances. Once the source of toxicity is identified, the Discharger must take all reasonable steps to reduce the toxicity to meet the chronic toxicity performance goal identified in section IV.A.2 of this Order.

The above accelerated monitoring (a minimum of ~~four~~~~six~~ succeeding tests performed at 14-day intervals) is based on the probability of encountering at least one toxicity exceedance assuming a true, but unknown level of occurrence.

Within 30 days of completion of the TRE, the Discharger must submit the results of the TRE, including a summary of the findings, data generated, a list of corrective actions taken or planned to achieve consistent compliance with the toxicity performance goal of this Order and prevent recurrence of exceedances of the ~~these~~ performance goal, and a time schedule for implementation of any planned corrective actions. The Discharger must implement any planned corrective actions in the TRE Final Report in accordance with the specified time schedule, unless otherwise directed in writing by the San Diego Water Board. The

⁶ See (a) *TRE Guidance for Municipal Wastewater Treatment Plants* (EPA 833-B-99-002, 1999); (b) *Generalized Methodology for Conducting Industrial Toxicity Reduction Evaluations* (EPA/600/2-88/070); *Toxicity Identification Evaluation, Phase I* (EPA/600/6-91/005F); (c) *Methods for Aquatic Toxicity Identification Evaluations, Phase II* (EPA/600/R-92/080); (d) *Methods for Aquatic Toxicity Identification Evaluations, Phase III* (EPA/600/R-92/081); and (e) *Marine Toxicity Identification Evaluation (TIE): Phase I Guidance Document* (EPA/600/R-96-054, 1996).

corrective actions and time schedule must be modified at the direction of the San Diego Water Board.

Refer to section III.B of the MRP (Attachment E).

B. Receiving Water Monitoring

The receiving water and sediment monitoring requirements set forth below are designed to measure the effects of the OOO discharge on the receiving water. These monitoring requirements will remain in effect on an interim basis, pending development of a new and updated monitoring and assessment program.

Refer to section IV of the MRP (Attachment E).

1. Surf Zone Water Quality Monitoring Requirements

Surf zone water quality monitoring is required to determine if the effluent is causing or contributing to exceedances of the water quality standards in the surf zone, the area where the ocean surface waves come closer to shore and break. For monitoring locations S1 through S5, monitoring for enterococcus bacteria has been changed to monitoring for enterococci bacteria; weekly monitoring a minimum of five samples for fecal coliform and enterococcus are required to be collected within a rolling 30-day period~~has been increased to five times per month~~; and weekly monitoring for total coliform has been decreased to three times per month. These changes reflect the new bacterial provisions contained in the 2018 amendment to the Ocean Plan. The 2018 amendment to the Ocean Plan also removes the requirement to conduct repeat sampling if a single sample exceeds any of the bacterial single sample maximum standards. Thus, this repeat sampling requirement has not been carried over from Order No. R9-2012-0004. Monitoring locations S6 and S7 were not specified in Order No. R9-2012-0004 and will be determined by the San Diego Water Board based on the results of the Plume Tracking Monitoring Program.

Refer to section IV.A of the MRP (Attachment E).

2. Nearshore and Offshore Water Quality Monitoring Requirements

Nearshore and offshore water quality monitoring is required to determine if the effluent is causing or contributing to exceedances of the water quality standards outside of the ZID, to determine the fate of the effluent plume, evaluate the contribution of the discharge to ocean acidification, and to gather data for future permit reissuances. Nearshore and offshore monitoring requirements have been carried over from Order No. R9-2011-0016 with the following exceptions:

- a. For nearshore monitoring requirements, this Order adds monitoring requirements for temperature, depth, dissolved oxygen, light transmittance, pH, colored dissolved organic matter (CDOM), and salinity to better understand the characteristics of the wastewater plume, evaluate compliance with receiving water limitations, and incorporate the requirements of Ocean Plan. ~~This Order also adds monitoring requirements for nitrogen and phosphorus to evaluate the contribution of nutrients to the receiving water,~~

~~which has implications for ocean acidification, hypoxia, and harmful algal blooms.~~

- b. For offshore monitoring requirements, this Order changes the pH monitoring from a grab sample at the surface to profile monitoring; the temperature, dissolved oxygen, and light transmittance monitoring from surface, mid-depth, and bottom to profile monitoring; and the conductivity surface, mid-depth, and bottom monitoring to salinity profile monitoring. These changes are also made to better understand the characteristics of the wastewater plume. ~~This Order also adds monitoring requirements for nitrogen and phosphorus to evaluate the contribution of nutrients to the receiving water, which has implications for ocean acidification, hypoxia, and harmful algal blooms.~~
- b.c. ~~For offshore monitoring requirements, this Order adds a monitoring requirement for CDOM to be consistent with Appendix III section 10.1 of the Ocean Plan.~~
- e.d. For nearshore and offshore monitoring requirements, monitoring for enterococcus bacteria has been changed to monitoring for enterococci and monitoring requirements for total coliform has been removed. These changes reflect the new bacterial provisions contained in the 2018 amendment to the Ocean Plan.
- e. This Order requires the Discharger to collect samples for the human associated Human Marker HF183 genetic marker (HF183) concurrently with samples collected for fecal coliform at the offshore monitoring locations. The ~~Human Marker~~ HF183, derived from the 16S rRNA gene of *Bacteroides*, has been widely used to identify sewage pollution in coastal waters. For this Order, monitoring for the ~~Human Marker~~ HF183 is used to confirm the presence of human fecal material when the single sample maximum receiving water limitation for fecal coliform is exceeded. Analysis of the ~~Human Marker~~ HF183 is only required if the sample for fecal coliform exceeds the single sample maximum receiving water limitation. Results for the ~~Human Marker~~ HF183 is used for informational purposes only, there is no receiving water limitation for the ~~Human Marker~~ HF183. This requirement was included due to approximately 73 exceedances of bacteria receiving water limitations between the years 2011 through 2019 at the offshore monitoring locations located near the OOO (i.e., monitoring locations A1 through A5). During 2011 to 2019, only six exceedances of bacteria receiving water limitations occurred at the reference stations (i.e., monitoring locations B1 and B2).
- f. This Order recommends the Discharger monitor for pH by spectrophotometric technique and total alkalinity at a subset of offshore monitoring locations, as recommended by SCCWRP to provide more precise measure of pH for compliance with the pH receiving water limitation in section V.A 3.b. of this Order and to evaluate contributions of POTWs on ocean acidification. Measurements of pH by spectrophotometric technique

and total alkalinity is used provide a more precise measure of pH in the receiving water and to calibrate the pH measurements collected by potentiometric sensors (i.e., glass electrodes) attached to conductivity-temperature-depth (CTD) profile samplers utilized during routine receiving water monitoring. Section V.A.3.b of this Order requires that pH shall not be changed at any time more than 0.2 standard units from that which occurs naturally. The imprecision of pH measurement technology (e.g., glass electrodes) has been well documented in the scientific literature. The margin of error associated with using dated technology to measure pH can be greater than 0.2 pH units, which makes it impossible to achieve the precision required to measure compliance with the pH receiving water limitation in section V.A.3.b of this Order. However, calibrating glass electrodes with measurements of pH by spectrophotometric technique and total alkalinity in the laboratory can increase the precision of the glass electrodes measurements collected in the field.

In addition to imprecision of glass electrodes, emerging evidence suggests that monitoring parameters other than pH, especially aragonite saturation state (relevant to shell-building in calcifying organisms) and partial pressure of carbon dioxide (relevant to fish behavior and navigation) is needed to assess ocean acidification effects (see *The West Coast Ocean Acidification and Hypoxia Science Panel, Major Findings, Recommendations, and Actions*, Appendix G at Pgs. 26-27 available at: https://ftp.sccwrp.org/pub/download/DOCUMENTS/TechnicalReports/926_WestCoastOAHSciencePanel.pdf). While the main driver of ocean acidification is due to atmospheric carbon dioxide, the discharge of anthropogenic nutrients from wastewater treatment plants may exacerbate ocean acidification, especially on smaller spatial scales. A recent study⁷ suggests that nutrients from wastewater effluent can provide a significant source of nitrogen for nearshore productivity in Southern California waters, and may be equivalent to upwelling on smaller spatial scales that are more relevant to algal blooms. Anthropogenic nutrients from wastewater effluent may increase algal blooms. As these algal blooms die off, the decay promotes bacterial respiration resulting in increased carbon dioxide, lower pH, and decreases in oxygen (e.g., hypoxia). This Order recommends and requests the Discharger calculate aragonite saturation state to evaluate the potential effects of the discharge on ocean acidification.

d.g. Monitoring frequency at nearshore and offshore stations has been reduced from monthly to quarterly to help offset the costs of additional monitoring requirements and the development of a Plume Tracking Monitoring Program.

⁷ Howard, M.D.A., Sutula, M., Caron, D., Chao, Y., Farrara, J.D., Frenzel, H., Jones, B.H., Robertson, G., McLaughlin, K., Sengupta, A., 2012. Comparison of Natural and Anthropogenic Nutrient Sources in the Southern California Bight, Southern California Coastal Water Research Project Annual Report, Costa Mesa, CA.

e-h. Monitoring locations N6 and N7 were not specified in Order No. R9-2011-0016 and will be determined by the San Diego Water Board based on the results of the Plume Tracking Monitoring Program.

Refer to sections IV.B of the MRP (Attachment E).

3. Benthic Monitoring Requirements

Sediments integrate constituents that are discharged to the ocean. Most particles that come from the OOO discharge, and any associated contaminants, will eventually settle to the seafloor where they are incorporated into the existing sediments. Sediments can accumulate these particles over the years until the point where sediment quality has degraded and beneficial uses are impaired.

Consistent with Appendix III, section 6 of the Ocean Plan, section IV.C of the MRP (Attachment E) requires periodic assessment of sediment quality to evaluate potential effects of the OOO discharge and compliance with narrative water quality standards specified in the Ocean Plan. The required assessment consists of the measurement and integration of three lines of evidence: 1) physical and chemical properties of seafloor sediments, 2) seafloor sediment toxicity to assess bioavailability and toxicity of sediment contaminants, and 3) ecological status of the biological communities (benthos) that live in or on the seafloor sediments.

The benthic community is strongly affected by sediment composition (e.g., sand, silt, and clay distributions), sediment quality (e.g., chemistry, toxicity), and water quality. Because benthic macroinvertebrates (e.g., infauna) are dependent on their surroundings, they often serve as important biological indicators that reflect the overall conditions of the marine environment.

Benthic monitoring requirements have been updated from Order No. R9-2012-0004 to provide more detail on monitoring frequency, methods, and analyses. This Order adds monitoring requirements for dissolved sulfide, acid volatile sulfides, selenium, iron, tin, aluminum, total nitrogen, total organic carbon, pesticides, DDT, PCBs, and PAHs to help determine if concentrations of pollutants in marine sediments are at levels that would degrade the benthic community. This Order removes the monitoring requirements for BOD, chemical oxygen demand, cyanide, sulfides, and radioactivity. While the Ocean Plan requires annual monitoring for sediment chemistry, the San Diego Water Board reduced the frequency to once per permit term to offset costs of the Plume Tracking Monitoring Program and other monitoring requirements.

This Order also requires sediment toxicity analyses at each offshore station. Sediment toxicity is a measure of the response of invertebrates exposed to surficial sediments under controlled laboratory conditions. The sediment toxicity line of evidence is used to assess both pollutant-related biological effects and exposure and provides a measure of exposure to all pollutants present, including non-traditional or unmeasured chemicals.

To help facilitate collaboration with other agencies conducting receiving water sediment monitoring (e.g., dischargers to the Encina Ocean Outfall and/or San Elijo Ocean Outfall) or participation in a regional monitoring program, receiving water sediment monitoring may be conducted anytime within the permit term, with the results due no later than 180 days before the expiration of this Order. This Order requires the development of a Benthic Monitoring Work Plan, which includes a schedule for completion of sediment sampling and submission of the results, protocols for sediment sample collection and processing, and the proposed methods for analyzing the sediment data and integrating the three lines of evidence.

Refer to section IV.C of the MRP (Attachment E).

4. Fish and Macroinvertebrate Monitoring Requirements

Marine aquatic invertebrates are excellent indicators of ecosystem health because they are ubiquitous, abundant, diverse, and typically sedentary. The growth, survival, and reproduction of aquatic invertebrates are all sensitive to declines in environmental health, making analysis of assemblage structure a good ecosystem monitoring tool. Additionally, many pollutants discharged into receiving waters have the potential to bioaccumulate and persist in the tissues of aquatic organisms, including marine fishes. Chemical pollutants that bioaccumulate tend to magnify in concentration as they pass through the aquatic food chain. Fish monitoring data is required to assess the human health risks for individuals who may consume fish and to assess trends of contaminants levels in the receiving water over time.

Fish and invertebrate monitoring requirements have been updated from Order No. R9-2012-0004. This Order replaces the requirement to conduct diver surveys with a requirement to conduct benthic trawls at three trawling locations once per permit term. Benthic trawls are a standard method to evaluate the impact of ocean outfall discharges. Unlike diver surveys, benthic trawls are not restricted by depth, which allows evaluation of the benthic community at the same depth contour as the outfall. Additionally, benthic trawls allow for greater data comparability to other ocean outfalls in the San Diego region and the Southern California Bight.

Consistent with appendix III, section 9 of the Ocean Plan, this Order also requires the analyses of fish tissue samples once per permit term to determine if concentrations of pollutants are bioaccumulating to levels that are harmful to human health or degrade marine communities. The fish targeted for tissue analysis are: 1) flatfish, which have greater exposure to pollutants sources such as sediment and sediment-dwelling prey; and 2) rockfish, which are commonly targeted by commercial and recreational fishers. Flatfish tissue samples will be collected at or near the trawl stations. The three trawl stations are classified into three zones for the purpose of collecting sufficient numbers of fish for tissue analyses. Trawl Zone 1 represents the nearfield zone, defined as the area within a 0.5-km radius of trawl station T3; Trawl Zone 2 is considered the down coast zone, defined as the area within a 1-km radius of trawl station T4; and Trawl Zone

3 is considered the up coast zone, and is defined as the area centered within a 1-km radius of trawl station T5. The fish shall be collected by hook and line or by setting baited lines from within zones surrounding rig fishing stations RF1, RF2, and RF3 listed in Table E-1. Rig Fishing Zone 2 is the nearfield area centered within a 0.5-km radius of station RF2; Rig Fishing Zone 1 represents the northern farfield area centered within 1-km radius of station RF1; and Rig Fishing Zone 3 represents the southern farfield area centered within a 1-km radius of station RF3.

5. Receiving Water Monitoring Reporting Requirements

a. California Environmental Data Exchange Network

The California Environmental Data Exchange Network (CEDEN) is a central database to find and share information about the State's water bodies, including streams, lakes, rivers, and the coastal ocean. Many groups in the State monitor water quality, aquatic habitat, and wildlife health to ensure good stewardship of our ecological resources. CEDEN aggregates this data and makes it accessible to environmental managers and the public. CEDEN requires a minimum level of quality assurance and quality control measures to ensure the data reported are of sufficient quality. When developing integrated reports for listing impaired water bodies pursuant to Clean Water Act section 303(d), Regional Water Quality Control Boards must rely on data submitted to CEDEN to make determinations of water quality. Data not included in CEDEN are generally not permissible to be included in the integrated reports.

This Order requires the Discharger submit applicable receiving water monitoring data to CEDEN or an equivalent database that is linked to CEDEN to ensure the monitoring data is available to the San Diego Water Board for integrated reports and is easily accessible to the public and other environmental managers. Additionally, the requirement to submit receiving water monitoring data to CEDEN is consistent with the San Diego Water Board's *A Framework for Monitoring and Assessment in the San Diego Region*, endorsed by the San Diego Water Board in December 2012 through Resolution No. R9-2012-0069. CEDEN currently does not accept all receiving water monitoring data collected pursuant to this Order; however, CEDEN is periodically updated to accept additional data. The Discharger is only required to submit data that is accepted by CEDEN until such time CEDEN is updated to accept the data. The Discharger is required to submit an annual certification that all applicable receiving water monitoring data has been timely uploaded to CEDEN.

b. Receiving Water Monitoring Report

The Receiving Water Monitoring Report requirements included in this Order have been updated from the previous Order, Order No. R9-2012-0004. This Order reduces the frequency of the Receiving Water Monitoring Report from annually to once per permit term and provides additional details on what is

required to be included in the Receiving Water Monitoring Report. The main objectives of the Receiving Water Monitoring Report is to 1) evaluate compliance with the receiving water limitations of this Order including California Ocean Plan water quality objectives and water-contact bacteriological standards; 2) identify any biological or chemical changes in the receiving water that may be associated with the wastewater discharge and 3) answer the key regulatory questions posed in the MRP that the receiving water monitoring program was designed to answer. The assessment and evaluation of the receiving water monitoring data in the Receiving Water Monitoring Report documents any effects of wastewater discharge, other anthropogenic influences (e.g., storm water discharge, urban runoff), or natural factors (e.g., climate changes) on coastal water quality, seafloor sediment conditions, and local marine organisms. The Receiving Water Monitoring Report helps educate the public about potential water quality impacts resulting from the discharge.

c. State of the Ocean

This Order requires the Discharger present a State of the Ocean oral report to the San Diego Water Board at a public Board meeting. The State of the Ocean oral report summarizes the conclusions of the Receiving Water Monitoring Report, including a description of the monitoring efforts completed, the status and trends of receiving water quality conditions, and plans for future monitoring efforts. Similar to the Receiving Water Monitoring Report, the State of the Ocean oral report helps educate the public about potential water quality impacts resulting from the discharge in a more concise and approachable manner. In addition to educating the public, the State of the Ocean oral report also allows for public participation by providing a forum for the public to comment on the findings presented by the Discharger.

5.6. Groundwater – Not Applicable

C. Regional Monitoring Requirements

Regional ocean water monitoring provides information about the sources, fates, and effects of anthropogenic contaminants in the coastal marine environment necessary to make assessments over large areas. The large-scale assessments provided by regional monitoring describe and evaluate cumulative effects of all anthropogenic inputs and enable better decision-making regarding protection of beneficial uses of ocean waters. Regional monitoring data assists in the interpretation of core monitoring studies by providing a more accurate and complete characterization of reference conditions and natural variability. Regional monitoring also leads to methods standardization and improved quality control through inter-calibration exercise. The coalitions implementing regional monitoring enable sharing of technical resources, trained personnel, and associated costs. Focusing these resources on regional issues and developing a broader understanding of pollutants effects in ocean waters enables the development of more rapid and effective response strategies. Based on all of

these considerations the San Diego Water Board supports regional approaches to monitoring ocean waters.

The Discharger shall, as directed by the San Diego Water Board, participate with other regulated entities, other interested parties, and the San Diego Water Board in development and implementation of new and improved monitoring and assessment programs for ocean waters in the San Diego Region and discharges to those waters.

Refer to section V of the MRP (Attachment E).

1. Kelp Bed Canopy Monitoring Requirements

Kelp consists of a number of species of brown algae. Along the central and southern California coast, giant kelp (*Macrocystis pyrifera*) is the largest species colonizing rocky, and in some cases sandy, subtidal habitats. Giant kelp is an important component of coastal and island communities in southern California, providing food and habitat for numerous animals.

Refer to section V.A of the MRP (Attachment E).

2. Southern California Bight Regional Monitoring Program Participation Requirements

The Southern California Bight (Bight), defined as the concave bend of the shoreline extending from Point Conception to Punta Colonet in Mexico, is host to unique, biologically diverse marine ecosystems that have long been vulnerable to the impacts of human activity. The coastal zone of the Bight hosts nearly 22 million U.S. residents that engage in a wide variety of industrial, military, and recreational activities. Approximately 5,600 miles of watersheds, half of which is highly developed, drain into the Bight. The Southern California Bight Regional Monitoring Program brings together researchers and water-quality managers to pool their resources and work together to investigate the condition of marine ecosystems both spatially and temporally, and extend greater protections to the Bight's diverse habitats and natural resources.

The Discharger is required to participate in the Southern California Bight Regional Monitoring Program coordinated by SCCWRP, or any other coordinator named by the San Diego Water Board, pursuant to Water Code sections 13267 and 13383, and 40 CFR section 122.48. The intent of the Southern California Bight Regional Monitoring Program is to maximize the efforts of all monitoring partners using a more cost-effective monitoring design and to best utilize the pooled scientific resources of the Bight.

During these coordinated sampling efforts, the Discharger's receiving water sampling and analytical effort, as defined in section IV of the MRP (Attachment E), may be reallocated to provide a regional assessment of the impact of the discharge of municipal wastewater to the Bight. In that event, the San Diego Water Board shall notify the Discharger in writing that the requirement to perform the receiving water sampling and analytical effort defined in section IV of the MRP (Attachment E) is suspended for the duration of the reallocation. Anticipated modifications to the monitoring program will be coordinated so as to provide a

more comprehensive picture of the ecological and statistical significance of monitoring results and to determine cumulative impacts of various pollution sources. The level of resources in terms of sampling and analytical effort redirected from the receiving water monitoring program required under section IV of the MRP (Attachment E) shall equal the level of resources provided to implement the regional monitoring and assessment program, unless the San Diego Water Board and the Discharger agree otherwise. The specific scope and duration of the receiving water monitoring program reallocation and redirection shall be determined and set by the San Diego Water Board, in consultation with the Discharger.

Refer to section V.B of the MRP (Attachment E).

D. Special Studies Requirements

1. **Climate Change Action Plan.** Changing climate conditions may fundamentally alter the way wastewater facilities are designed and operated. Climate change research indicates the overarching driver of change is increased atmospheric carbon dioxide (CO₂) from human activity. The increased CO₂ emissions trigger changes to climatic patterns, which increase the intensity of sea level rise and coastal storm surges (Δ Sea Level), lead to more erratic rainfall and local weather patterns (Δ Weather Patterns), trigger a gradual warming of freshwater and ocean temperatures (Δ Water Temperature), and trigger changes to ocean water chemistry (Δ Water pH). The changes to the sea level and weather patterns may affect the Facilities (e.g., flooding, wildfires, increased influent flows during wet weather, and heat waves). The changes to the water temperature and pH may affect how the receiving waters reacts to the discharges.

The California Public Resources Code (Public Resources Code) recognizes that anthropogenic greenhouse gas emissions responsible for climate change are also driving major shifts in the chemical properties of the world's oceans (Public Resources Code section 35630(c)). Furthermore, Governor Newsom's Executive Order N-10-1920 directs state agencies to prepare a water resiliency portfolio that meets the needs of California's communities, economy, and environment. The State Water Board's Resolution No. 2017-0012, Comprehensive Response to Climate Change, and the San Diego Water Board's Resolution No. R9-2018-0051, Addressing Threats to Beneficial Uses from Climate Change, also require a proactive approach to climate change in all state and regional actions.

Based on all of these considerations, this Order requires the Discharger to prepare and submit a Climate Change Action Plan (CCAP) within three years of the effective date of this Order.

2. **Plume Tracking Monitoring Program.** This Order includes a requirement to implement a Plume Tracking Monitoring Program. Plume tracking is necessary to determine if the plume is moving towards the shore or surface where it may encroach upon water recreation areas. Additionally, plume direction and mixing have a direct effect on sediment loading as the direction of the plume determines where the discharged particles will eventually settle. While nearshore and

offshore monitoring can identify if the plume is encroaching upon the water recreation area during typical oceanographic conditions, infrequent sampling at preset intervals is unable to capture atypical oceanographic conditions that may lead to abnormal plume behavior. The Plume Tracking Monitoring Program will be used to evaluate whether the monitoring methods and locations established in Order Nos. R9-2005-0136 and R9-2011-0016 and reinstated in this Order are still appropriate and applicable.

E. Other Monitoring Requirements

Discharge Monitoring Report-Quality Assurance (DMR-QA) Study Program.

Under the authority of section 308 of the CWA (33 U.S.C. section 1318), USEPA requires major and selected minor permittees under the NPDES Program to participate in the annual DMR-QA Study Program. The DMR-QA Study evaluates the analytical ability of laboratories that routinely perform or support self-monitoring analyses required by NPDES permits. There are two options to satisfy the requirements of the DMR-QA Study Program: (1) The Discharger can obtain and analyze a DMR-QA sample as part of the DMR-QA Study; or (2) Per the waiver issued by USEPA to the State Water Board, the Discharger can submit the results of the most recent Water Pollution Performance Evaluation Study from its own laboratories or its contract laboratories. A Water Pollution Performance Evaluation Study is similar to the DMR-QA Study. Thus, it also evaluates a laboratory's ability to analyze wastewater samples to produce quality data that ensure the integrity of the NPDES Program. The Discharger shall ensure that the results of the DMR-QA Study or the results of the most recent Water Pollution Performance Evaluation Study are submitted annually to the State Water Board. The State Water Board's Quality Assurance Program Officer will send the DMR-QA Study results or the results of the most recent Water Pollution Performance Evaluation Study to USEPA's DMR-QA Coordinator and Quality Assurance Manager.

Refer to section I.H of the MRP (Attachment E)

VIII. PUBLIC PARTICIPATION

The San Diego Water Board has considered the issuance of WDRs that will serve as an NPDES permit for the Facilities. As a step in the WDR adoption process, the San Diego Water Board staff has developed tentative WDRs and has encouraged public participation in the WDR adoption process by providing a period of a minimum of 30 days for public review and comment on the Tentative Order.

A. Notification of Interested Parties

The San Diego Water Board notified the Discharger and interested agencies and persons of its intent to prescribe WDRs for the discharge and provided an opportunity to submit written comments and recommendations. Notification was provided by posting a Notice of Public Hearing and Comment and the tentative WDRs on the San Diego Water Board's website for the duration of the public comment period. The Tentative Order was posted on the San Diego Water Board website and emailed to the Discharger and all known interested parties on September 27, 2019.

The public also had access to the meeting agenda including all supporting documents and any changes in meeting dates and locations through the San Diego Water Board's website at: <https://www.waterboards.ca.gov/sandiego/>
<http://www.waterboards.ca.gov/sandiego/><https://www.waterboards.ca.gov/sandiego/>.

B. Written Comments

Interested persons were invited to submit written comments concerning tentative WDRs as provided through the notification process. Comments were due either in person or by mail to the Executive Office at the San Diego Water Board at 2375 Northside Drive, Suite 100, San Diego, CA 92108.

To be fully responded to by staff and considered by the San Diego Water Board, the written comments were due at the San Diego Water Board office by 5:00 p.m. on October 28, 2019.

C. Public Hearing

The San Diego Water Board held a public hearing on the tentative WDRs during its regular Board meeting on the following dates and time and at the following location:

Date: December 11, 2019 and February 12, 2020

Time: 9:00 AM

Location: San Diego Regional Water Quality Control Board, San Diego Water Board Meeting Room, 2375 Northside Drive, Suite 108, San Diego, California 92108

Interested persons were invited to attend. At the public hearing, the San Diego Water Board heard testimony, pertinent to the discharge, WDRs, and permit. For accuracy of the record, important testimony was requested in writing.

D. Reconsideration of Waste Discharge Requirements

Any person aggrieved by this action of the San Diego Water Board may petition the State Water Board to review the action in accordance with Water Code section 13320 and CCR, title 23, sections 2050. The State Water Board must receive the petition by 5:00 p.m., within 30 calendar days of the date of adoption of this Order at the following address, except that if the thirtieth day following the date of this Order falls on a Saturday, Sunday, or State holiday, the petition must be received by the State Water Board by 5:00 p.m. on the next business day. Petitions may be sent in as follows:

By mail:

State Water Resources Control Board
Office of Chief Counsel
P.O. Box 100, 1001 I Street
Sacramento, CA 95812-0100

In Person:

State Water Resources Control Board
Office of Chief Counsel
1001 I Street
Sacramento, California 95814

By email: waterqualitypetitions@waterboards.ca.gov

By fax:
(916) 341-5199

For instructions on how to file a petition for review, see:

http://www.waterboards.ca.gov/public_notices/petitions/water_quality/wqpetition_instr.shtml
https://www.waterboards.ca.gov/public_notices/petitions/water_quality/wqpetition_instr.shtml

E. Information and Copying

The ROWD, other supporting documents, and comments received are on file and may be inspected at the address above at any time between 8:30 a.m. and 4:45 p.m., Monday through Friday. Copying of documents may be arranged through the San Diego Water Board by calling (619) 516-1990.

F. Register of Interested Persons

Any person interested in being placed on the mailing list for information regarding the WDRs and NPDES permit should contact the San Diego Water Board, reference the Facilities, and provide a name, address, and phone number.

G. Additional Information

Requests for additional information or questions regarding this Order should be directed to Joann Lim by email at Joann.Lim@waterboards.ca.gov or by phone at (619) 521-3362.

ATTACHMENT G – DISCHARGE PROHIBITIONS CONTAINED IN THE OCEAN PLAN AND BASIN PLAN

I. Ocean Plan Discharge Prohibitions

- A. The Discharge of any radiological chemical, or biological warfare agent or high-level radioactive waste into the ocean is prohibited.
- B. Waste shall not be discharged to designated Areas of Special Biological Significance except as provided in chapter III.E. of the Ocean Plan.
- C. Pipeline discharge of sludge to the ocean is prohibited by federal law; the discharge of municipal and industrial waste sludge directly to the ocean, or into a waste stream that discharges to the ocean, is prohibited. The discharge of sludge digester supernatant directly to the ocean, or to a waste stream that discharges to the ocean without further treatment, is prohibited.
- D. The by-passing of untreated wastes containing concentrations of pollutants in excess of those of Table 3 or Table 4 [of the Ocean Plan] to the ocean is prohibited, except as allowed by Federal Standard Provisions I.G and I.H (Attachment D).
- E. The discharge of trash to surface waters of the State or the deposition of trash where it may be discharged into surface waters of the State is prohibited.

II. Basin Plan Discharge Prohibitions

- A. The discharge of waste to waters of the State in a manner causing, or threatening to cause a condition of pollution, contamination or nuisance as defined in Water Code section 13050, is prohibited.
- B. The discharge of waste to land, except as authorized by WDRs of the terms described in Water Code section 13264 is prohibited.
- C. The discharge of pollutants or dredged or fill material to waters of the United States except as authorized by an NPDES permit or a dredged or fill material permit (subject to the exemption described in Water Code section 13376) is prohibited.
- D. Discharges of recycled water to lakes or reservoirs used for municipal water supply or to inland surface water tributaries thereto are prohibited, unless the San Diego Water Board issues an NPDES permit authorizing such a discharge; the proposed discharge has been approved by the State of California Department of Public Health and the operating agency of the impacted reservoir; and the discharger has an approved fail-safe long-term disposal alternative.
- E. The discharge of waste to inland surface waters, except in cases where the quality of the discharge complies with applicable receiving water quality objectives, is prohibited. Allowances for dilution may be made at the discretion of the San Diego Water Board. Consideration would include streamflow data, the degree of treatment provided and safety measures to ensure reliability of facility performance. As an example, discharge of secondary effluent would probably be permitted if streamflow provided 100:1 dilution capability.
- F. The discharge of waste in a manner causing flow, ponding, or surfacing on lands not owned or under the control of the discharger is prohibited, unless the discharge is authorized by the San Diego Water Board.

- G.** The dumping, deposition, or discharge of waste directly into waters of the State, or adjacent to such waters in any manner which may permit it's being transported into the waters, is prohibited unless authorized by the San Diego Water Board.
- H.** Any discharge to a storm water conveyance system that is not composed entirely of storm water is prohibited unless authorized by the San Diego Water Board. [The federal regulations, 40 CFR section 122.26(b)(13), define storm water as storm water runoff, snow melt runoff, and surface runoff and drainage. 40 CFR section 122.26(b)(2) defines an illicit discharge as any discharge to a storm water conveyance system that is not composed entirely of storm water except discharges pursuant to an NPDES permit and discharges resulting from firefighting activities.] [section 122.26 amended at 56 FR 56553, November 5, 1991; 57 FR 11412, April 2, 1992].
- I.** The unauthorized discharge of treated or untreated sewage to waters of the State or to a storm water conveyance system is prohibited.
- J.** The discharge of industrial wastes to conventional septic tank/subsurface disposal systems, except as authorized by the terms described in Water Code section 13264, is prohibited.
- K.** The discharge of radioactive wastes amenable to alternative methods of disposal into the waters of the State is prohibited.
- L.** The discharge of any radiological, chemical, or biological warfare agent into waters of the State is prohibited.
- M.** The discharge of waste into a natural or excavated site below historic water levels is prohibited unless the discharge is authorized by the San Diego Water Board.
- N.** The discharge of sand, silt, clay, or other earthen materials from any activity, including land grading and construction, in quantities which cause deleterious bottom deposits, turbidity or discoloration in waters of the State or which unreasonably affect, or threaten to affect, beneficial uses of such waters is prohibited.